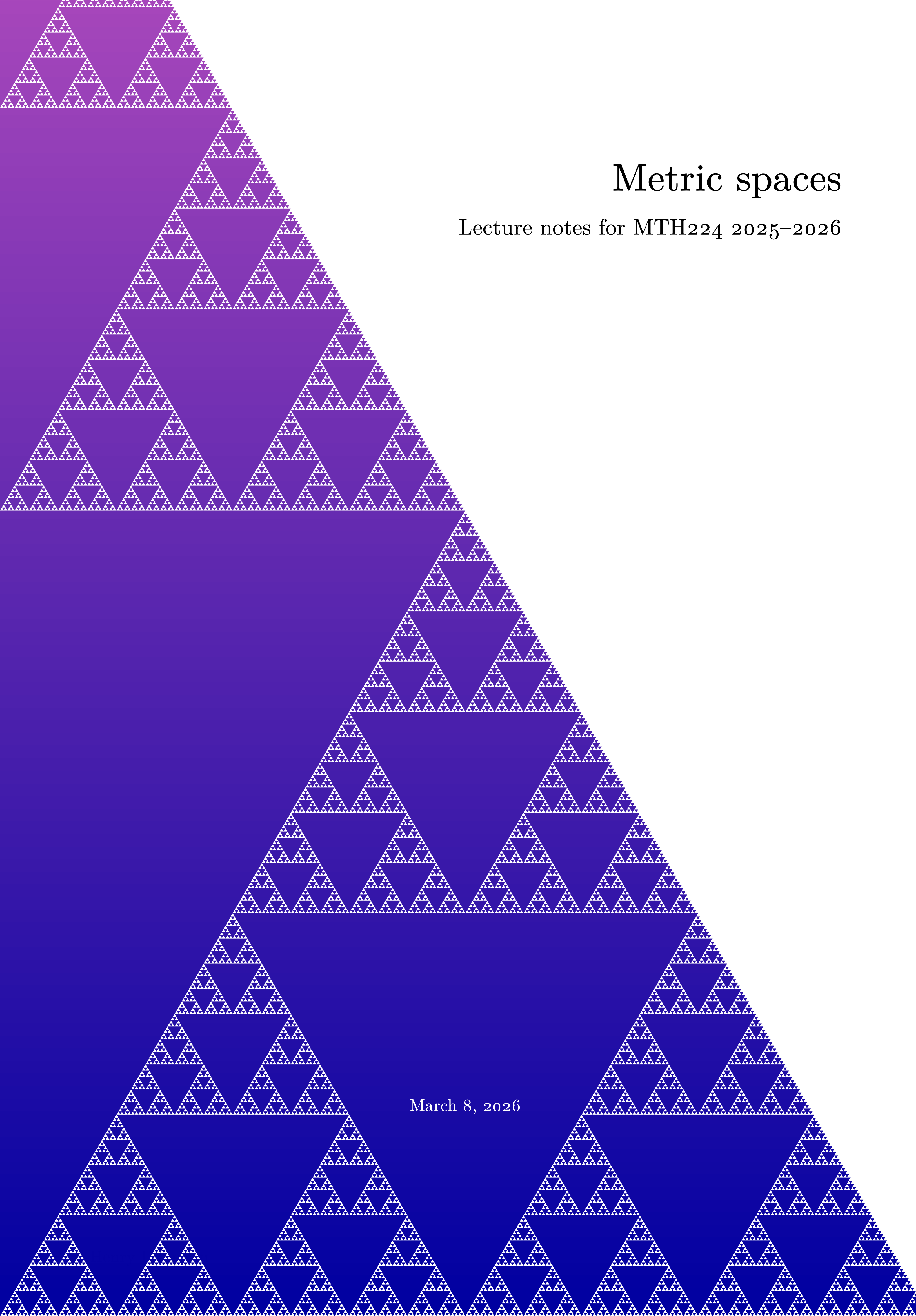
Metric spaces

Lecture notes for MTH224 2025–2026



March 8, 2026

FOREWORD

These notes are a work in progress. They are still plenty of typos (and hopefully few mistakes). If you see some, please let me know by email or directly via learning mall and I will correct them.

Things might be presented in a slightly different way than what was done in class, or in a different order.

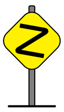
Theorem 0.1.

Blue boxes are for theorems.

Definition 0.2.

Green boxes are for definitions.

Remark 0.3.



Red boxes are for remarks.
Some passages are designed to forewarn the readers against serious errors, where they risk falling; these passages are indicated in the margin with the “dangerous bend” sign.

To go further

In these notes, you will see a few black boxes like this one. These are meant for the interested reader that wants to know more and can be skipped without harm. They are not necessarily be meant to be read in a first reading, but are here to provide more details if you come back to read these notes in the future.
The content that is written inside these “To go further” boxes will **NOT** be in the exam.

Alastair Darby and Gaoming Zhang contributed to reduce the number of typos. These notes would not have been the same without their comments.

CONTENTS

SYMBOLS

$\mathbf{N}$	natural integers excluding 0: $\mathbf{N} = \{1, 2, 3, \dots\}$
$\mathbf{Z}$	relative integers: $\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
$\mathbf{Q}$	rational numbers: $\mathbf{Q} = \left\{ \frac{p}{q} \;\middle|\; p, q \in \mathbf{Z}, q \neq 0 \right\}$
$\mathbf{R}$	real numbers
$\mathbf{C}$	complex numbers: $\mathbf{C} = \{a + b\mathrm{i} \mid a, b \in \mathbf{R}, \mathrm{i}^2 = -1\}$
$\mathbf{F}$	field (for example: $\mathbf{Q}$, $\mathbf{R}$ or $\mathbf{C}$)

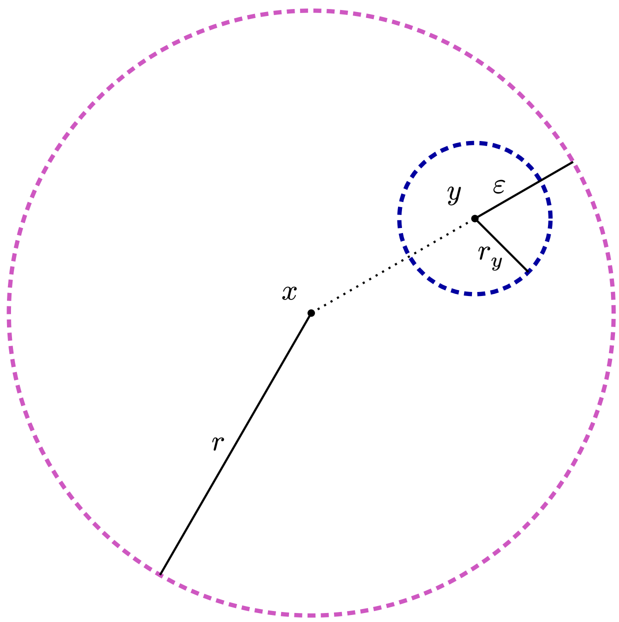
Table 1: Important sets.

α, A	alpha	ν, N	nu
β, B	beta	$\xi, \varXi$	xi
$\gamma, \varGamma$	gamma	o, O	omicron
$\delta, \varDelta$	delta	$\pi, \varPi$	pi
ε, E	epsilon	ρ, P	rho
ζ, Z	zeta	$\sigma, \varSigma$	sigma
η, H	eta	τ, T	tau
$\theta, \varTheta$	theta	$\upsilon, \varUpsilon$	upsilon
ι, I	iota	$\varphi, \varPhi$	phi
κ, K	kappa	χ, X	chi
$\lambda, \varLambda$	lambda	$\psi, \varPsi$	psi
μ, M	mu	$\omega, \varOmega$	omega

Table 2: Greek letters.

CHAPTER 1

CONTINUITY



1.1 Definitions & Examples

As the name of the module hints, we will study metric spaces. These are sets that are enriched with a *distance* that measure how far apart any two points lie. We hence need to define what a distance is. We want it to generalise the natural Euclidean distance in $\mathbf{R}^n$ (which is the usual length of the segment between two points), but also other "natural" notions of distance.

Before introducing an axiomatic definition of a distance, let us go through some real word notions of distance between two points, let us say between cities in China. For two cities A and B, one can measure a "distance" between these two points as the length of the shortest path from A to B, the length of the shortest road from A to B, but also as the minimum price of a train ticket from A to B. All these different notions of distance satisfies some natural commun properties that we formalise in the next definition.

Definition 1.1.1 (Metric Space)**.**

A **metric space** is a pair (X, d) consisting of a non-empty set X and a function, called **distance** or **metric**,

$$d : X \times X \to \mathbf{R}$$

that, for all $x, y, z \in X$, satisfies:

(M0) $d(x, y) \geq 0$;

(M1) $d(x, x) = 0$;

(M2) $d(x, y) = 0 \implies x = y$; *(positivity)*

(M3) $d(x, y) = d(y, x)$; *(symmetry)*

(M4) $d(x, z) \leq d(x, y) + d(y, z)$. *(triangle inequality)*

Axiom (M2) is called *positivity* because it is equivalent to: $x \neq y \implies d(x, y) > 0$. Axiom (M4) is called *triangle inequality* because it says that in a "triangle", the length of any side is smaller than the sum of the lengths of the other sides.

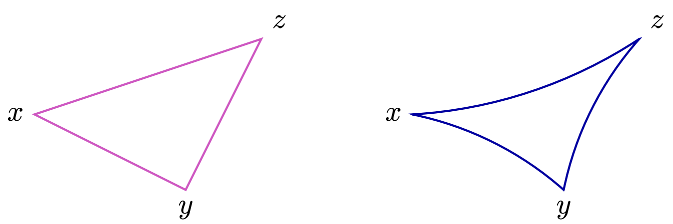


Figure 1.1: Triangles in the Euclidean plane (on the left) and in the hyperbolic plane (on the right). In both cases the length of a side is smaller than the sum of the lengths of the two other sides, one hence have $d(x, z) \leq d(x, y) + d(y, z)$.

Observe that symmetry means that "the shortest time taken to go from A to B" is not a good notion of distance.

We will see in Tutorial 1 Question 1 that the axiom (M0) is redundant: any function $d\colon X \times X \to \mathbf{R}$ satisfying (M1)–(M4) automatically satisfies (M0). However, axiom (M0) is the only superfluous axiom: the axioms (M1)–(M4) are independent. This means that for any $i \in \{1, 2, 3, 4\}$ it is possible to find a set X and a function $d\colon X \times X \to \mathbf{R}$ not that does not satisfy axiom (Mi) but satisfies all the (Mj) for $j \in \{1, 2, 3, 4\} \setminus \{i\}$. Can you find such functions?

Let us start exploring the notion of metric space with a few classical examples you already have encountered in your studies.

Example 1.1.2 (The Real Line)**.** On $\mathbf{R}$ we can define $d_{\mathbf{R}}(x, y) := |x - y|$ and we know from Analysis 1 that this satisfies the axioms (M0)–(M4) of being a metric. So $(\mathbf{R}, d)$ forms a metric space. When we consider $\mathbf{R}$ as a metric space the metric $d_{\mathbf{R}}$ will be understood unless otherwise stated.

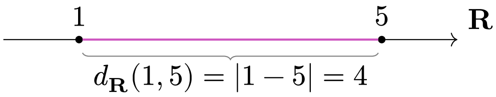


Figure 1.2: Distance between two points of the real line. The purple line is the unique shortest path between the two points.

Example 1.1.3 (Euclidean n-space)**.** We generalise Example 1.1.2 to n-dimensional **Euclidean space**[1] $\mathbf{R}^n$. For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbf{R}^n$ we set

$$d_2(x, y) := \left((x_1 - y_1)^2 + \dots + (x_n - y_n)^2\right)^{\frac{1}{2}} = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}, \tag{1.1}$$

where we take the positive square root. It is easy to check that the axioms (M0)–(M3) of Definition 1.1.1 are satisfied. We will check (M4), which follows from *Cauchy's*[2] *Inequality*, on Tutorial 1 Question 2.

When we consider $\mathbf{R}^n$ as a metric space the **Euclidean metric** (as defined in equation (1.1)) will be understood unless otherwise stated.

Remark 1.1.4.

For $n = 1$ we have the real line $\mathbf{R}$ where the metric

$$d_2(x, y) = \sqrt{(x - y)^2} = |x - y|$$

is the usual notion of *length* as in Example 1.1.2. For the cases $n = 2$ and 3 we recover the usual 2 and 3-dimensional Euclidean spaces with their normal distance functions.

[1]Named after Εὐκλείδης (fl. 300 BC).
[2]Augustin-Louis Cauchy (1789–1857).

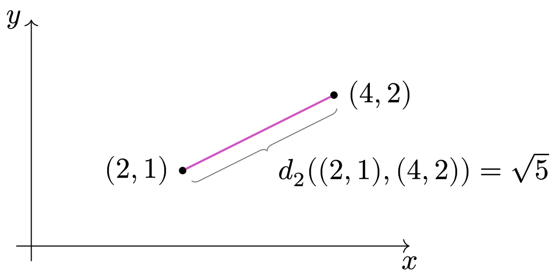


Figure 1.3: Euclidean distance between two points in the plane. The purple line is the unique shortest path between the two points.

It is possible to have different metrics on the same underlying set as we will see below with $\mathbf{R}^n$.

Example 1.1.5 (Taxicab Metric)**.** On $\mathbf{R}^n$, we define the **taxicab metric** (sometimes called the **Manhattan metric**) by

$$d_1(x, y) := |x_1 - y_1| + \cdots + |x_n - y_n| = \sum_{i=1}^{n} |x_i - y_i|.$$

It is easy to check that (M0)–(M3) hold. For (M4) we write $r_i = x_i - y_i$ and $s_i = y_i - z_i$, where $1 \leq i \leq n$. So we need to prove that

$$\sum_{i=1}^{n} |r_i + s_i| \leq \sum_{i=1}^{n} |r_i| + \sum_{i=1}^{n} |s_i|,$$

which is true since $|r_i + s_i| \leq |r_i| + |s_i|$, for all $1 \leq i \leq n$.

Note that on $\mathbf{R}$ the Euclidean and taxicab metrics are the same.

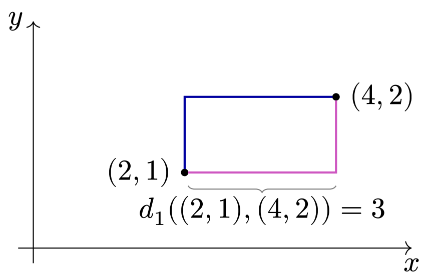


Figure 1.4: Taxicab distance between two points in the plane. The purple and blue lines are two shortest paths between the two points.

While the taxicab metric is not the usual Euclidean metric, it is not entirely new to you as it comes from a norm on the vector space $\mathbf{R}^n$, see MTH107 Advanced Linear Algebra and Section 1.2 for norms on vector space.

Definition 1.1.6 $((\mathbf{R}^n, d_p))$**.**

The metrics d_2 and d_1 on $\mathbf{R}^n$ from Examples 1.1.3 and 1.1.5 can be generalised, which explains the choice of subscripts, as follows. Let $p \geq 1$ and define

$$d_p(x, y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}},$$

for $x, y \in \mathbf{R}^n$ and where we take the non-negative real p-th root. The triangle inequality can be proved for d_p in general by using a generalisation of Cauchy's Inequality called *Minkowski's Inequality.*

Definition 1.1.7 $((\mathbf{R}^n, d_\infty))$**.**

We can define another metric on $\mathbf{R}^n$:

$$d_\infty(x, y) := \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}.$$

We leave the checking of the axioms as an exercise (Tutorial 1 Question 3).

Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be two points in $\mathbf{R}^2$. If $p_1 < p_2 < \dots$ is an unbounded increasing sequence of real numbers, one can show that $d_{p_i}(x, y)$ converges to $\max(|x_1 - y_1|, x_2 - y_2) = d_\infty(x, y)$. The same is true for pair of points in $\mathbf{R}^n$. The metric d_∞ is therefore, in some sense, the limit of the metrics d_p for $p \to \infty$. Hence the name.

We have seen that a given set X might admit more than one metric. The following shows that any non-empty set can be turned into a metric space.

Definition 1.1.8 (Discrete Metric Spaces)**.**

Let X be a non-empty set and define

$$d_0(x, y) := \begin{cases} 1, & x \neq y; \\ 0, & x = y. \end{cases}$$

Clearly the axioms (M0)–(M3) are satisfied. For (M4) observe that $d_0(x, y) + d_0(y, z)$ is either 0, 1, or 2, and is only 0 when $x = y = z$ in which case $d_0(x, z) = 0$ also. The metric d_0 is called the **discrete metric**, and (X, d_0) a **discrete metric space**.

Discrete metric spaces often serve as counter-examples to ideas, suggested by geometric intuition, that may not actually hold in general metric spaces.

Before going further, let us quickly compare the examples of metric we have seen so far.

Example 1.1.9. Taking points $P = (2,1)$ and $Q = (4,2)$ in $\mathbf{R}^2$ we calculate their distances according to some of the different metrics that we have looked at on $\mathbf{R}^2$:

$$\begin{aligned}
d_0(P,Q) &= 1, \quad \text{since } P \neq Q \\
d_1(P,Q) &= |2-4| + |1-2| = 3 \\
d_2(P,Q) &= \sqrt{(2-4)^2 + (1-2)^2} = \sqrt{5} \approx 2.236 \\
d_5(P,Q) &= \sqrt[5]{(2-4)^5 + (1-2)^5} = \sqrt[5]{33} \approx 2.012 \\
d_\infty(P,Q) &= \max\{|2-4|, |1-2|\} = 2
\end{aligned}$$

where we write d_0 for the discrete metric on $\mathbf{R}^2$. This shows that all the above metrics are distinct.

Remark 1.1.10.

Let d be a metric on a set X. It directly follows from the definition that for any positive real number c, scaling the metric by c (that is looking at the function $(x,y) \mapsto c \cdot d(x,y)$) gives a new metric.
The reader can verify that the metrics d_0, d_1, d_2 and d_∞ of the above example are not scalar multiples one of each other. The same is true for the metrics d_p and d_q, $p \neq q$, from Definition 1.1.6.

Examples 1.1.2, 1.1.3 and 1.1.5 and Definitions 1.1.6 and 1.1.7 can easily be adapted to the complex case. In particular, we recover the standard metric on $\mathbf{C}$.

Example 1.1.11. The standard metric on the complex numbers $\mathbf{C}$ is given by

$$d_{\mathbf{C}}(z_1, z_2) := |z_1 - z_2|,$$

where $|z|$ is the **modulus** of the complex number z (that is, $|z| = r$, where $z = r\mathrm{e}^{\mathrm{i}\theta}$). The axioms of Definition (1.1.1) hold by analogy with those for the Euclidean metric on $\mathbf{R}$.

We know (from MTH107 Advanced Linear Algebra) that a vector subspace is a subset that is itself a vector space for the induced operations. A similar definition holds for subgroups (see MTH122 Introduction to Abstract Algebra). We will mimic this for metric subspace.

Definition 1.1.12 (Metric Subspaces).

Let (X,d) be a metric space. A **metric subspace** is a subset $W \subseteq X$ endowed with the restriction of d to W:

$$d|_W : W \times W \to \mathbf{R},$$

i.e. $d|_W(w_1, w_2) := d(w_1, w_2)$.

Observe that we made a slight abuse of notation when writing $d|_W$ to describe the restriction $d|_{W\times W}$ of d to $W \times W \subseteq X \times X$.

As the name hints, a metric subspace $(W, d|_W)$ of a metric space (X, d) is itself a metric space. That is, for any subset $W \subseteq X$ of a metric space (X, d), the function $d|_W$ is automatically a metric on W.

When studying mathematical objects (vector spaces, groups, ...) we are not only interested in those objects, but also in "good" maps between them. We start by defining what it means for two metric spaces to be "identical as metric spaces".

Definition 1.1.13 (Isometry).

An **isometry**[a] between two metric spaces (X, d_X) and (Y, d_Y) is a bijective function $f\colon X \to Y$ such that

$$d_Y(f(x), f(y)) = d_X(x, y), \quad \forall x, y \in X.$$

Two metric spaces are said to be **isometric** if there exists an isometry between them.

[a] From the Greek ἴσος μέτρον, meaning equal measure.

Remark 1.1.14.

If $f\colon X \to Y$ is an isometry, then f^{-1} is also an isometry. So, an isometry is a bijective map such that both f and f^{-1} "preserve" the metric space structure. This is similar to the definition of isomorphisms of vector spaces or of group isomorphisms.

It is easily shown that isometry is an equivalence relation (reflexive, symmetric and transitive) on metric spaces. The proof is left as an exercise (Tutorial 3 Question 9). Two metric spaces are isometric if and only if they are indistinguishable as metric spaces. They can however be quite different for some other properties.

Example 1.1.15. Let $f\colon \mathbf{R}^2 \to \mathbf{C}$ be given by $f(x, y) = x + iy$. Then f is an isometry from $\mathbf{R}^2$ with the Euclidean metric to $\mathbf{C}$ with its standard metric (see Examples 1.1.3 and 1.1.11). This means that $\mathbf{R}^2$ with the Euclidean metric and $\mathbf{C}$ with the standard metric are indistinguishable as metric spaces, despite being quite different as fields.

For algebraic structures (like vector spaces or groups), the homomorphisms are the maps preserving the structure. For metric spaces, asking to preserve the metric is usually too much. Indeed, if $f\colon X \to Y$ is a function such that $d_Y(f(x), f(y)) = d_X(x, y)$ for all $x, y \in X$, then it is automatically injective. We will therefore only ask our maps to behave well for the notion of "proximity".

Definition 1.1.16 (Continuous Function)**.**

Given two metric spaces (X, d_X) and (Y, d_Y) and a point $a \in X$ we say that a function $f\colon X \to Y$ is **continuous at** a (with respect to the metrics d_X and d_Y) if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } d_X(x, a) < \delta \implies d_Y(f(x), f(a)) < \varepsilon.$$

If we need to be precise, we will say that f is (d_X, d_Y)-continuous at $a \in X$.
If f is continuous at every point $a \in X$, then we say that f is **continuous**.

Remark 1.1.17.

In Analysis 1 we were concerned with functions $f\colon A \to \mathbf{R}$, where $A \subseteq \mathbf{R}$. Note that Definition 1.1.16 coincides with that given in Analysis 1 where we consider $\mathbf{R}$ with the metric $d_{\mathbf{R}}$ defined in Example 1.1.2 and consider $(A, d|_A)$ as a metric subspace of $(\mathbf{R}, d_{\mathbf{R}})$.

Continuous functions possesses some important properties that make them suitable to use as "nice functions" between metric spaces.

Proposition 1.1.18. *Composition of continuous functions is still continuous. More precisely, if* $f\colon (X, d_X) \to (Y, d_Y)$ *and* $g\colon (Y, d_Y) \to (Z, d_Z)$ *are two continuous functions, then the composition* $g \circ f\colon (X, d_X) \to (Z, d_Z)$ *is also continuous.*

Composition of continuous functions has identities. More precisely, the identity function $\mathrm{Id}_X\colon (X, d) \to (X, d)$ *is continuous.*

Composition of continuous functions is associative. That is, if $f\colon (X, d_X) \to (Y, d_Y)$, $g\colon (Y, d_Y) \to (Z, d_Z)$ *and* $h\colon (Z, d_Z) \to (W, d_W)$ *are continuous functions, then* $h \circ (g \circ f) = (h \circ g) \circ f$.

Proof. For the first statement, use the continuity of f at x and of g at $f(x)$. See Tutorial 1 Question 8 for the details.

The second statement is trivial, see also Tutorial 1 Question 7.

The last statement is true for any functions between sets. □

To go further

The above proposition says that metric spaces with continuous functions form a **category**. Category theory is the branch of mathematics that study "objects and nice maps between them" from an abstract viewpoint. Other examples of categories are: vector spaces with linear maps and groups with group homomorphisms.

1.2 Metrics from Norms

In MTH107 Advanced Linear Algebra we studied inner product on vector spaces. We saw that each inner product $\langle \cdot, \cdot \rangle$ gives raise to a norm $\|\cdot\|$ satisfying some nice properties. We will take these properties as the axiomatic definition of a norm.

Definition 1.2.1.

A **norm** on a vector space V (over a field $\mathbf{F} \subseteq \mathbf{C}$) is a function

$$\|\cdot\|: V \to \mathbf{R}$$

that, for all $x, y \in V$ and all $\lambda \in \mathbf{F}$, satisfies

(N0) $\|x\| \geq 0$;

(N1) $\|0\| = 0$;

(N2) $\|x\| = 0 \implies x = 0$; *(positivity)*

(N3) $\|\lambda x\| = |\lambda| \|x\|$; *(homogeneity)*

(N4) $\|x + y\| \leq \|x\| + \|y\|$. *(triangle inequality)*

The pair $(V, \|\cdot\|)$ is called a **normed vector space**.

As for the definition of distance, axiom (N0) is superfluous: if $\|\cdot\|$ satisfies (N1) to (N4), it automatically satisfies (N0). Indeed, for every $x \in V$ we have $0 = \|0\| = \|x - x\| \leq \|x\| + |-1|\|x\| = 2\|x\|$.

The following elementary examples are inspired by the metrics defined in Section 1.1.

Example 1.2.2 (Norms on $\mathbf{R}^n$ or $\mathbf{C}^n$)**.** The following are norms on $\mathbf{R}^n$ and $\mathbf{C}^n$:

1. Taxicab norm: $\|x\|_1 := \sum_{i=1}^n |x_i|$;
2. Euclidean norm: $\|x\|_2 := \sqrt{(\sum_{i=1}^n |x_i|^2}$;
3. L_p norm for $p \geq 1$: $\|x\|_p := \sqrt[p]{(\sum_{i=1}^n |x_i|^p}$;
4. Max norm: $\|x\|_\infty := \max\{|x_1|, \dots, |x_n|\}$.

Remark 1.2.3.

In the above examples, the only norm that comes from an inner product is $\|\cdot\|_2$. Indeed, none of the other norm satisfies the parallelogram identity $2\|x\|^2 + 2\|y\|^2 = \|x - y\|^2 + \|x + y\|^2$.

The following result explains why we are interested in norms in this module.

Lemma 1.2.4. *If $(V, \|\cdot\|)$ is a normed vector space, then $d(x, y) := \|x - y\|$ is a metric on V.*

Proof. One check that for each $i \in \{1, \dots, 4\}$ the axiom (Ni) implies the corresponding axiom (Mi). Only the triangle inequality needs an argument:

$$\begin{aligned} d(x, z) &= \|x - z\| = \|(x - y) + (y - z)\| \\ &\leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z). \end{aligned}$$

□

Norms from Example 1.2.2 generate the corresponding metrics from Section 1.1. Observe that the discrete metric on $\mathbf{R}$ does not come from a norm as this would contradict (N3).

1.3 Function Spaces

In this section we will define various metrics on spaces of functions. The set $\mathbf{R}^{[a,b]}$ of all real functions $f\colon [a,b] \to \mathbf{R}$ is in some sense too big to put a nice metric on it. We will therefore restrict ourself to subspaces of $\mathbf{R}^{[a,b]}$.

Recall that for a subset W of $\mathbf{R}$, a function $f\colon W \to \mathbf{R}$ is **bounded** if there exists $K \in \mathbf{R}$ such that $|f(x)| \leq K$, for all $x \in W$.

Example 1.3.1 (Function Spaces)**.** Let X be the set of all bounded functions $f\colon W \to \mathbf{R}$. Given two points $f, g \in X$ we define

$$d_{\sup}(f,g) := \sup_{x \in W} |f(x) - g(x)|.$$

This is well-defined (that is, the right-hand side exists) since, as f and g are bounded we can write

$$|f(x)| \leq K \quad \text{and} \quad |g(x)| \leq L$$

for some $K, L \in \mathbf{R}$ say, then

$$|f(x) - g(x)| \leq |f(x)| + |g(x)| \leq K + L, \quad \forall x \in W,$$

so the supremum exists. We will now check that $d_{\sup}$ is a metric on X:

"(M0)" Is clear since it is the supremum of a set of non-negative numbers.

"(M1) and (M2)" The supremum $\sup_{x \in W} |f(x) - g(x)|$ is 0 if and only if $f(x) = g(x)$ for all $x \in W$, that is if and only if $f = g$ as functions from W to $\mathbf{R}$.

"(M3)" Since $|f(x) - g(x)| = |g(x) - f(x)|$, for all $x \in W$, it follows that $d_{\sup}(f,g) = d_{\sup}(g,f)$.

"(M4)" Let $f, g, h \in X$. For any $x \in [a,b]$,

$$\begin{aligned} |f(x) - h(x)| &\leq |f(x) - g(x)| + |g(x) - h(x)| \\ &\leq \sup_{x \in W} |f(x) - g(x)| + \sup_{x \in W} |g(x) - h(x)| \\ &= d_{\sup}(f,g) + d_{\sup}(g,h). \end{aligned}$$

So $d_{\sup}(f,g) + d_{\sup}(g,h)$ is an upper bound for the set $S = \{|f(x) - g(x)| \mid x \in [a,b]\}$. Hence, $\sup S = d_{\sup}(f,h) \leq d_{\sup}(f,g) + d_{\sup}(g,h)$.

This metric is usually called the **sup metric** and we usually write $\mathscr{B}(W) := (X, d_{\sup})$ for this metric space.

If $f\colon [a,b] \to \mathbf{R}$ is continuous, then we know from Analysis 1 that f is bounded. Let Y denote the set of all continuous functions $f\colon [a,b] \to \mathbf{R}$. Then $\mathcal{C}[a,b] := (Y, d_{\sup})$ forms a metric subspace of $\mathscr{B}[a,b]$.

Example 1.3.2. The polynomial functions $f(x) = x^4$ and $g(x) = 1$ are continuous on $[0,1]$ and therefore $f, g \in \mathscr{C}[0,1]$. We find that

$$d_{\text{sup}}(f,g) = d_{\text{sup}}(x^4,1) = \sup_{x\in[0,1]} |x^4 - 1| = |0^4 - 1| = 1.$$

Therefore, $d_{\text{sup}}(x^4, 1) = 1$ in $\mathscr{C}[0,1]$ and $\mathscr{B}[0,1]$.

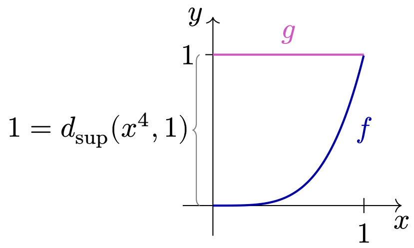


Figure 1.5: Graphs of the polynomial functions $f(x) = x^4$ and $g(x) = 1$. The d_{sup} distance between the functions is the supremum of the vertical distance between their graphs.

On the set of all continuous functions $f\colon [a,b] \to \mathbf{R}$ we can define

$$d_1(f,g) := \int_a^b |f(x) - g(x)|\, \mathrm{d}x. \tag{1.2}$$

Notice that $|f(x) - g(x)| \geq 0$, for all $x \in [a,b]$, so

$$\int_a^b |f(x) - g(x)|\, \mathrm{d}x = 0 \iff \forall x \in [a,b] : f(x) = g(x) \tag{1.3}$$

Lemma 1.3.3. *The function d_1 from (1.2) is a metric on the set Y of all continuous functions $f\colon [a,b] \to \mathbf{R}$.*

Proof. The axiom (M0) clearly holds and for (M2) we know by (1.3) that $f = g$ exactly when $d_1(f,g) = 0$. Clearly (M3) holds.

For (M4), take continuous functions $f, g, h\colon [a,b] \to \mathbf{R}$ and $x \in [a,b]$. Then

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|$$

so

$$\int_a^b |f(x) - h(x)|\, \mathrm{d}x \leq \int_a^b |f(x) - g(x)|\, \mathrm{d}x + \int_a^b |g(x) - h(x)|\, \mathrm{d}x. \qquad \square$$

This metric is known as the L_1 **metric** and we denote the space of all continuous functions $f\colon [a,b] \to \mathbf{R}$ with the L_1 metric by $\mathscr{L}_1[a,b] := (Y, d_1)$.

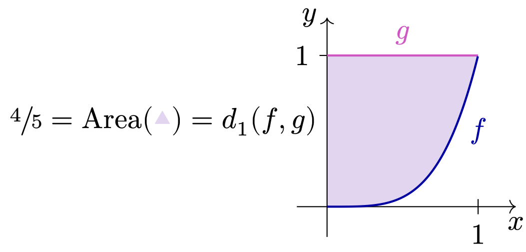


Figure 1.6: Graphs of the polynomial functions $f(x) = x^4$ and $g(x) = 1$. The d_1 distance between the functions is the area of the surface between their graphs.

Example 1.3.4. The polynomials x^4 and 1 both lie in $\mathscr{L}_1[0,1]$ and now

$$d_1(x^4, 1) = \int_0^1 |x^4 - 1| \, \mathrm{d}x = \left[x - x^5/5\right]_0^1 = 4/5.$$

Therefore, $d_1(x^4, 1) = 4/5$ in $\mathscr{L}_1[0,1]$.

There are other metrics on the set of all continuous functions $f\colon [a,b] \to \mathbf{R}$, such as

$$d_p(f,g) := \left(\int_a^b |f(x) - g(x)|^p \, \mathrm{d}x\right)^{\frac{1}{p}}$$

where $p \geq 1$, which is known as the L_p **metric**, which results in a metric space $\mathscr{L}_p[a,b] := (Y, d_p)$.

All the metrics from this section come from norms. Let us see that in details for $d_{\sup}$ and d_1.

Example 1.3.5. The set $X = \{f\colon [a,b] \to \mathbf{R} \mid f \text{ is bounded}\}$ forms a real vector space by defining

$$(f+g)(x) := f(x) + g(x) \quad \text{and} \quad (\lambda f)(x) := \lambda f(x),$$

for all $f, g \in X$, $x \in [a,b]$ and $\lambda \in \mathbf{R}$.

We can then define a norm $\|\cdot\|_{\sup}$ on X by $\|f\|_{\sup} := \sup\{|f(x)| \mid x \in [a,b]\}$ which leads to the $d_{\sup}$ metric on X and the metric space $\mathscr{B}[a,b]$.

The set $Y = \{f\colon [a,b] \to \mathbf{R} \mid f \text{ is continuous}\}$ is a vector subspace of X and we can define a different norm $\|\cdot\|_1$ on Y defined by $\|f\|_1 := \int_a^b |f(x)| \, \mathrm{d}x$ which leads to the metric d_1 and the metric space $\mathscr{L}_1[a,b]$.

1.4 Sequence Spaces

Sequence spaces are discrete versions of the function spaces that we discussed in the previous section. Recall that an (infinite) sequence of real numbers is a function $a\colon \mathbf{N} \to \mathbf{R}$ where we usually write $a_n := a(n)$. The set of all these sequences $\mathbf{R}^{\mathbf{N}}$ forms a real vector space by defining

$$(a+b)_n := a_n + b_n \quad \text{and} \quad (\lambda a)_n := \lambda a_n,$$

for $a, b \in \mathbf{R}^{\mathbf{N}}$ and $\lambda \in \mathbf{R}$.

Definition 1.4.1.

A sequence a is **bounded** if there exists $K \in \mathbf{R}$ such that $|a_n| \leq K$, for all $n \in \mathbf{N}$. The space of bounded sequence is denoted by $\mathscr{X}$.

Proposition 1.4.2. *The map* $\|\cdot\|_\infty \colon \mathscr{X} \to \mathbf{R}$ *defined by* $\|a\|_\infty := \sup_{n\in\mathbf{N}} |a_n|$ *is a norm on* $\mathscr{X}$.

Proof. The proof is an easy verification and is left as an exercise (Tutorial 2 Question 5). □

Observe that, by Lemma 1.2.4, $\ell^\infty := (\mathscr{X}, d_\infty)$ forms a metric space, where $d_\infty(a, b) := \|a - b\|_\infty$.

Definition 1.4.3.

A sequence a is **absolutely convergent** if $\sum_{n\in\mathbf{N}} |a_n| < \infty$. The space of absolutely convergent sequences is denoted by $\mathscr{Y}$.

Proposition 1.4.4. *The map* $\|\cdot\|_1 \colon \mathscr{Y} \to \mathbf{R}$ *defined by* $\|a\|_1 := \sum_{n\in\mathbf{N}} |a_n|$ *is a norm on* $\mathscr{Y}$.

Proof. The proof is an easy verification and is left as an exercise (Tutorial 2 Question 6). □

We call the metric space that this forms $\ell^1 := (\mathscr{Y}, d_1)$, where $d_1(a, b) := \|a - b\|_1$.

Remark 1.4.5.

Although this is beyond the scope of this module we will note that, for all $0 < p < \infty$, we can consider the vector subspace of $\mathbf{R}^{\mathbf{N}}$ consisting of all sequences a where $\sum_{n\in\mathbf{N}} |a_n|^p < \infty$. For $p \geq 1$ we can define on this subspace a norm $\|\cdot\|_p$ given by $\|a\|_p := \left(\sum_{n\in N} |a_n|^p\right)^{1/p}$, which gives us the metric space ℓ^p. For $0 < p < 1$ there is no such norm, but we can still define a metric given by $d_p(a, b) := \sum_{n\in N} |a_n - b_n|^p$.

1.5 Open Sets

Having a metric, we can define open and closed subsets, which generalise the notion of open and closed subsets of $\mathbf{R}^n$. We start by defining the simplest open subsets, the open balls, which generalise open intervals of $\mathbf{R}$, open disks in $\mathbf{R}^2$ and open balls in $\mathbf{R}^3$.

Definition 1.5.1 (Open Ball)**.**

Given a metric space (X, d), the **open ball** of radius $r > 0$ around a point $x \in X$ is the set

$$B_r(x) = B_r(x; d) := \{y \in X \mid d(x, y) < r\} \subseteq X.$$

As said above, for $n \leq 3$ we recover the classical notions of open intervals/disks/balls in the in the Euclidean space $\mathbf{R}^n$. Examples of open balls in $\mathbf{R}^2$ for the metrics d_1, d_2 and d_∞ are given in Tutorial 2 Question 7, while Tutorial 2 Question 9 shows what happens for the British Railway metric from Tutorial 1 Question 5.

Example 1.5.2 (Euclidean Metric)**.** In $\mathbf{R}$, $B_r(x) = (x - r, x + r)$. In $\mathbf{R}^2$, $B_r(x)$ is the interior of a disc of radius r centred around x. In $\mathbf{R}^3$, $B_r(x)$ is the interior of a solid ball centred on x.

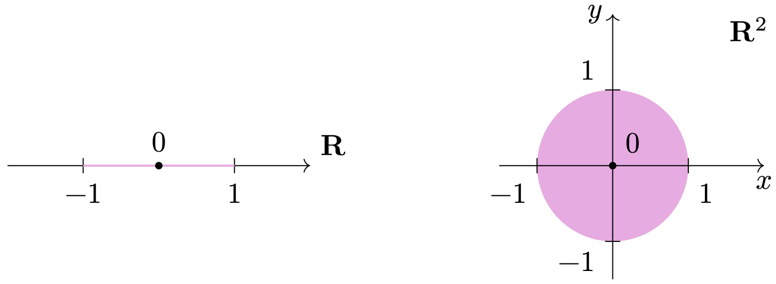


Figure 1.7: Open balls of radius 1 around the origin for the Euclidean metric on $\mathbf{R}$ (on the left) and on $\mathbf{R}^2$.

As usual, it is enlightening to consider the case of discrete spaces.

Example 1.5.3 (Discrete Metric)**.** In a discrete metric space (X, d),

$$B_r(x) = \begin{cases} \{x\}, & \text{if } r \leq 1; \\ X, & \text{if } r > 1. \end{cases}$$

It is important to keep in mind that the definition of an open ball depends heavily on the ambiant space, as demonstrated by the following example.

Example 1.5.4. Let $I = [0, 1] \subseteq \mathbf{R}$ with its standard metric $d_\mathbf{R}$ on $\mathbf{R}$. Then $B_1(1; d_\mathbf{R}) = (0, 2)$ whilst $B_1(1; d_\mathbf{R}|_I) = (0, 1]$, where $(I, d_\mathbf{R}|_I) \subseteq (\mathbf{R}, d_\mathbf{R})$ is a metric subspace.

More generally, if $(Y, d|_Y) \subseteq (X, d)$ is a subspace, then the open balls of Y are exactly the $B \cap Y$ for B an open ball of X. Indeed, we have

$$B_r(y; d|_Y) = \{z \in Y \mid d|_Y(z, y) < r\} = \{z \in Y \mid d(z, y) < r\} = B_r(y; d) \cap X.$$

We can now express the continuity of a function using the language of open balls:

Definition 1.5.5 (Continuous Function)**.**

A function $f\colon X \to Y$ between metric spaces is **continuous at** $a \in X$ if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } f(B_\delta(a)) \subseteq B_\varepsilon(f(a)).$$

This is just a translation of Definition 1.1.16, but it is often more useful.

Using open balls, one can define general open sets. To do so, we first need to define what is the interior of a set.

Definition 1.5.6 (Interior)**.**

Let W be a subset of a metric space (X, d). We say that $w \in W$ is an **interior point** of W if there exists $\varepsilon > 0$ such that $B_\varepsilon(w) \subseteq W$. The **interior of** W is the set W° of all interior points of W.

Observe that $W^\circ \subseteq W$.

Interior points obviously generalise the natural notion of interior points of intervals of $\mathbf{R}$.

We can now define the main notion of this section.

Definition 1.5.7 (Open Set)**.**

A subset $U \subseteq (X, d)$ is **open in** X if every $u \in U$ is an interior point, that is if $U^\circ = U$. Equivalently, U is open if

$$\forall u \in U, \exists \varepsilon > 0 \text{ such that } B_\varepsilon(u) \subseteq U.$$

Remark 1.5.8.

Be aware that in the above definition ε depends on u and can take different values for different points in U. A simple example of this fact is given by the open interval $(0, 1)$ in $\mathbf{R}$ for which all $u_n = {}^1\!/n$ are interior points but there no given ε can work for all of them as we necessarily have $0 < \varepsilon_n < {}^1\!/n$.

Now that we have defined both open balls and open sets, we need to verify that the names are not misleading.

Lemma 1.5.9. *Every open ball $B_r(x)$ in a metric space (X, d) is open in X.*

Proof. Suppose we have a point $y \in B_r(x)$ and let $\varepsilon := r - d(x, y)$, which is positive. If $z \in B_\varepsilon(y)$, then $d(y, z) < \varepsilon$ and

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \varepsilon = r,$$

so $z \in B_r(x)$. Therefore, for every $y \in B_r(x)$ there exists $\varepsilon > 0$ such that $B_\varepsilon(y) \subseteq B_r(x)$, and so $B_r(x)$ is open in X. $\square$

See the chapter illustration on page 1 for an illustration of the above proof (The illustration shows the ball $B_{r_y}(y)$ for some $0 < r_y < \varepsilon$).

The converse of Lemma 1.5.9 does not hold.

Remark 1.5.10.

Not every open set is an open ball, for example consider the subset $(0,1) \times (0,1)$ in $\mathbf{R}^2$. This is an open set with the Euclidean metric but not an open ball.

However, we have a weak converse to Lemma 1.5.9. Namely:

Lemma 1.5.11. *An open subset U of a metric space is a union of open balls.*

Proof. The proof is left to the reader (see Tutorial 3 Question 2). □

The example of $(0,1) \times (0,1)$ in $\mathbf{R}^2$ shows that it is not always possible to write an open set as a *finite* union of open balls.

Below are a few examples of open sets. The last one shows that being open is not an intrinsic property of a subset of X, but depends on the metric we have put on X.

Example 1.5.12.

1. By Lemma 1.5.9 all open balls in a metric space are open.
2. The subset $[a,b] \subseteq \mathbf{R}$ is not open in $(\mathbf{R}, d_{\mathbf{R}})$, but $[a,b]$ is open in itself, that is, in $([a,b], d_{\mathbf{R}}|_{[a,b]})$.
3. In a discrete metric space X, every subset of X is open in X.
4. The set $\{0\}$ is open in $\mathbf{R}$ with the discrete metric but not with the usual metric.

The operation of taking the interior satisfies some interesting properties that are summarised below.

Proposition 1.5.13. *Given any two subsets $U, V \subseteq (X,d)$ of a metric space we have*

1. *$U \subseteq V \implies U^\circ \subseteq V^\circ$;*
2. *$(U^\circ)^\circ = U^\circ$;*
3. *U° is the largest open subset of X that is contained in U.*

Proof. The proof is left to the reader (see Tutorial 3 Question 1). □

The collection of open subsets satisfies some nice closure properties.

Proposition 1.5.14. *Let (X,d) be a metric space. Then*

1. *The empty set $\varnothing$ and the whole space X are open in X;*
2. *If $U_1, \ldots, U_m$ are open sets, then $\bigcap_{i=1}^m U_i$ is open in X;*

3. *The union of any collection of open sets is open in X.*

Proof. Since the empty has no elements, the condition that all its points are interior points is empty and thus true. The fact that all points of X are interior point is also trivially true.

Let $x \in \bigcap_{i=1}^{m} U_i$. Then $x \in U_i$, for every $i = 1, \dots, m$, and U_i is open in X which implies that there exists $\varepsilon_i > 0$ such that $B_{\varepsilon_i}(x) \subseteq U_i$. Take $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_m\}$. Then $B_\varepsilon(x) \subseteq B_{\varepsilon_i}(x) \subseteq U_i$, for $i = 1, \dots, m$, so $B_\varepsilon(x) \subseteq \bigcap_{i=1}^{m} U_i$.

Let $x \in \bigcup_{i \in I} U_i$, where I is some indexing set, and each U_i is open in X. Then $x \in U_k$, for some k, and since U_k is open there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U_k \subseteq \bigcup_{i \in I} U_i$. $\square$

Combining the above proposition and Lemma 1.5.11, one obtain a characterisation of open sets.

Corollary 1.5.15. *Let $W \subseteq (X, d)$ be a subset of a metric space. Then W is open if and only if it is a union of open balls.*

Remark 1.5.16.

It is not necessarily true that the intersection of an infinite number of open sets is open. For example, in $\mathbf{R}$ consider $U_i := (-1/i, 1/i)$, for $i \in \mathbf{N}$. Then each U_i is open in $\mathbf{R}$ but $\bigcap_{i=1}^{\infty} U_i = \{0\}$ which is not open in $\mathbf{R}$.

Topology[3] is the branch of mathematics that studies the notion of open sets. While the exact definition of a topology is not part of this module, we will sometimes talk about "topological properties" meaning all properties following uniquely from Proposition 1.5.14 without needing to use the metric.

To go further

A **topological space** is a couple $(X, \mathcal{T})$ of a set and a collection $\mathcal{T}$, called a **topology**, of subsets of X satisfying Proposition 1.5.14. More precisely, a collection of subsets of X is a topology if it contains $\varnothing$ and X and is closed under finite intersections and arbitrary unions.

Using the characterisation of open sets as union of open balls, one can describe open sets of subspaces.

Proposition 1.5.17. *Let (X, d) be a metric space and $(Y, d|_Y)$ be a subspace. Then*

1. *The open subsets of Y are exactly the $U \cap Y$ for U open in Y;*

2. *If Y is open and $U \subseteq Y$ is open in Y, then U is open in Y.*

[3]From the Greek τόπος and λογία, meaning place/location and study.

Proof. "1" Let y be any point of $Y \cap U$. There exists $\varepsilon > 0$ such that $B_\varepsilon(y; d) \subseteq U$. Since $B_\varepsilon(y; d|_Y) = B_\varepsilon(y; d) \cap Y$ we conclude that $B_\varepsilon(y; d|_Y)$ is contained in $Y \cap U$. This shows that $Y \cap U$ is open in Y.

For the other direction, let U be an open subset of Y, so U is a union of balls by Lemma 1.5.11: $U = \bigcup_{i \in I} B_{r_i}(y_i; d|_Y)$. Then $\tilde{U} = \bigcup_{i \in I} B_{r_i}(y_i; d)$ is open in X and $\tilde{U} \cap Y = U$.

"2" We have $U = Y \cap \tilde{U}$ for some open subset $\tilde{U}$ of X. If Y is open in X, then so is U as the intersection of two open subsets. ☐

Using open sets, one obtain a third equivalent characterisation of continuity. This characterisation does not use the metric, only the notion of open sets, hence its name.

Remark 1.5.18.

Recall that for a function $f\colon X \to Y$ (not necessarily injective) and a subset $W \subseteq Y$ that

$$f^{-1}(W) := \{x \in X \mid f(x) \in W\} \subseteq X.$$

Theorem 1.5.19 (Topological Continuity)**.**

Suppose that $f\colon X \to Y$ is a function between metric spaces. Then f is continuous if and only if for every open set U in Y, $f^{-1}(U)$ is open in X.

Proof. "⇒" Suppose that f is continuous and let U be open in Y. We have to show that $f^{-1}(U)$ is open in X. Let $x \in f^{-1}(U)$. Then $f(x) \in U$ and, since U is open in Y, there exists $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq U$. Since f is continuous, by Definition 1.5.5, there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. So $f(B_\delta(x)) \subseteq U$, which implies that $B_\delta(x) \subseteq f^{-1}(U)$ and we have proved that $f^{-1}(U)$ is open in X.

"⇐" Assume the condition on open sets holds. We need to prove that f is continuous at every point $x \in X$. Given $\varepsilon > 0$, then $B_\varepsilon(f(x))$ is open in Y by Lemma 1.5.9. So by the assumption $f^{-1}(B_\varepsilon(f(x)))$ is open in X. Also, $x \in f^{-1}(B_\varepsilon(f(x)))$ so by the definition of open set, there exists $\delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Then $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$ which proves that f is continuous at x according to Definition 1.5.5. ☐

Remark 1.5.20.

Be careful that the topological characterisation of continuity is about the preimage of open sets and *not* about the image of open sets.

A function between two metric spaces is said to be **open** if the image of any open set is open. There exist bijective continuous but non-open functions, as well as bijective open but non-continuous functions. See Example 1.5.21.

Example 1.5.21. Let $([0, 1), d|_{[0,1)})$ where $d|_{[0,1)}$ is the restriction of the standard metric on $\mathbf{R}$. Let $S^1 = \{z \in \mathbf{C} \mid |z| = 1\}$ be the unit circle with metric $d_{S^1}(z, w) = \theta$, where $\theta \in [0, \pi)$ is the angle zOw (or wOz if zOw is bigger than π). Then the function

$f\colon [0,1) \to S^1, x \mapsto \mathrm{e}^{2\pi \mathrm{i} x}$ is a bijection. One can verify that f is continuous, but not open. Indeed, the subset $U = [0, 0.1) \subseteq [0, 0.1)$ is open, but $1 = e^{1\pi \mathrm{i} 0}$ belongs to $f(U) = \{\mathrm{e}^{2\pi \mathrm{i}\theta} \mid \theta \in [0, 0.1)\}$ but is not an interior point of it.

The inverse function $f^{-1}\colon S^{-1} \to [0,1)$ is open but not continuous.

The following result follows from Theorem 1.5.19 and the fact that any subset of a discrete metric space is open.

Lemma 1.5.22. *If (X, d_X) is a discrete metric space, then any function $f\colon X \to Y$ to another metric space (Y, d_Y) is automatically continuous.*

We conclude this section by one last example.

Example 1.5.23. The function $f\colon \mathbf{R} \to \mathbf{R}$ given by $f(x) = \begin{cases} x, & x \neq 0; \\ 1, & x = 0, \end{cases}$ is not continuous as can be shown using the result of Theorem 1.5.19 by observing that $f^{-1}(0, +\infty) = [0, +\infty)$ is not open in $\mathbf{R}$.

1.6 Equivalent Metrics

In Definition 1.1.13 we defined isometries between metric spaces (X, d) and (Y, d'), and we have seen that this is an equivalence relation. By taking $X = Y$ and the identity map, this also defines an equivalence relation on the set of all metric on a given set X. That is, two metrics d d' on the same set are isometric if $\mathrm{Id}(X, d) \to (X, d')$ is an isometry. In this section we will see two other, coarser, equivalence relations on this set.

Definition 1.6.1 (Lipschitz[4] Equivalence)**.**

Two metrics d and d' on a set X are **Lipschitz equivalent** if there exist $h, k \in (0, \infty)$ such that

$$hd'(x,y) \leq d(x,y) \leq kd'(x,y), \quad \forall x, y \in X.$$

Lemma 1.6.2. *Lipschitz equivalence is an equivalence relation on the collection of all metrics on a set X.*

Proof. The proof is left as an exercise (Tutorial 3 Question 9). □

It is clear that isometric metrics are Lipschitz equivalent (for $h = k = 1$). In other words, Lipschitz equivalence is a coarser equivalence relation than isometry. The converse is not true, as demonstrated by the following example.

[4]Named after Rudolf Otto Sigismund Lipschitz (1832–1903).

Example 1.6.3 (Metrics on $\mathbf{R}^n$)**.** In Examples 1.1.3 and 1.1.5 and Definition 1.1.7 we have the metrics d_1, d_2, d_∞ on $\mathbf{R}^n$. We can check (Tutorial 3 Question 4) that

$$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n d_\infty(x,y), \quad \text{for all } x,y \in \mathbf{R}^n.$$

This proves that they are all Lipschitz equivalent, despite not being isometric.

Besides isometry and Lipschitz equivalence, we can define a third interesting equivalence relation on the space of metrics on X.

Definition 1.6.4 (Topological Equivalence)**.**

Two metrics d and d' on the set X are **topologically equivalent** if they give rise to the same open sets, that means, a subset $U \subseteq X$ is open with respect to the metric d if and only if it is open with respect to the metric d'.

Lemma 1.6.5. *Topological equivalence defines an equivalence relation on the collection of all metrics on a set X.*

Proof. The proof is left as an exercise (Tutorial 3 Question 9). □

Topological equivalence is the coarser equivalence relation among our three relations (isometry, Lipschitz equivalence and topological equivalence), as demonstrated below.

Proposition 1.6.6. *Lipschitz equivalent metrics are topologically equivalent. In other words, topological equivalence is a coarser equivalence relation than Lipschitz equivalence.*

Proof. Suppose that d and d' are Lipschitz equivalent metrics on a set X, so we have $h, k \in (0,\infty)$ such that

$$hd'(x,y) \le d(x,y) \le kd'(x,y), \quad \text{for all } x,y \in X.$$

Suppose also that $U \subseteq X$ is open with respect to the metric d. Let $x \in U$. Then there exists $\varepsilon > 0$ such that $B_\varepsilon(x;d) \subseteq U$. Take $\delta := \varepsilon/k$. Then $d(x,y) \le kd'(x,y) < \varepsilon$, for all $y \in B_\delta(x;d')$. So $B_\delta(x;d') \subseteq B_\varepsilon(x;d) \subseteq U$ and U is open with respect to the metric d'.

Showing that U is open with respect to d if it is open with respect to d' is done in a similar way but now using the constant h. The details are left as an exercise. □

Remark 1.6.7.

It is not true that if two metrics are topologically equivalent then they are Lipschitz equivalent. See Tutorial 3 Question 7 for a counter-example.

Not all metrics on a given sets are topologically equivalent.

Example 1.6.8. Let Y be the set of continuous functions from $[a,b]$ to $\mathbf{R}$. It can be endowed with the $d_{\sup}$ metric (see page 10) as well as with the $d_{\sup}$ metric (see page 11). The metric spaces $\mathcal{C}[a,b] = (Y, d_{\sup})$ and $\mathscr{L}_1[a,b] = (Y, d_1)$ are not topologically equivalent. Let $f_0 \colon [a,b] \to \mathbf{R}$ denote the constant function with value 0. We will show that $B_1(f_0; d_{\sup})$ is not d_1-open. If it were then there would exist $\varepsilon > 0$ such that $B_\varepsilon(f_0; d_1) \subseteq B_1(f_0; d_{\sup})$. But there always exists a continuous function $f \colon [a,b] \to \mathbf{R}$ such that $d_1(f, f_0) < \varepsilon$ but where $d_{\sup}(f, f_0) = 1$ [Can you think of such a function?] so $f \in B_\varepsilon(f_0; d_1)$ but $f \notin B_1(f_0; d_{\sup})$.

It follows from Proposition 1.6.6 that the metrics d_1, d_2, d_∞ (and d_p for $p \geq 1$) on $\mathbf{R}^n$ are all topologically equivalent. However, the discrete metric d_0 on $\mathbf{R}^n$ has one-point sets as open sets so it cannot be equivalent (either topologically or Lipschitz) to d_1, d_2 or d_∞.

Remark 1.6.9.

Since the metrics d_1, d_2, d_∞ on $\mathbf{R}^n$ are all topologically equivalent, even though the open balls in these metrics are different, the open sets are the same.

When studying vector spaces and groups, you have seen products, which were operations on the base set $X \times Y$ satisfying some natural compatibility conditions with the original operations. We would like to do the same for metric spaces. That is, given (X, d_X) and (Y, d_Y) we want to find a natural metric on the set $X \times Y$. Sadly, for metric spaces the situation is not as well-behaved as for vector spaces (or groups) and we can define many such natural metrics on the product. Below are a few examples that generalise the ideas of different metrics on $\mathbf{R}^n$ (which is a product of n copies of $\mathbf{R}$) to products of metric spaces in general.

Definition 1.6.10 (Cartesian Product)**.**

A **Cartesian product**[5] of two metric spaces (X, d_X) and (Y, d_Y) is the set $X \times Y$ with one of the following metrics:

- $d_a((x_1, y_1), (x_2, y_2)) = d_X(x_1, x_2) + d_Y(y_1, y_2)$;
- $d_b((x_1, y_1), (x_2, y_2)) = (d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2)^{1/2}$;
- $d_c((x_1, y_1), (x_2, y_2)) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}$,

for any $(x_1, y_1), (x_2, y_2) \in X \times Y$. The proofs that d_a, d_b and d_c satisfy the axioms (M0)–(M4) are derived directly from those for $X = Y = \mathbf{R}$ of Examples 1.1.3 and 1.1.5 and Definition 1.1.7 respectively, where $d_a = d_1$, $d_b = d_2$ and $d_c = d_\infty$.

Proposition 1.6.11. *The metrics d_a, d_b and d_c on $X \times Y$ are Lipschitz equivalent, and hence topologically equivalent.*

Proof. It follows directly from Definition 1.6.10 that

$$d_c(\alpha, \beta) \leq d_b(\alpha, \beta) \leq d_a(\alpha, \beta) \leq 2d_c(\alpha, \beta)$$

for every $\alpha, \beta \in X \times Y$. So Definition 1.6.1 applies to each pair in turn. □

[5]Named after René Descartes (1596–1650).

These results may be extended to n-fold products, and particularly $\mathbf{R}^n$ which we looked at in Example 1.6.3.

Remark 1.6.12.

For any $p \geq 1$, it is also possible to define a metric $d_{a_p}((x_1, y_1), (x_2, y_2)) = (d_X(x_1, x_2)^p + d_Y(y_1, y_2)^p)^{1/p}$ on $X \times Y$ similarly to what we did in Definition 1.1.6. This metric is Lipschitz equivalent to d_a (and hence also to d_b and d_c). We also call it *a* Cartesian product.
More generally, one say that a metric d on $X \times Y$ is a Cartesian product if it is topologically equivalent to one of the metric d_b from Definition 1.6.10. That is, a Cartesian product is a metric on $X \times Y$ that has the same open sets as d_b.

1.7 Equivalent Metric Spaces

We now extend the notions of Lipschitz and topological equivalences between metrics on the same set to metric spaces in general.

Definition 1.7.1.

A **Lipschitz equivalence** $f\colon (X, d_X) \to (Y, d_Y)$ between two metric spaces X and Y is a bijection $f\colon X \to Y$ such that there exist $h, k \in (0, \infty)$ such that

$$hd_Y(f(x), f(y)) \leq d_X(x, y) \leq kd_Y(f(x), f(y))$$

for all $x, y \in X$.
If there is a Lipschitz equivalence between two metric spaces we call them **Lipschitz equivalent**.

It is easy to see that an isometry is a special kind of Lipschitz equivalence (by taking $h = k = 1$), and the identity map $\mathrm{Id}_X\colon (X, d) \to (X, d')$ is a Lipschitz equivalence if and only if d and d' are Lipschitz equivalent metrics on X.

Definition 1.7.2 (Homeomorphism)**.**

A **homeomorphism**[a] (or **topological equivalence**) $f\colon (X, d_X) \to (Y, d_Y)$ between two metric spaces X and Y is a bijection $f\colon X \to Y$ such that both f and f^{-1} are continuous.
If there is a homeomorphism between two metric spaces we call them **homeomorphic**.

[a] From the Greek ὅμοιος μορφή: similar shape.

Like with isometries, Lipschitz equivalences and homeomorphisms define an equivalence relation on the collection of all metric spaces, see Tutorial 3 Question 9. The entire subject of topology focuses on properties of spaces that do not change under homeomorphism.

Example 1.7.3. All open intervals are homeomorphic. More precisely, we have

1. Any open interval $(a, b) \subseteq \mathbf{R}$ is homeomorphic to any other open interval $(c, d) \subseteq \mathbf{R}$ via the homeomorphism $f\colon (a, b) \to (c, d)$ given by
$$f(x) := \frac{d-c}{b-a}(x-a) + c.$$
This is a continuous function and its inverse is also seen to be continuous.

2. The function $x \mapsto x/(1-|x|)$ maps the interval $(-1, 1)$ homeomorphically onto $\mathbf{R}$.

3. The function
$$x \mapsto \begin{cases} x, & \text{if } x \leq 0 \\ 1/1-x & \text{if } x \geq 0 \end{cases}$$
maps the interval $(-\infty, 1)$ homeomorphically onto $\mathbf{R}$.

Therefore, every open interval is homeomorphic to $\mathbf{R}$.

It is evident that isometry is a finer equivalence relation than Lipschitz equivalence. The forthcoming proposition, which is similar to Proposition 1.6.6, shows that homeomorphism is an even coarser equivalence relation.

Proposition 1.7.4. *Lipschitz equivalent metric spaces are homeomorphic.*

Proof. Suppose that $f\colon X \to Y$ is a Lipschitz equivalence. Then the following hold:
$$f(B_{h\varepsilon}(x)) \subseteq B_\varepsilon(f(x)) \quad \text{and} \quad f^{-1}(B_{\frac{\varepsilon}{k}}(y)) \subseteq B_\varepsilon(f^{-1}(y)),$$
see Tutorial 3 Question 10.

Then f is continuous by the first condition and f^{-1} is continuous by the second. □

So we have the following chain of implications:
$$\text{isometry} \implies \text{Lipschitz equivalence} \implies \text{homeomorphism}.$$

Proposition 1.7.5. *Two metrics d and d' on a set X are topologically equivalent if and only if the identity function $\mathrm{Id}_X\colon (X, d) \to (X, d')$ is a homeomorphism.*

Proof. "$\Rightarrow$" Let $U \subseteq (X, d')$ be open. Then $\mathrm{Id}_X^{-1}(U) = U$ is open in (X, d) as the metrics are topologically equivalent. Therefore Id_X is continuous. Proving that the inverse is continuous works similarly.

"$\Leftarrow$" Suppose that $U \subseteq X$ is open with respect to the metric d'; then $\mathrm{Id}_X^{-1}(U) = U$ is open with respect to d since Id_X is continuous. Similarly, if $W \subseteq X$ is open with respect to d, then it is open with respect to d' because Id_X^{-1} is continuous. □

1.8 Closed Sets

Closed sets can be thought of as "*dual*" to open sets. This section will thus be similar to Section 1.5 and we will also start with the definition of the simplest closed sets: the closed balls.

Definition 1.8.1 (Closed Ball)**.**

Given a metric space (X, d), the **closed ball** of radius $r > 0$ around a point $x \in X$ is the set
$$\overline{B}_r(x) = \overline{B}_r(x; d) := \{y \in X \mid d(x, y) \leq r\} \subseteq X.$$

For $n \leq 3$, we recover the classical notions of closed intervals/disks/balls in the Euclidean space $\mathbf{R}^n$.

Similarly to what happens for open balls, if $(Y, d|_Y) \subseteq (X, d)$ is a subspace, then the closed balls of Y are exactly the $\overline{B} \cap Y$ for $\overline{B}$ a closed ball of X.

Example 1.8.2. In a discrete metric space, all open balls are closed balls, but not necessarily for the same radius.

We defined open sets via the notion of interior points. The dual notion of interior point is closure points.

Definition 1.8.3 (Closure)**.**

Let W be a subset of a metric space (X, d). We say that $x \in X$ is a **closure point** of W if for every $\varepsilon > 0$, $B_\varepsilon(x) \cap W \neq \varnothing$. The **closure** of W is the set $\overline{W}$ of all closure points.

Observe that $W \subseteq \overline{W}$.

Remark 1.8.4.

Even if we are defining closure points, the definition is still made using *open* balls and not closed ones!

We can finally define closed sets.

Definition 1.8.5 (Closed Set)**.**

A subset $V \subseteq (X, d)$ is **closed in** X if it contains all of its closure points, that is, $\overline{V} = V$.

Example 1.8.6. The open ball $B_1(0)$ is not closed in the Euclidean plane $\mathbf{R}^2$, because every point x with $|x| = 1$ is a closure point.

The following proposition formalise the fact that closed and open sets are dual notions.

Proposition 1.8.7. *A set V is closed in X if and only if its complement $U := X \setminus V$ is open.*

Proof. "$\Rightarrow$" Suppose that V is closed. Since $\overline{V} = V$, no point of $u \in U$ can have $B_\varepsilon(u) \cap V \neq \varnothing$, for every $\varepsilon > 0$. So there exists some $\delta > 0$ such that $B_\delta(u) \cap V = \varnothing$. So $B_\delta(u) \subseteq U$.

"$\Leftarrow$" Suppose U is open and that there is an element $u \in \overline{V}$ such that $u \notin V$. Then $u \in U$ so $\exists \varepsilon > 0$ such that $B_\varepsilon(u) \subseteq U$. So $B_\varepsilon(u) \cap V = \varnothing$ which is a contradiction and hence $u \in V$ and V is closed. □

Note that this proposition also says that U is open in X if and only if its complement $X \setminus U$ is closed in X.

Duality between open and closed subsets implies the dual of Proposition 1.5.17

Proposition 1.8.8. *Let (X, d) be a metric space and $(Y, d|_Y)$ be a subspace. Then*

1. *The closed subsets of Y are exactly the $V \cap Y$ for V closed in Y;*
2. *If Y is closed and $V \subseteq Y$ is closed in Y, then V is open in Y.*

Proof. "1" A subset $V \subseteq Y$ is closed in Y if and only if $Y \setminus V$ is open, if and only if there exists an open subset $U \subseteq X$ such that $U \cap Y = Y \setminus V$, if and only if $X \setminus U) \cap Y = V$.

"1" We have $V = Y \cap \tilde{V}$ for some closed subset $\tilde{V}$ of X. If Y is closed in X, then so is V as the intersection of two clsoed subsets. □

Closure and interior are also dual notions.

Proposition 1.8.9.

For a subset $W \subseteq (X, d)$ of a metric space we have

1. $W^\circ = X \setminus (\overline{X \setminus W})$;
2. $\overline{W} = X \setminus (X \setminus W)^\circ$.

Proof. The proof is left to the reader (see Tutorial 4 Question 5). □

To better understand Proposition 1.8.9 it is interesting to start with $W = [0, 1) \subseteq \mathbf{R}$ and to compute all the sets $X \setminus W$, $\overline{X \setminus W}$ and $X \setminus \overline{X \setminus W}$.

To go further

Interaction between interior and closure is not straightforward. On the positive side, for any subset $W \subseteq (X, d)$ of a metric space we have

$$\overline{W^\circ} = \overline{(\overline{W^\circ})^\circ} \quad \text{and} \quad \overline{W}^\circ = (\overline{\overline{W}^\circ})^\circ .$$

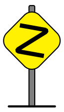
Since $(W^\circ)^\circ = W^\circ$ and $\overline{(\overline{W})} = \overline{W}$, it follows that starting from a subset W and taking interior and closure it is possible to obtain at most 7 different sets. These subsets are: W, W°, $\overline{W^\circ}$, $(\overline{W^\circ})^\circ$, $\overline{W}$, $\overline{W}^\circ$ and $(\overline{\overline{W}^\circ})$. The set $W = (0, 1) \cup (0, 2) \cup \{3\} \cup (\mathbf{Q} \cap [4, 5]) \subseteq \mathbf{R}$ shows that all the above subsets can indeed be pairwise different.

Similarly to Lemma 1.5.9, we verify that the names closed sets and closed balls are not misleading.

Corollary 1.8.10. *Every closed ball $\overline{B}_r(x)$ is closed in X.*

Proof. The complement $U := X \setminus \overline{B}_r(x)$ of a closed ball is given by $\{y \in X \mid d(x,y) > r\}$. So for any $y \in U$ with $d(x,y) = t$ we have $B_{t-r}(y) \subseteq U$. So U is open in X. $\square$

Since the complement $U := X \setminus \overline{B}_r(x)$ of a closed ball, given by $\{y \in X \mid d(x,y) > r\}$, is open in X we know that no point $x \in \mathbf{R}^2$ with $|x| > 1$ is a closure point of $B_1(0)$. So $\overline{B_1(0)} = \overline{B}_1(0)$ in $\mathbf{R}^2$. This explains the notation for closed balls.



Remark 1.8.11.

Beware, for general metric spaces it is not always true that $\overline{B_r(x)} \neq \overline{B}_r(x)$. Can you find a counterexample? *Hint: think of the discrete metric.* See Tutorial 4 Question 6

Closure satisfies properties dual to the ones from interior, see Proposition 1.5.13.

Proposition 1.8.12. *Given any two subsets $U, V \subseteq (X,d)$ of a metric space we have*

1. $U \subseteq V \implies \overline{U} \subseteq \overline{V}$;
2. $\overline{(\overline{V})} = \overline{V}$;
3. *$\overline{V}$ is the smallest closed subset of X that contains V.*

Proof. The proof is left to the reader (see Tutorial 4 Question 3). $\square$

The collection of closed subsets satisfies some nice closure properties, dual to the ones satisfied by open sets from Proposition 1.5.14.

Proposition 1.8.13. *Let (X,d) be a metric space. Then*

1. *The empty set $\varnothing$ and the whole space X are closed in X;*
2. *If $V_1, \dots, V_m$ are closed sets, then $\bigcup_{i=1}^m V_i$ is closed in X;*
3. *The intersection of any collection of closed sets is closed in X.*

Proof. All these properties follow from the dual properties for open sets and the following facts:

$$X \setminus X = \varnothing, \quad X \setminus \varnothing = X, \quad X \setminus \Big(\bigcup_{i=1}^m V_i\Big) = \bigcap_{i=1}^m (X \setminus V_i), \quad X \setminus \Big(\bigcap_{i\in I} V_i\Big) = \bigcup_{i=1}^m (X \setminus V_i). \quad \square$$

There are subsets of a metric space that can be neither open nor closed:

Example 1.8.14. Let x be a point in a metric space (X,d) and let P be a **proper subset** of $\{y \in X \mid d(x,y) = r\}$, that is, P is not the empty set nor the whole set. The set $P_r(x) := B_r(x) \cup P$ is known as a **partially open ball**.

A partially open ball in $\mathbf{R}^2$ is neither open nor closed. The points $y \in P$ are not in the interior of $P_r(x)$ and the points $y \notin P$, where $d(x,y) = r$ are closure points of $P_r(x)$.

We have seen in Theorem 1.5.19 that continuity can be characterised using open sets. It is thus not a surprise that we can give another characterisation of it using closed sets.

Theorem 1.8.15.

A function between metric spaces is continuous if and only if the preimage of every closed set is closed.

Proof. "$\Rightarrow$" Let $f\colon X \to Y$ be continuous and $V \subseteq Y$ be closed. Then

$$X \setminus f^{-1}(V) = f^{-1}(Y \setminus V)$$

is open, since $Y \setminus V$ is open, which implies that $f^{-1}(V)$ is closed.

"$\Leftarrow$" Suppose that $f^{-1}(V)$ is closed in X for every closed subset $V \subseteq Y$. Suppose that U is open in Y. Then $Y \setminus U$ is closed, which implies that $f^{-1}(Y \setminus U)$ is closed in X and consequently that

$$X \setminus f^{-1}(Y \setminus U) = f^{-1}(U)$$

is open in X. Thus, f is continuous. $\square$

Using closure, one can define what it means for a subset U to be dense in X. Intuitively, this means that every point in X is either in U or arbitrarily close to a point in U, or that points of U are tightly clustered.

Definition 1.8.16.

A subset $W \subseteq (X, d)$ of a metric space is **dense in** X if $\overline{W} = X$.

From what we learned in Analysis 1, both $\mathbf{Q}$ and $\mathbf{R} \setminus \mathbf{Q}$ are dense in $\mathbf{R}$.

The subset $\mathbf{Q} \cap [0, 1]$ is not dense in $\mathbf{R}$, but it is still "locally dense" in the sense that $\overline{\mathbf{Q} \cap [0, 1]} = [0, 1]$. For a strong negation of denseness, the correct notion is of being nowhere dense.

Definition 1.8.17.

A subset $W \subseteq (X, d)$ of a metric space is **nowhere dense in** X if $(\overline{W})^\circ = \varnothing$.

Remark 1.8.18.

When checking if a subset W is nowhere dense, do not forget to first take the closure of W before taking the interior! It is indeed possible that $W^\circ = \varnothing$ but W is not nowhere dense. This happens for example with $\mathbf{Q}$ in $\mathbf{R}$.

Example 1.8.19. The subset $\left\{\frac{1}{n} \,\middle|\, n \in \mathbf{N}_{\geq 1}\right\}$ is nowhere dense in $\mathbf{R}$.

The following shows that being nowhere dense is, sort of, a dual notion of being dense.

Proposition 1.8.20. *A subset $W \subseteq (X, d)$ is nowhere dense in X if and only if $X \setminus \overline{W}$ is dense in X.*

Proof.

$$\begin{aligned}(\overline{W})^\circ = \varnothing &\iff \text{ no } x \in X \text{ is an interior point of } \overline{W},\\ &\iff \forall \varepsilon > 0, B_\varepsilon(x) \not\subseteq \overline{W}, \quad \text{for all } x \in X,\\ &\iff \forall \varepsilon > 0, B_\varepsilon(x) \cap (X \setminus \overline{W}) \neq \varnothing, \quad \text{for all } x \in X,\\ &\iff x \in \overline{(X \setminus \overline{W})}, \quad \text{for all } x \in X,\\ &\iff X = \overline{(X \setminus \overline{W})}.\end{aligned}$$ □

Corollary 1.8.21. *A closed subset $V \subseteq (X,d)$ is nowhere dense in X if and only if $X \setminus V$ is dense in X.*

Example 1.8.22. The above corollary is not true for an arbitrary, not necessarily closed, subset V. For example, $\mathbf{R} \setminus \mathbf{Q}$ is dense in $\mathbf{R}$, but $\mathbf{Q}$ is far of being nowhere dense as $(\overline{\mathbf{Q}})^\circ = \mathbf{R}$.

We will conclude this section by a proposition that justifies the name nowhere dense. Before that we prove a small technical but useful lemma.

Lemma 1.8.23. *Let (X,d) be a metric space. Let $U \subseteq X$ be an open set and $W \subseteq X$ be any subset. Then $U \cap W \neq \varnothing$ if and only if $U \cap \overline{W} \neq \varnothing$.*

Proof. The left-to-right implication is trivial as $W \subseteq \overline{W}$.

For the other implication, let x be any element in $U \cap \overline{W} \neq \varnothing$. On one hand, since x is in the open set U, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$. On the other hand, x being a closure point of W, the intersection $B_\varepsilon(x) \cap W \subseteq U \cap W$ is not empty. □

Proposition 1.8.24. *For a subset $W \subseteq (X,d)$ of a metric space the following are equivalent:*

I. W is nowhere dense: $(\overline{W})^\circ = \varnothing$;

II. $\overline{W}$ contains no non-empty open subset;

III. For any non-empty open subset $U \subseteq X$, the subset $W \cap U$ is not dense in U (i.e. $U \not\subseteq \overline{W \cap U}$).

Proof. "I $\iff$ II" This follows from the fact that V° is the biggest open subset of V.

"II $\implies$ III" We do the proof by contrapositive. Suppose that there exists a non-empty open subset U with $W \cap U$ dense in U. Then $U \subseteq \overline{U \cap W} \subseteq \overline{W}$.

"III $\implies$ II" Let U be an arbitrary non-empty open set. Since U is not contained in $\overline{U \cap W}$, the set $V := U \setminus \overline{U \cap W}$ is a non-empty open subset of U such that $V \cap (\overline{U \cap W}) = \varnothing$. We conclude that $V \cap (U \cap W) = V \cap W$ is empty and, by the previous lemma, that $V \cap \overline{W} = \varnothing$. This implies that V is contained in $X \setminus \overline{W}$, and in particular that U is not contained in $\overline{W}$ (as witnessed by any point of V), which finishes the proof. □

1.9 Boundedness

Having a notion of distance between points is enough to define a notion of "size" of subsets. We start by defining what it means for a subset to not be too big.

Definition 1.9.1 (Bounded Subset)**.**

A subset W of a metric space (X, d) is **bounded** if $W \subseteq \overline{B}_r(x)$ for some closed ball $\overline{B}_r(x)$ in X.

If W is a bounded subset of (X, d), witnessed by $\overline{B}_r(x)$, then for any $w_1, w_2 \in W$,

$$d(w_1, w_2) \leq d(w_1, x) + d(x, w_2) \leq 2r.$$

Therefore, the following definition makes sense:

Definition 1.9.2 (Diameter)**.**

If W is a non-empty bounded subset of a metric space (X, d), then the **diameter** of W is

$$\operatorname{diam} W := \sup\{d(w_1, w_2) \mid w_1, w_2 \in W\}.$$

Alternatively, we could have first use the formula in Definition 1.9.2 to define the diameter $d(W) \in \mathbf{R}_{\geq 0} \cup \{+\infty\}$ of any subset W of X and then define a bounded subset to be a subset of finite diameter.

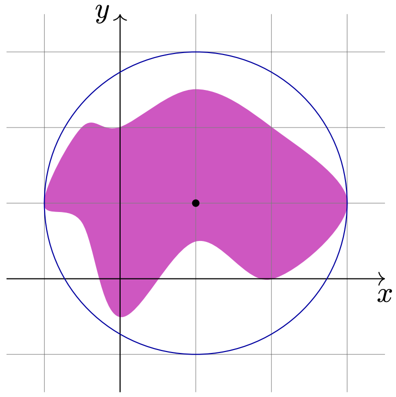


Figure 1.8: A subset of $\mathbf{R}^2$ of diameter 2.

Remark 1.9.3.

It is in general not true that a bounded subset W is contained in a closed ball of radius $r = {}^1\!/_2 \operatorname{diam}(W)$. For example, in a discrete metric space if W has at least two elements then its diameter is 1, but any ball containing W has radius at least 1.

Bounded subsets are preserved by Lipschitz equivalences. Namely,

Lemma 1.9.4. *Let $f\colon (X, d_X) \to (Y, d_Y)$ be a Lipschitz equivalence. Then a subset W of X is bounded if and only if $f(W)$ is bounded.*

Proof. There exists $h, k \in (0, \infty)$ such that $hd_Y(f(x), f(y)) \leq d_X(x, y) \leq kd_Y(f(x), f(y))$ for all $x, y \in X$. It follows that $h \operatorname{diam}_{d_Y} f(W) \leq \operatorname{diam}_{d_X} W \leq k \operatorname{diam}_{d_Y} f(W)$. □

The above statement is not true anymore if we replace Lipschitz equivalence by topological equivalence. In fact, if (X, d) is a metric space, then there exists a topologically equivalent metric d' such that (X, d') is bounded, see Tutorial 4 Question 7.

The collection of bounded subsets satisfies some nice closure properties.

Proposition 1.9.5. *Let X be a metric space. Then the following holds:*

If $Y \subseteq Z$ are two subsets of X with Z bounded, then Y is also bounded;

Any point x in X belong to some bounded subset;

The union of any finite number of bounded subsets of a X is bounded.

Proof. "Proposition 1.9.5" The proof is left to the reader (see Tutorial 5 Question 1).

"Proposition 1.9.5" We have $x \in \{x\} = B_0(x)$.

"Proposition 1.9.5" We only need to prove this for two subsets as the result follows by induction. Suppose that W_1 and W_2 are bounded subsets of a metric space (X, d) and let $x_1, x_2 \in X$ and $r_1, r_2 \in \mathbf{R}$ be such that $d(w, x_i) \leq r_i$, for all $w \in W_i$ $(i = 1, 2)$. Put $x = x_1$ say and $r = \max\{r_1, r_2 + d(x_1, x_2)\}$. Then for any $w \in W_1 \cup W_2$, either $w \in W_1$ and $d(w, x) \leq r_1 \leq r$, or $w \in W_2$ and $d(w, x) = d(w, x_1) \leq d(w, x_2) + d(x_2, x_1) \leq r_2 + d(x_2, x_1) \leq r$. So $W_1 \cup W_2 \subseteq \overline{B}_r(x)$. □

An arbitrary union of bounded subsets is not necessary bounded. A counter-example is given by $\bigcup_{n\geq 1}[-n, n]$ in $\mathbf{R}$.

To go further

A **bornological space** is a couple $(X, \mathcal{B})$ of a set and a collection $\mathcal{B}$, called a **bornology**, of subsets of X satisfying Proposition 1.9.5. More precisely, a collection $\mathcal{B}$ of subsets of X is a bornology if it is closed by taking subsets and finite unions and if $X = \bigcup_{B\in\mathcal{B}} B$.

Using bounded subsets, one can generalise the classical notion of real bounded functions, see Section 1.3, to functions with values in an arbitrary metric space.

Definition 1.9.6 (Bounded Function).

If $f\colon A \to X$ is a function into a metric space (X, d), then f is **bounded** if the set

$$f(A) := \{f(a) \mid a \in A\} \subseteq X$$

is a bounded subset of (X, d).

Observe that in the definition of a bounded function $f\colon A \to X$ we do not require A to be a metric space.

1.10 Boundary

In $\mathbf{R}$ with the standard metric, all four intervals $[0,1]$, $[0,1)$, $(0,1]$ and $(0,1)$ have the same boundary points: 0 and 1. We can generalise the notion of boundary points to subsets of arbitrary metric spaces. We do this by defining a boundary point as a point that is both arbitrarily close to W and arbitrarily close to its complement $X \setminus W$.

Definition 1.10.1.

Suppose $W \subseteq (X,d)$. An element $x \in X$ is a **boundary point** of W if, for every $\varepsilon > 0$,

$$B_\varepsilon(x) \cap W \neq \varnothing \quad \text{and} \quad B_\varepsilon(x) \cap (X \setminus W) \neq \varnothing.$$

The **boundary** ∂W of W is the set of all such boundary points.

Remark 1.10.2.

It follows directly from the definition that $\partial W = \partial(X \setminus W)$.

Interior and closure of a subset satisfies $W^\circ \subseteq W \subseteq \overline{W}$. We can use this to use an alternative definition of the boundary.

Proposition 1.10.3. *The boundary of any subset $W \subseteq (X,d)$ is the set $\overline{W} \setminus W^\circ$.*

Proof. Suppose that $x \in \partial W$. Then every $B_\varepsilon(x)$ meets W, so $x \in \overline{W}$; but $B_\varepsilon(x)$ meets $X \setminus W$, so $x \notin W^\circ$. Hence $\partial W \subseteq \overline{W} \setminus W^\circ$.

Conversely, suppose that $x \in \overline{W} \setminus W^\circ$. Then $x \in \overline{W}$, so every $B_\varepsilon(x)$ meets W; and $x \notin W^\circ$, so $B_\varepsilon(x)$ meets $X \setminus W$. Therefore, $B_\varepsilon(x) \cap W$ and $B_\varepsilon(x) \cap (X \setminus W)$ are both non-empty and $x \in \partial W$. Hence $\overline{W} \setminus W^\circ \subseteq \partial W$, as required. $\square$

Boundaries are not any subsets, they are always closed.

Corollary 1.10.4. *Let $W \subseteq (X,d)$. Then ∂W is closed in X.*

Proof. We set $\partial W = \overline{W} \setminus W^\circ = \overline{W} \cap (X \setminus W^\circ)$ is the intersection of two closed subsets. It is therefore closed. $\square$

If W is not closed, then it has closure points not in W, so $\partial W \not\subseteq W$. If W° is empty, then $\partial W = \overline{W}$.

The following characterisation of boundary explains the naming.

Lemma 1.10.5. *For any subset $W \subseteq (X,d)$ of a metric space we have $\partial W = \overline{W} \cap \overline{(X \setminus W)}$.*

Proof. The proof is left to the reader (see Tutorial 5 Question 3). $\square$

We conclude this section with some examples.

Example 1.10.6.

1. In $\mathbf{R}^n$, the closed ball $\overline{B}_r(0)$ is closed and its interior is the open ball $B_r(0)$. So $\partial\overline{B}_r(0) = \{x \in \mathbf{R}^n \mid |x| = r\}$ is the sphere of radius r. Similarly, $B_r(0)$ is open and its closure is the closed ball $\overline{B}_r(0)$. So $\partial B_r(0) = \{x \in \mathbf{R}^n \mid |x| = r\}$ also!

2. The subset $\mathbf{Q} \subseteq \mathbf{R}$ is dense but $\mathbf{Q}^\circ$ is empty. Therefore $\partial\mathbf{Q} = \mathbf{R}$.

3. The subset $S := \mathbf{R}^n \setminus \{0\}$ of $\mathbf{R}^n$ is open, because $\{0\}$ is closed. Moreover, $\overline{S} = \mathbf{R}^n$, because, for every $\varepsilon > 0$, $B_\varepsilon(0) \cap S \neq \varnothing$. Therefore, $\partial S = \mathbf{R}^n \setminus S = \{0\}$.

1.11 Sequences

We know from Analysis 1 that a sequence of real number converges to some limit l if the points in the sequence become arbitrary close to l. Since this definition only use the distance, we can generalise it to any metric space.

Remind that for any set X a **sequence** in X is a function $s\colon \mathbf{N} \to X$. It is usual to write terms of the sequence $s(n)$ as x_n and denote the whole sequence by $(x_n)_{n\in\mathbf{N}}$ or simply by (x_n).

Definition 1.11.1.

A sequence (x_n) in a metric space **converges** to the point $x \in X$ if

$$\forall\varepsilon > 0, \exists N \in \mathbf{N} \text{ such that } n \geq N \implies d(x, x_n) < \varepsilon.$$

The point x is known as the **limit** of (x_n).

We can write the convergence of (x_n) to x as $\lim_n x_n = x$, or $x_n \to x$ (as $n \to \infty$).

Remark 1.11.2.

We write $\lim_n x_n = x$ using equality, but not all sequences are convergent so the left hand side might not exist. Can you find a counterexample? *Hint: remember Analysis 1.*

In terms of open balls we have:

$$\lim_n x_n = x \iff \forall\varepsilon > 0, \exists N \in \mathbf{N}, \forall n \geq N : x_n \in B_\varepsilon(x).$$

Lemma 1.11.3. *In a metric space (X, d), $\lim_n x_n = x$ if and only if $d(x, x_n) \to 0$ in $\mathbf{R}$.*

Proof. Simple consequence of Definition 1.11.1. □

Example 1.11.4.

1. In $\mathbf{R}$, Definition 1.11.1 coincides with the usual definition of $\lim_n x_n = x$ as we learnt in Analysis 1.

2. In a discrete metric space (X, d), a ball $B_\varepsilon(x) = \{x\}$ for any $\varepsilon \leq 1$. So $x_n \to x$ if and only if the sequence is eventually constant (i.e. there exists $N \in \mathbf{N}$ such that $x_n = x$, for all $n \geq N$).

Example 1.11.5. In $\mathscr{C}[0,1]$ the sequence of continuous functions $f_n := x^n$ does not converge to the constant zero function since $\sup_{x\in[0,1]}|x^n| = 1$. But on the other hand, in $\mathscr{L}_1[0,1]$

$$d_1(x^n, 0) = \int_0^1 |x^n| \,\mathrm{d}x = {}^1\!/\!_{(n+1)} \to 0, \quad \text{as } n \to \infty,$$

so (f_n) converges to the constant zero function in $\mathscr{L}_1[0,1]$ but not in $\mathscr{C}[0,1]$.

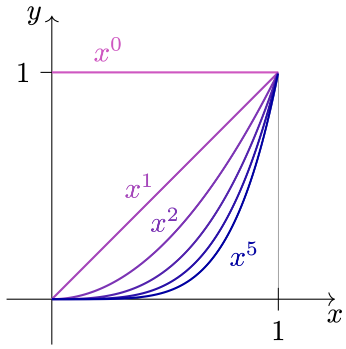


Figure 1.9: Graphs of the polynomial functions $x \mapsto x^n$ for $n \in \{0, \dots, 5\}$. The area below the curve tends to 0, but the supremum of the function is always 1.

In Definition 1.11.1 we talked about *the* limit (and not *a* limit) of a sequence. This is justified by the following result.

Proposition 1.11.6. *In any metric space (X, d) the limit of a convergent sequence is unique.*

Proof. Suppose that $x_n \to x$ and $x_n \to x'$, where $x \neq x'$. Then $d := d(x, x') > 0$. Choose $\varepsilon = {}^d\!/\!_2$. So $B_\varepsilon(x) \cap B_\varepsilon(x') = \varnothing$. On the other hand, $x_n \in B_\varepsilon(x)$, for all $n \geq N$, and $x_n \in B_\varepsilon(x')$, for all $n \geq N'$. So $x_n \in B_\varepsilon(x) \cap B_\varepsilon(x')$, for any $n \geq \max\{N, N'\}$, which is a contradiction. Hence, $x = x'$. $\square$

Limit points can be used to characterise the closure of a subset.

Proposition 1.11.7. *Suppose that $W \subseteq (X, d)$ and $w \in X$. Then $w \in \overline{W}$ if and only if there exists a sequence (w_n) in W such that $\lim_n w_n = w$.*

Proof. "$\Rightarrow$" Let $w \in \overline{W}$. Then $B_{1/n}(w) \cap W$ is non-empty for any integer $n \geq 1$. Choose a point $w_n \in B_{1/n}(w) \cap W$ for every $n \in \mathbf{N}$. Moreover, for any $\varepsilon > 0$, there exists $n \in \mathbf{N}$, such that $0 < 1/n < \varepsilon$; so $w_n \in B_\varepsilon(w)$. Hence, $\lim_n w_n = w$.

"$\Leftarrow$" Suppose that $\lim_n w_n = w$. Then $w_n \in B_\varepsilon(w)$ for all sufficiently large n and any $\varepsilon > 0$. But $w_n \in W$, so $B_\varepsilon(w) \cap W \neq \varnothing$ and so $w \in \overline{W}$. □

Using the above result, one can characterise closed sets in terms of converging sequences. We can also characterise open sets in term of converging sequences.

Proposition 1.11.8. *Let $W \subseteq X$ be a subset of a metric space. Then*

1. *W is closed if and only for all $(x_n) \subseteq W$, if the $\lim_n x_n$ exists (in X) then it belongs to W;*

2. *W is open if and only if for every sequence (x_n) of X that converges to a point in W, there exists $K \in \mathbf{N}$ such that $x_k \in W$, for all $k \geq K$.*

Proof. "1" This follows from Proposition 1.11.7 and the fact that W is closed if and only if $\overline{W} = W$.

"2" The proof is left to the reader (see Tutorial 5 Question 6). □

Example 1.11.9. The subset $W := \{1/n \mid n \in \mathbf{N}\}$ of $\mathbf{R}$ does not contain 0. But the sequence $(w_n = 1/n)$ lies in W, and has limit 0; so $0 \in \overline{W}$. Moreover, any convergent sequence in W must tend to some $1/m$ or 0. So $\overline{W} = W \cup \{0\}$.

We can use convergence to give yet another characterisation of continuity.

Theorem 1.11.10.

A function $f : X \to Y$ between two metric spaces is continuous at $x \in X$ if and only if

$$(x_n) \text{ converges to } x \text{ in } X \implies f(x_n) \text{ converges to } f(x) \text{ in } Y.$$

Proof. "$\Rightarrow$" Suppose that f is continuous at $x \in X$ and that (x_n) is a sequence in X that converges to x. Then, for any $\varepsilon > 0$, there exists $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$ and there also exists a $N \in \mathbf{N}$ such that $n \geq N$ implies $x_n \in B_\delta(x)$. Therefore, for $n \geq N$, $f(x_n) \in B_\varepsilon(f(x))$, and $f(x_n)$ converges to $f(x)$.

"$\Leftarrow$" Now suppose that $f(x_n)$ converges to $f(x)$ in Y for every (x_n) that converges to x in X. For some $\varepsilon > 0$ assume that there does not exist a $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. Then for each $n \geq 1$, there exists a $x_n \in B_{1/n}(x)$ with $f(x_n) \notin B_\varepsilon(f(x))$. So (x_n) converges to x by construction, but $f(x_n)$ does not converge to $f(x)$. This is a contradiction and therefore such a $\delta > 0$ does exist, and therefore f is continuous at x. □

Corollary 1.11.11. *A function $f : X \to Y$ is continuous if and only if*

$$(x_n) \text{ converges to } x \text{ in } X \implies f(x_n) \text{ converges to } f(x) \text{ in } Y,$$

for every $x \in X$.

Example 1.11.12. The function $e\colon \mathbf{R} \to \mathbf{R}^2$ given by $e(x) = (\cos x, \sin x)$ is continuous [can you prove it?]. We know that the sequence $(x_n = \pi/n)$ converges to $0 \in \mathbf{R}$ so $(e(x_n))$ converges to $e(0) = (1,0) \in \mathbf{R}^2$. It is interesting to compare this with the sequence $(w_n = (2n + 1/n)\pi)$ which does not converge in $\mathbf{R}$ but $(e(w_n) = (\cos(\pi/n), \sin(\pi/n)))$ also converges to $(1,0)$ since $e(x_n) = e(w_n)$.

Given a subset W of a metric space, we know from Proposition 1.11.7 that $\overline{W}$ is the set of all limit points of all sequences of elements of W. But any w in W is such a limit point for the trivial reason that $\lim_n w = w$. We thus might want to be more precise and to look at all points of X that can be "approximated" by sequences of elements of W. That is, we are interested at limits of not ultimately constant sequences of elements of W.

Definition 1.11.13.

Let $W \subseteq (X,d)$ be a subset of a metric space. A point $x \in X$ is an **accumulation point** of W if, for every $\varepsilon > 0$, $B_\varepsilon(x) \cap W \setminus \{x\} \neq \varnothing$. We denote by $\mathrm{Acc}(W)$ the set of all accumulation points of W.

Note that for any subsets of X we have $A \cap (B \setminus C) = (A \cap B) \setminus C$. So $B_\varepsilon(x) \cap W \setminus \{x\}$ is not ambiguous.

Example 1.11.14. In $\mathbf{R}$ we have $\mathrm{Acc}([0,1) \cup \{2\}) = [0,1]$.

Accumulation points are related to the closure of W in the following way.

Proposition 1.11.15. *For any subset $W \subseteq X$ of a metric space we have $\overline{W} = W \cup \mathrm{Acc}(W)$.*

Proof. The proof is left as an exercise (Tutorial 4 Question 7). □

Combining Propositions 1.11.7 and 1.11.15 we obtain

Corollary 1.11.16. *We have* $\mathrm{Acc}(W) = \{\lim w_n \mid \forall n : w_n \in W \setminus \{x\}\}$.

The following concept is the dual of accumulation points.

Definition 1.11.17.

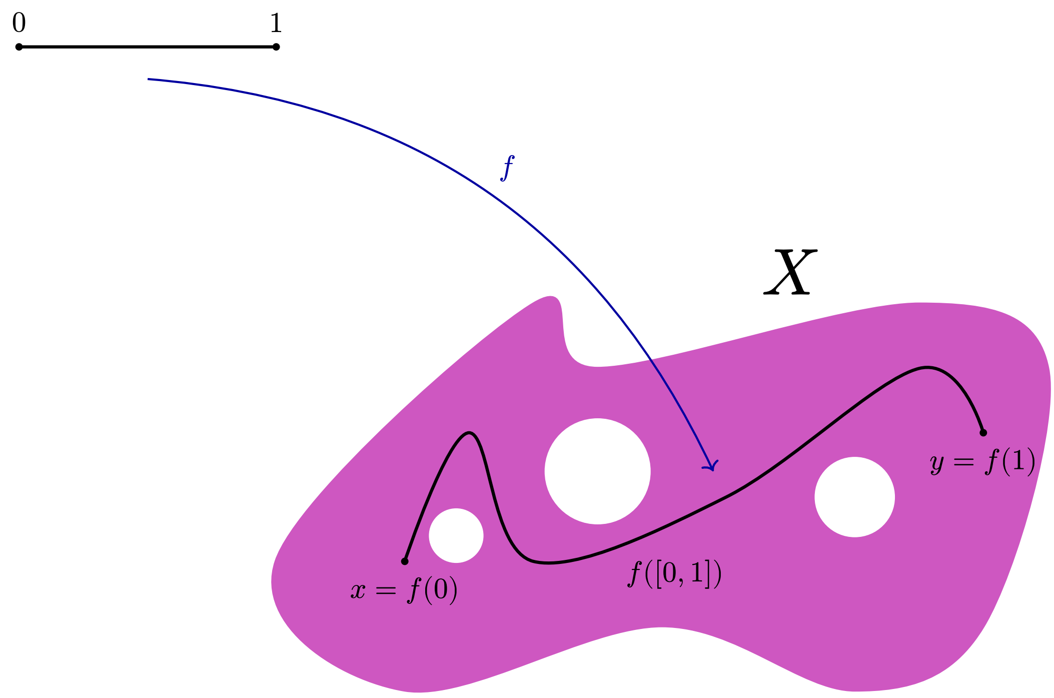
Let $W \subseteq (X,d)$ be a subset of a metric space. A point $w \in W$ is an **isolated point** of W if there exists $\varepsilon > 0$ such that $B_\varepsilon(w) \cap W = \{w\}$. We write $\mathrm{Iso}(W)$ for the set of all isolated points of W.

Proposition 1.11.18. *For any subset $W \subseteq X$ of a metric space we have* $\mathrm{Iso}(W) = W \setminus \mathrm{Acc}(W)$.

Proof. The proof is left as an exercise (Tutorial 4 Question 7). □

CHAPTER 2

CONNECTEDNESS



2.1 Connectedness

A common intuition is that a subset W of $\mathbf{R}^n$ is connected if for any two points x and y in W it is possible to "draw a path" from x to y that stays in W. Whilst this gives an interesting notion, which we will study in Section 2.2, it is sometimes not general enough as there are subsets from $\mathbf{R}^2$ that we would like to be connected but do not satisfy this definition. The intuition for the forthcoming general definition of connexity is the following. First of all, in $\{0,1\}$ with the discrete metric, 0 should not be connected to $\{1\}$, so $\{0,1\}$ is not connected. Then we want the fact of being connected to be preserved by continuous functions. So, a subset W of X has at least two "connected components" if we can colour elements of W in purple an blue such that elements coloured in purple stay away from elements coloured in blue.

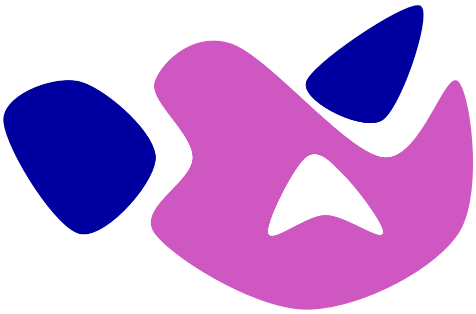


Figure 2.1: A non-connected subset of $\mathbf{R}^2$, with at least two connected components.

Definition 2.1.1 (Connected)**.**

A metric space (X,d) is **connected** if there does not exist a surjective continuous function from X onto a discrete two-point space.
A subset $W \subseteq X$ is **connected** if $(W, d|_W)$ is connected.

Observe that with this definition, the empty set $\varnothing$ is connected.

In Figure 2.1 we can define a map $f\colon W \to \{0,1\}$ sending purple points to 0 and blue points to 1. This map is surjective and continuous if we endow $\{0,1\}$ with the discrete metric.

Definition 2.1.2 (Partition)**.**

A **partition** $A \mid B$ of a metric space X is a pair of non-empty subspaces $A, B \subseteq X$ such that $A \cup B = X$, $A \cap B = \varnothing$, and both A and B are open in X.
We say that $A \mid B$ **partitions** X.

Note that A and B are also both closed in X when $A \mid B$ partitions X.

Theorem 2.1.3.

A metric space X is connected if and only if it admits no partition.

Proof. "$\Rightarrow$" Let D denote $\{0, 1\}$ with the discrete metric. Suppose that $A \mid B$ partitions X. Define $f: X \to D$ by

$$f(x) := \begin{cases} 0, & x \in A; \\ 1, & x \in B. \end{cases}$$

Since A and B are non-empty, f is surjective. The only open sets of D are: $\varnothing, \{0\}, \{1\}, D$. Now

$$f^{-1}(\varnothing) = \varnothing, \quad f^{-1}(0) = A, \quad f^{-1}(1) = B, \quad f^{-1}(D) = X.$$

So f is continuous as the preimage of every open set in D is open in X. So by definition, X is not connected.

"$\Leftarrow$" If X is not connected there exits a surjective continuous function $f: X \to D$ and it is easy to show that $f^{-1}(0) \mid f^{-1}(1)$ partitions X. $\square$

To go further

One can generalise Definitions 2.1.1 and 2.1.2 and Theorem 2.1.3 in the following way. Let (X, d) be a metric space and $W \subseteq$ be a subspace. We say that W has **at least n connected components** if there exists a surjective map $f: X \to \{1, \ldots, n\}$ which is continuous for the discrete metric on $\{1, \ldots, n\}$. We say that W admits a **n-partition** if there exists n non-emtpy open pairswise disjoint subsets U_i such that $W = \bigcup_i U_i$. The **number of connected components** of W is the minimal n, if it exists, such that W has at least n connected component, equivalently such that there exists a n-partition. If no such n exists, we say that W has infinitely many components. For example, the subset of $\mathbf{R}^2$ depicted in Figure 2.1 has 3 connected components.

One can even generalise more and define for any, possibly infinite, cardinal κ the fact to have at least κ-connected components, which is equivalent to admitting a κ-partition.

It is easy to prove that a set is connected if and only if it has at most 1 connected component.

The characterisation of connectedness in terms of the absence of partition is sometimes easier to use as demonstrated in the following example.

Example 2.1.4. The rational numbers $\mathbf{Q}$ are not connected. For if $\alpha \in \mathbf{R} \setminus \mathbf{Q}$ is any irrational number, then $(\mathbf{Q} \cap (-\infty, \alpha)) \mid (\mathbf{Q} \cap (\alpha, +\infty))$ partitions $\mathbf{Q}$.

The following corollary of Theorem 2.1.3 gives us a third characterisation of connectedness.

Theorem 2.1.5.

A metric space is connected if and only if the only subsets of X which are both open and closed in X are $\varnothing$ and X.

Proof. If $A \mid B$ partitions X, then A (and B) is an open and closed subset of X which is neither $\varnothing$ nor X. Conversely, if there exists an open and closed set A in X other than $\varnothing$ and X, then $A \mid (X \setminus A)$ partitions X. $\square$

By an **interval** in $\mathbf{R}$ we mean subsets of the form:

$$[a,b],\ (a,b),\ (a,b],\ [a,b),\ (a,+\infty),\ [a,+\infty),\ (-\infty,b),\ (-\infty,b],\ (-\infty,+\infty)=\mathbf{R},$$

where $a \leq b$ two real numbers. Observe that $a=b$ gives $[a,a]=\{a\}$ in the first interval and $(a,a)=(a,a]=[a,a)=\varnothing$ in intervals 2 to 4.

We will shows that intervals are exactly the connected subsets of $\mathbf{R}$. To do so, we will use the following characterisation in term of "betweeness".

Lemma 2.1.6. *A non-empty subset $A \subseteq \mathbf{R}$ is an interval if and only if it satisfies the following property: if $x, y \in A$ and $z \in \mathbf{R}$ are such that $x < z < y$, then $z \in A$.*

Proof. If A is an interval it clearly has the stated property. Conversely, suppose A has the stated property. Let

$$\begin{aligned} a &= \inf A \quad (\text{or } a = -\infty \text{ if } A \text{ is not bounded below}),\\ b &= \sup A \quad (\text{or } b = +\infty \text{ if } A \text{ is not bounded above}). \end{aligned}$$

We shall prove that $(a,b) \subseteq A \subseteq [a,b]$.

First, if $z \in (a,b)$, then $z > a$, so by definition of a there exists x in A with $x < z$. Similarly, there exists $y \in A$ with $y > z$. So by the hypothesis, $z \in A$. This proves that $(a,b) \subseteq A$, and the inclusion $A \subseteq [a,b]$ follows from the definitions of a and b. $\square$

Lemma 2.1.7. *No subspace of $\mathbf{R}$ other than an interval is connected.*

Proof. Suppose $A \subseteq \mathbf{R}$ is not an interval. Then by Lemma 2.1.6 there exist $x, y \in A$ and $z \in \mathbf{R} \setminus A$ such that $x < z < y$. Then $(A \cap (-\infty, z)) \mid (A \cap (z, +\infty))$ partitions A. Indeed both $A \cap (-\infty, z)$ and $A \cap (z, +\infty)$ are open in A, they are non-empty since they contain x and y respectively, they are clearly disjoint and their union is A since $z \notin A$. $\square$

Lemma 2.1.8. *Any interval $I \subseteq \mathbf{R}$ is connected.*

Proof. Suppose that $A \mid B$ partitions I. Let $a \in A$ and $b \in B$ and suppose that $a < b$ (otherwise change the names of A and B). Since $a, b \in I$ and I is an interval $[a,b] \subseteq I$. Let $A' = A \cap [a,b]$ and $B' = B \cap [a,b]$. Since A, B are closed in I, A' and B' are closed in $[a,b]$. Moreover, since $[a,b]$ is closed in $\mathbf{R}$, A' and B' are closed in $\mathbf{R}$.

Since $B' \neq \varnothing$ and B' is bounded below (by a for example) $b' = \inf B'$ exists by the completeness axiom, and $b' \geq a$. Since B' is closed in $\mathbf{R}$, $b' \in B'$. Since $a \in A'$ and

$A' \cap B' = \varnothing$, it follows that $b' > a$. Now let $A'' = A' \cap [a, b']$. Arguing as for B', we get that $a'' = \sup A''$ exists, $a'' \in A''$ and that $a'' < b'$. Now $(a'', b') \cap A = \varnothing$, since otherwise a'' is not an upper bound for A''. Similarly, $(a'', b') \cap B' = \varnothing$, otherwise b' is not a lower bound for B'. But $A' \cup B' = [a, b]$ and $(a'', b') \subseteq [a, b]$. Therefore, we have a contradiction. $\square$

As a corollary of the previous two lemmas, we directly obtain:

Proposition 2.1.9. *A subspace of* R *is connected if and only if it is an interval.*

We now show that connectedness is preserved by continuous maps.

Proposition 2.1.10. *The continuous image of a connected metric space is connected. That is, if* $f: X \to Y$ *is a continuous function between metric spaces where* X *is connected, then* $f(X)$ *is connected.*

Proof. Using the above notation, if $A \mid B$ partitions $f(X)$, then $f^{-1}(A) \mid f^{-1}(B)$ partitions X. $\square$

As a corollary we have:

Theorem 2.1.11.

Connectedness is a topological property. That is, if X *and* Y *are homeomorphic, then* X *is connected if and only if* Y *is connected.*

Proposition 2.1.10 is an useful tool to show that some space as connected, as demonstrated below.

Example 2.1.12. The unit circle $S^1 = \{(x, y) \in \mathbf{R}^2 \mid x^2 + y^2 = 1\}$ is connected since $e: [0, 2\pi] \to \mathbf{R}^2$, $e(x) = (\cos x, \sin x)$, is continuous and $e([0, 2\pi]) = S$.

Another consequence of Proposition 2.1.10 (and of the description of connected subsets of $\mathbf{R}$ of Lemma 2.1.7) is:

Corollary 2.1.13. *If* X *is a connected space and* $f: X \to \mathbf{R}$ *is continuous, then* $f(X)$ *is an interval.*

Using the above, we recover an important results from Analysis:

Corollary 2.1.14 (Intermediate Value Theorem)**.** *If* $f: X \to \mathbf{R}$ *is a continuous function from a connected metric space* X *and* $y \in (\inf f(X), \sup f(X))$*, then there exists* $x \in X$ *such that* $f(x) = y$*. (If* $f(X)$ *is not bounded below then we take* $\inf f(X) = -\infty$ *and similarly if* $f(X)$ *is not bounded above.)*

Proof. Since X is connected, $f(X)$ is a connected subspace of $\mathbf{R}$ and therefore an interval. So $y \in (\inf f(X), \sup f(X))$ implies that $y \in f(X)$. $\square$

A real function $f\colon \mathbf{R} \to \mathbf{R}$ is often identified with its graph, that is with a subset of $\mathbf{R}^2$. Studying the graph of f is useful to understand better the function f. The same thing can be done for arbitrary function. Let $f\colon X \to Y$ be a function between two sets. The **graph of** f is the set

$$G_f := \{(x, y) \in X \times Y \mid f(x) = y\}.$$

The fact that the graph of a continuous real function $f\colon \mathbf{R} \to \mathbf{R}$ is a curve in $\mathbf{R}^2$ can be generalised to metric spaces.

Proposition 2.1.15. *If $f\colon X \to Y$ is a continuous function between metric spaces, then G_f is homeomorphic to X.*

Proof. Let $i\colon G_f \to X \times Y$ be the inclusion map and let $p_1\colon X \times X \to X$ be the projection onto the first factor. Let $\theta\colon G_f \to X$ and $\psi\colon X \to G_f$ be defined respectively by $\theta = p_1 \circ i$ and $\psi(x) = (x, f(x))$ for all $x \in X$. It is easily checked that θ and ψ are mutually inverse. Continuity of θ follows from continuity of p_1 and i. For the continuity of ψ note that we can write $\psi = (\mathrm{Id} \times f) \circ \Delta$, where $\Delta\colon X \to X \times X$ is the continuous function $\Delta(x) = (x, x)$. Continuity of Δ is evident and continuity of $\mathrm{Id} \times f$ follows from the continuity of Id and f. $\square$

Corollary 2.1.16. *If $f\colon X \to Y$ is continuous with X connected, then G_f is a connected subset of Y.*

Proposition 2.1.17. *If W is a connected subspace of a metric space X and if $W \subseteq Y \subseteq \overline{W}$, then Y is connected.*

Proof. Let $f\colon Y \to D$ be any continuous function, where D is the discrete space $\{0, 1\}$. Since W is connected and $f|_W$ is continuous, $f(W) = \{0\}$ or $\{1\}$. Suppose, without loss of generality, that $f(W) = \{0\}$ and suppose, to get a contradiction, that there exists $y \in Y$ such that $f(y) = 1$. By continuity, $f^{-1}(1)$ contains a small ball $B_\varepsilon(y) \subseteq Y$. Since Y is in the closure of W there exists a point $x \in B_\varepsilon(y) \cap W$. But then $f(x) = 0$ since x is in W and $f(x) = 1$ since x belongs to $f^{-1}(1)$. This is the desired contradiction. $\square$

The similar statement for the interior is not true as demonstrated by the following example.

Example 2.1.18. Let $W = \overline{B}_1(-1) \cup \overline{B}_1(1) \subseteq \mathbf{R}^2$. Then W is connected, but $W^\circ = B_1(-1) \cup B_1(1)$ is not.

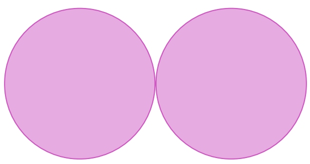


Figure 2.2: A connected set with non-connected interior.

We conclude this section by constructing an example of a pathological connected subset of $\mathbf{R}^2$.

Example 2.1.19 (Topologist's Sine Curve)**.** Let G be the graph of $\sin {}^1\!/_x$ for $x > 0$ and let $J := \{(0, y) \in \mathbf{R}^2 \mid -1 \leq y \leq 1\}$. Since $(0, +\infty)$ is connected then, by Corollary 2.1.16,

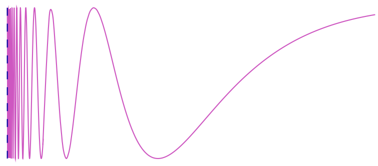


Figure 2.3: The graph G of $\sin({}^1\!/_x)$ in purple and the segment J in blue.

G is connected. To show that $G \cup J$ is connected we only need to show that $J \subseteq \overline{G}$. Let $p = (0, y) \in J$. We need to show that for any $\varepsilon > 0$, $B_\varepsilon(p) \cap G \neq \varnothing$. Choose $n \in \mathbf{N}$ such that ${}^1\!/_{2n\pi} < \varepsilon$. Since $\sin(\frac{1}{2}(4n+1)\pi) = 1$ and $\sin(\frac{1}{2}(4n+3)\pi) = -1$, by the Intermediate Value Theorem $\sin {}^1\!/_x$ takes on every value between -1 and 1 in the interval $[{}^2\!/_{(4n+3)\pi}, {}^2\!/_{(4n+1)\pi}]$. In particular, $\sin {}^1\!/_{x_0} = y$ for some x_0 in this interval. The distance between $(0, y)$ and $(x_0, \sin {}^1\!/_{x_0}) = (x_0, y)$ is $x_0 < \varepsilon$, so $(x_0, \sin {}^1\!/_{x_0}) \in B_\varepsilon(p) \cap G$ as required.

2.2 Path-connectedness

In this section, we will study a stronger notion of connectedness, which corresponds better to the intuition that two points are connected if one can draw a path from the first one the second.

We will write I for the closed interval $[0, 1] \subseteq \mathbf{R}$.

To make formal the notion of "connected by a path", we need to define what is a path.

Definition 2.2.1 (Path)**.**

Given two points x, y in a metric space X, a **path** from x to y is a continuous function $f : [0, 1] \to X$ such that $f(0) = x$ and $f(1) = y$.
We say that x and y are **joined** by the path f.

See the chapter illustration on page 36 for an illustration of a path between two points.

Definition 2.2.2 (Path-connected)**.**

A metric space X is **path-connected** if any two points in X can be joined by a path in X.

Example 2.2.3. For all $n \in \mathbf{N}$, the Euclidean space $\mathbf{R}^n$ is path-connected. If $n \neq 1$, then $\mathbf{R}^n \setminus \{0\}$ is also path-connected, but $\mathbf{R} \setminus \{0\}$ is not path connected.

Intervals in $\mathbf{R}$ are path-connected, but the same is true more generally for balls in $\mathbf{R}^n$.

Lemma 2.2.4. *Balls (open or closed) in the Euclidean space $\mathbf{R}^n$ are path-connected.*

Proof. Let x and y be two distinct points in a ball $B \subseteq \mathbf{R}^n$. Then the line segment P between x and y is contained in B. We have $P = \{\lambda x + (1-\lambda)y \mid \lambda \in [0,1]\}$, so $f\colon [0,1] \to P, t \mapsto tx + (1-t)y$ is a path from x to y. $\square$

As announced, path-connectedness is a stronger notion than connectedness.

Theorem 2.2.5.

A path-connected metric space X is connected.

Proof. Suppose by contradiction that X is path-connected but not connected, and let $f\colon X \to \{0,1\}$ be a surjective continuous function where $\{0,1\}$ is endowed with the discrete metric. Let $x, y \in X$ with $f(x) = 0$ and $f(y) = 1$. Let $g\colon I \to X$ be a path from x to y. Then $f \circ g\colon I \to D$ is a continuous and surjective function contradicting the connectedness of I. So X is connected. $\square$

Similarly to what happens for connectedness (Proposition 2.1.10 and Theorem 2.1.11), path-connectedness is preserved by homeomorphism.

Theorem 2.2.6.

The continuous image of a connected metric space is connected. It follows that path-connectedness is a topological property. That is, if X and Y are homeomorphic, then X is path-connected if and only if Y is connected.

Proof. The proof is left to the reader (see Tutorial 6 Question 10). $\square$

One easily verify that intervals in $\mathbf{R}$ are path-connected. It thus follows from Proposition 2.1.9 that a subset of $\mathbf{R}$ is connected if and only if it is path connected. However, in general path-connectedness is a strictly stronger notion than connectedness as demonstrated by the next example.

Example 2.2.7. The topologist's sine curve $G \cup J$ from Example 2.1.19 is connected. We will show that it is not path-connected by proving that any continuous path $f\colon I \to G \cup J$ which begins at $f(0) = (0,0) \in J \subseteq \mathbf{R}^2$ will never leave J and therefore cannot reach any point of G.

Let $i\colon G \cup J \to \mathbf{R}^2$ be the inclusion and $p_i\colon \mathbf{R}^2 \to \mathbf{R}$ be the projection maps for $i = 1, 2$. Then

$$f_1 := p_1 \circ i \circ f\colon I \to \mathbf{R} \quad \text{and} \quad f_2 := p_2 \circ i \circ f\colon I \to \mathbf{R}$$

are both continuous as they are the composition of continuous functions. We also have $f_i(0) = 0$ for $i = 1, 2$.

Let $\alpha := \sup\{x \in [0,1] \mid f_1([0,x]) = 0\}$. We know that $\alpha \in [0,1]$. If $\alpha = 1$, then $f_1(x) = 0$, for all $x \in [0,1]$, which implies that $f(x) \in J$, for all $x \in [0,1]$, which is what we want to prove.

Suppose $\alpha < 1$. Note that $f_1(\alpha) = 0$ by continuity and $f_2(\alpha) \in [-1, 1]$. By the continuity of f_2 at α, there exists $\delta > 0$ such that

$$|f_2(x) - f_2(\alpha)| < 1, \quad \text{for all } x \in (\alpha - \delta, \alpha + \delta) \cap [0, 1]. \tag{2.1}$$

By definition of α, there exists $x_0 \in (\alpha, \alpha + \delta) \cap [0, 1]$ such that $f_1(x_0) > 0$. By the Intermediate Value Theorem, $[0, f_1(x_0)] \subseteq f_1([\alpha, x_0])$.

Suppose that $f_2(\alpha) \geq 0$. Then $f_2(\alpha) - 1 \in [-1, 0]$. So there exists $t \in [0, f_1(x_0)]$ such that $\sin(1/t) = f_2(\alpha) - 1$. Also, there exists $\tilde{x} \in [\alpha, x_0] \subseteq (\alpha - \delta, \alpha + \delta) \cap [0, 1]$ such that $t = f_1(\tilde{x}) > 0$. So $f_2(\tilde{x}) = \sin(1/f_1(\tilde{x})) = \sin(1/t) = f_2(\alpha) - 1$. This implies that $|f_2(\tilde{x}) - f_2(\alpha)| = 1$ which contradicts Equation (2.1).

The case for $f_2(\alpha) \leq 0$ follows similarly by observing that $f_2(\alpha) + 1 \in [0, 1]$.

If we can go with a path f from x to y and from y to z, it seems natural that we should be able to go from x to z by concatenating the two paths. This basic idea almost works, except that it will gives a continuous function $h\colon [0, 2] \to X$, so we need to rescale our two paths.

Lemma 2.2.8. *Suppose $f, g\colon I \to X$ are paths in a metric space X such that $f(1) = g(0)$. Then*

$$h(x) := \begin{cases} f(2x), & x \in [0, 1/2]; \\ g(2x - 1), & x \in [1/2, 1], \end{cases}$$

is a path in X from $f(0)$ to $g(1)$.

Proof. The proof is left to the reader (see Tutorial 6 **??** 11). □

We have seen that while a subset W of $\mathbf{R}$ is connected if and only if it is path-connected, this is not true anymore for subsets of $\mathbf{R}^2$. This is however still true if we restrict ourselves to open subsets of $\mathbf{R}^n$.

Proposition 2.2.9. *A connected open subset U of $\mathbf{R}^n$ is path-connected.*

Proof. Let $u \in U$ and $A \subseteq U$ be the subset of points that can be joined to u by a path in U, and let $B = U \setminus A$. We shall prove that if B were non-empty, then $A \mid B$ would partition U contradicting its connectedness.

First we show that A is open. Let $x \in A$. Then there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$. Given any $y \in B_\varepsilon(x)$, there is a path (straight line) g in $B_\varepsilon(x)$ from x to y. But since $x \in A$, there is a path f in U joining u to x. The path h, as given in Lemma 2.2.8 is a path in U from u to y. So $y \in A$ and $B_\varepsilon(x) \subseteq A$ and A is open.

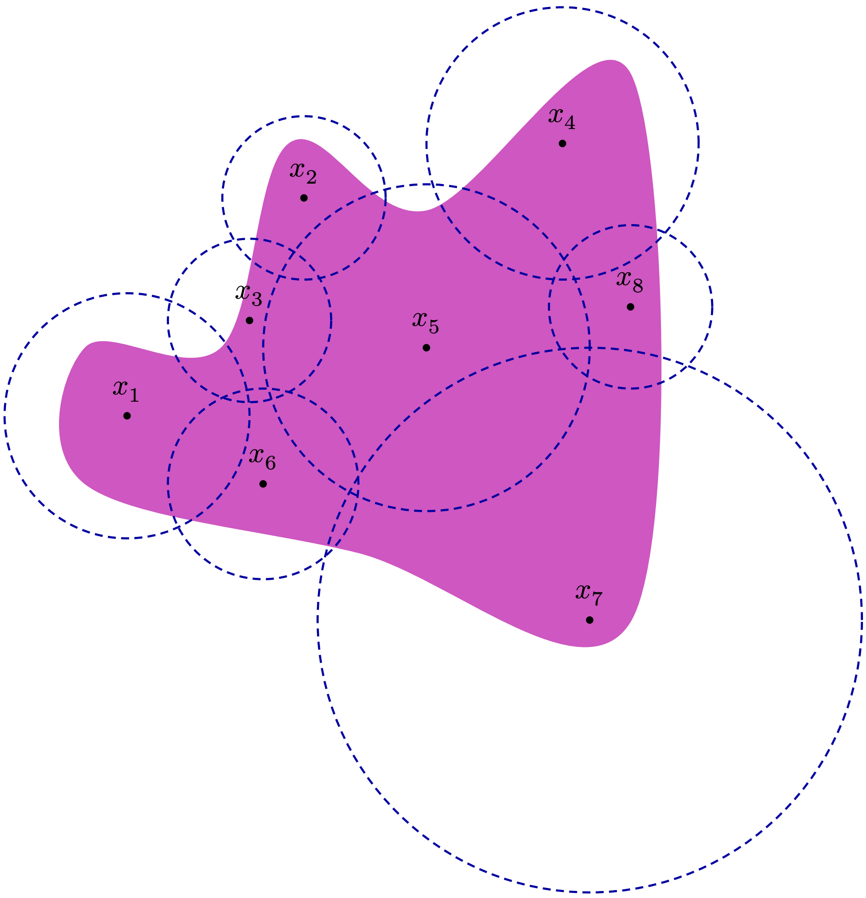
We now show that B is open. So let $b \in B \subseteq U$ and let $\varepsilon >$ be such that $B_\varepsilon(b) \subseteq U$. The ball $B_\varepsilon(b)$ cannot any element a of A, as otherwise we would have a path from b to a and hence from b to u, which is absurd. We conclude that $B_\varepsilon(b) \subseteq U \setminus A$ and thus that B is open.

Finally, $A \cap B = \varnothing$ and $A \cup B = U$ so, by the connectedness of U, $B = \varnothing$. □

One can shows that the topologist's sine curve from Example 2.1.19 is closed. Proposition 2.2.9 is thus not true for closed subsets.

CHAPTER 3

COMPACTNESS



3.1 Definition and first properties

The aim of this chapter is to study compact (sub)spaces, which are a particularly well-behaved kind of (sub)spaces. In Euclidean spaces, these are exactly the closed bounded of $\mathbf{R}^n$. However, taking this as a definition of compactness will fail as in a discrete metric space (X, d) every subset is both closed and bounded but not necessarily "well-behaved". We hence need a more precise definition of compactness.

Definition 3.1.1 (Cover).

Let (X, d) be a metric space and let $W \subseteq X$ be a subset. A **cover** of W is a collection $\mathscr{U} := \{U_i \subseteq X \mid i \in I\}$ of subsets such that

$$W \subseteq \bigcup_{i \in I} U_i;$$

a **subcover** of $\mathscr{U}$ is a subcollection $\{U_i \subseteq X \mid i \in J\}$ for some $J \subseteq I$ which also covers W. If I is a finite set, then $\mathscr{U}$ is called a **finite cover**. If every U_i is open in X, then $\mathscr{U}$ is called an **open cover** of W.

Remark 3.1.2.

Remember that the above definition also holds for $W = X$.

See the chapter illustration on the previous page for an illustration of a finite open cover by open balls in $\mathbf{R}^2$.

Using the notion of open cover, one can define compactness.

Definition 3.1.3 (Compact).

A subset $W \subseteq (X, d)$ is **compact** if every open cover of W has a finite subcover.

To better understand this definition, let us see how it can fail. Firstly, W can fail to be compact because is not closed (see Proposition 3.1.9).

Example 3.1.4. The open interval $(0, 1) \subseteq \mathbf{R}$ is not compact since the collection

$$\mathscr{U} := \{U_n = (1/n, 1) \mid n \geq 2\}$$

is an open cover of $(0, 1)$ but any finite subcollection $\{U_{n_1}, \dots, U_{n_j}\}$ contains a largest interval, let's say $(1/n_k, 1)$, so it fails to be a subcover since $1/2n_k$ does not lie in U_{n_i} for any $1 \leq i \leq j$.

Secondly, W can fail to be compact because is not bounded (see Proposition 3.1.9).

Example 3.1.5. The collection $\mathscr{V} := \{V_x = (-x, x) \mid x > 0\}$ is an open cover of $\mathbf{R}$ but any finite subcollection has a largest open interval and therefore $\mathbf{R}$ is not compact.

The next example shows that even if W is both closed and bounded, it is not necessarily compact.

Example 3.1.6. Let (X, d_0) be a discrete metric space. Then all subsets of X are closed and bounded, but a subset of X is compact if and only if it is finite (Tutorial 7 Question 1).

Compact (sub)spaces are small enough, or manageable, in some sense. In particular, finite subsets are compact.

Proposition 3.1.7. *If W is finite, then it is compact.*

Proof. Suppose that $\mathscr{U} = \{U_i \mid i \in I\}$ is an open cover of $W = \{w_1, \dots, w_N\}$. For each $1 \leq k \leq N$ it is possible to choose an i_k such that $w_k \in U_{i_k}$. Therefore, $W \subseteq \bigcup_{k=1}^{N} U_{i_k}$, and $\{U_{i_k} \mid 1 \leq k \leq N\}$ is a finite subcover. □

Proposition 3.1.8. *If $W \subseteq (X, d)$ is compact, then it is bounded.*

Proof. Choose $w \in W$ and consider the collection $\{B_n(w) \mid n \geq 1\}$. This is an open covering of W since $d(w, y)$ is finite, for all $y \in W$. So there must exist a finite subcover $\{B_{n_i}(w) \mid 1 \leq i \leq k\}$. This implies that $W \subseteq B_r(w)$ for $r = \max\{n_1, \dots, n_k\}$. So W is bounded. □

Proposition 3.1.9. *If $W \subseteq (X, d)$ is compact, then it is closed in X.*

Proof. Let $x \in X \setminus W$. We will show that $X \setminus W$ is open by proving that $x \in (X \setminus W)^\circ$. Define $d(w) := {}^{d(w,x)}/_{2}$, for each $w \in W$. Therefore, the open balls $B_{d(w)}(w)$ and $B_{d(w)}(x)$ are disjoint. The collection $\{B_{d(w)}(w) \mid w \in W\}$ is an open cover of W and therefore has a finite subcover which we can write as $\{B_{d(w_i)}(w_i) \mid 1 \leq i \leq k\}$ for some points $w_1, \dots, w_k \in W$. Let $\varepsilon = \min\{d(w_1), \dots, d(w_k)\}$. Then $B_\varepsilon(x)$ is disjoint from W and so $x \in (X \setminus W)^\circ$ which implies that W is closed in X. □

Theorem 3.1.10 (Heine–Borel[1] (1-dimensional))**.**

Any closed bounded interval $[a, b]$ in $\mathbf{R}$ is compact.

Proof. Let $\mathscr{U}$ be an open cover of $[a, b]$, and define

$$G := \{x \in \mathbf{R} \mid x \geq a \text{ and } [a, x] \text{ is covered by a finite subcollection of } \mathscr{U}\}.$$

We need to prove that $b \in G$.

First note that if $x \in G$, then $y \in G$ for all $a \leq y \leq x$.

We see that $a \in G$ since a must belong to some $U \in \mathscr{U}$, since $\mathscr{U}$ covers $[a, b]$. Since U is open, there exists $\delta > 0$ such that $[a, a + \delta) \subseteq U$ which implies that $[a, a + \delta) \subseteq G$ also.

Now we know that $G \neq \varnothing$, either G is bounded above or not. If G is not bounded above, then there exists $c \in G$ with $c > b$ which implies that $b \in G$. If G is bounded

[1] Heinrich Eduard Heine (1821–1881) and Félix Édouard Justin Émile Borel (1871–1956).

above, then its supremum $\gamma := \sup G$ exists. Note that $\gamma > a$. If $\gamma > b$, then there is some $c \in G$, with $c > b$, which implies that $b \in G$.

Now suppose that $\gamma \leq b$. There exists some $U_\gamma \in \mathscr{U}$ with $\gamma \in U_\gamma$, and some $\varepsilon > 0$ with $(\gamma - \varepsilon, \gamma + \varepsilon) \subseteq U_\gamma$. Moreover, some $x \in G$ must satisfy $x > a$ and $x > \gamma - \varepsilon$, by the definition of the supremum. But $x \in G$ implies that $[a, x]$ admits a finite subcover of $\mathscr{U}$, to which we may add U_γ and cover $[a, \gamma + \varepsilon/2]$. Thus, $\gamma + \varepsilon/2$ lies in G, which contradicts the assumption that $\gamma = \sup G$. □

3.2 Continuous functions on compact spaces

Proposition 3.2.1. *If $f\colon X \to Y$ is continuous and X is compact, then the image $f(X) \subseteq Y$ is also compact.*

Proof. Let $\mathscr{U}$ be an open cover for $f(X)$. Since f is continuous, $f^{-1}(U)$ is open in X, for each $U \in \mathscr{U}$. The collection $\{f^{-1}(U) \mid U \in \mathscr{U}\}$ is an open cover for X since

$$x \in X \implies f(x) \in U \text{ for some } U \in \mathscr{U} \implies x \in f^{-1}(U).$$

Since X is compact there exists a finite subcover $\{f^{-1}(U_1), \dots, f^{-1}(U_n)\}$ of X and the collection $\{U_1, \dots, U_n\}$ is a finite subcover of $f(X)$ since

$$X \subseteq \bigcup_{i=1}^{n} f^{-1}(U_i) \implies f(X) \subseteq f\Big(\bigcup_{i=1}^{n} f^{-1}(U_i)\Big) \implies f(X) \subseteq \bigcup_{i=1}^{n} U_i.$$

□

Example 3.2.2. The map $e\colon \mathbf{R} \to \mathbf{R}^2$ given by $e(x) = (\cos x, \sin x)$ is continuous and maps the closed interval $[0, 2\pi]$ onto the circle S of radius 1 around the origin. Therefore, S is a compact subset of $\mathbf{R}^2$.

Corollary 3.2.3. *Compactness is a topological property.*

This means that if two spaces X and Y are homeomorphic, then X is compact if and only if Y is compact.

Corollary 3.2.4. *Any continuous function from a compact space is bounded.*

Proof. This follows from Propositions 3.2.1 and 3.1.8. □

If a function $f\colon X \to \mathbf{R}$ is bounded, this means that $f(X)$ is a bounded subset of $\mathbf{R}$ which implies that $\sup f(X)$ and $\inf f(X)$ exist. These are bounds of f on X but they may not be in $f(X)$ themselves, that is, there does not exist $m, M \in X$ such that $f(M) = \sup f(X)$ and $f(m) = \inf f(X)$. If they are in X, then we say that f **attains its bounds** on X.

Corollary 3.2.5. *If $f\colon X \to \mathbf{R}$ is continuous and X is compact, then f attains its bounds on X.*

Proof. By Corollary 3.2.5 $f(X)$ is bounded which implies that $\sup f(X)$ and $\inf f(X)$ exist and are bounds for $f(X)$. Also, $\sup f(X)$ and $\inf f(X)$ are elements of $\overline{f(X)}$ (Exercise). Since $f(X)$ is compact it is closed, which implies that $\sup f(X), \inf f(X) \in f(X) = \overline{f(X)}$. □

Proposition 3.2.6. *Let V be a closed subset of a compact metric space (X, d). Then V is compact.*

Proof. Let $\mathscr{U}$ be an open cover of V. Since V is closed, $X \setminus V$ is open. If we add $X \setminus V$ to $\mathscr{U}$ we get an open cover of X. By the compactness of X we get a finite subcover of X, which is also a finite cover of V. If $X \setminus V$ is in this finite cover, then we can remove it to obtain a finite subcover of $\mathscr{U}$ for V. □

Theorem 3.2.7.

Suppose that $f \colon X \to Y$ is a continuous bijection where X is compact. Then f is a homeomorphism.

Proof. We need to show that $g := f^{-1} \colon Y \to X$ is continuous. We shall prove that if V is closed in X, then $g^{-1}(V)$ is closed in Y, which will imply that g is continuous.

Let V be closed in X. Since X is compact, V is compact by Proposition 3.2.6. So $f(V)$ is compact by Proposition 3.2.1, which implies that $f(V)$ is closed in Y. But $f(V) = g^{-1}(V)$ so $g^{-1}(V)$ is closed in Y. □

Corollary 3.2.8. *If $f \colon X \to Y$ is a continuous injective function where X is compact, then X is homeomorphic to $f(X)$.*

Example 3.2.9. If $f \colon [a, b] \to \mathbf{R}$ is a continuous strictly monotonic function, then it has a continuous inverse $f^{-1} \colon f([a, b]) \to [a, b]$.

3.3 The Heine–Borel Theorem

Theorem 3.3.1.

A product $X \times Y$ of metric spaces is compact if and only if both X and Y are compact.

Proof. "⇒" The projection maps $p_X \colon X \times Y \to X$ and $p_Y \colon X \times Y \to Y$ are continuous and so X and Y are both compact as the continuous images of $X \times Y$ by Proposition 3.2.1.

"⇐" Suppose X and Y are both compact and let $\mathscr{U}$ be an open cover of $X \times Y$. We shall say that $W \subseteq X$ is *good* if $W \times Y$ is covered by a finite subcollection of $\mathscr{U}$. We want to prove that X is good. We will split this proof up into three steps:

Change the presentation?

Step 1 If $W_1, W_2, \dots, W_r \subseteq X$ are good, then $\bigcup_{i=1}^r W_i$ is good.

Proof. Given any $i \in \{1, \dots, r\}$ there is a finite subcollection say $\mathcal{U}_i$ of $\mathcal{U}$ which covers $W_i \times Y$. Then $\bigcup_{i=1}^r W_i \times Y$ is covered by the finite subcollection

$$\mathcal{U}_1 \cup \mathcal{U}_2 \cup \dots \cup \mathcal{U}_r$$

of $\mathcal{U}$. ■

Step 2 For all $x \in X$ there exists an open subset $U(x) \subseteq X$ such that $x \in U(x)$ and $U(x)$ is good.

Proof. Fix $x \in X$. Then for every $y \in Y$, $(x, y) \in W(y)$, for some $W(y) \in \mathcal{U}$ since $\mathcal{U}$ covers $X \times Y$.

Lemma 3.3.2. *There are open subsets $U(y) \subseteq X$ and $V(y) \subseteq Y$ such that*

$$(x, y) \in U(y) \times V(y) \subseteq W(y).$$

Proof. Since $W(y)$ is open in $X \times Y$, there exists $\varepsilon > 0$ such that $B_\varepsilon((x, y)) \subseteq W(y)$. So far we have not specified exactly the metric we are considering on $X \times Y$ but we know that the metrics d_a, d_b, d_c on $X \times Y$ (from Definition 1.6.10) are all topologically equivalent and so we can choose which we take without loss of generality. For this proof we will take d_c. Then

$$B_\varepsilon((x, y); d_c) = \{(x', y') \in X \times Y \mid \max\{d_X(x, x'), d_Y(y, y') < r\}\},$$

where d_X is the metric on X and d_Y is the metric on Y. It is then a fairly easy exercise to check that

$$B_\varepsilon((x, y); d_c) = B_\varepsilon(x; d_X) \times B_\varepsilon(y; d_Y).$$

We can then take $U(y) = B_\varepsilon(x; d_X)$ and $V(y) = B_\varepsilon(y; d_Y)$ to complete the proof. ■

The collection $\{V(y) \mid y \in Y\}$ is an open cover of Y and so has a finite subcover $\{V(y_1), V(y_2), \dots, V(y_r)\}$. Let

$$U(x) := U(y_1) \cap U(y_2) \cap \dots \cap U(y_r).$$

Then for each $i \in \{1, \dots, r\}$ we have

$$U(x) \times Y = U(x) \times \bigcup_{i=1}^r V(y_i) = \bigcup_{i=1}^r (U(x) \times V(y_i)) \subseteq \bigcup_{i=1}^r W(y_i),$$

which implies that $U(x)$ is good. Also, $x \in U(x)$ and $U(x)$ is open in X as a finite intersection of open sets. ■

Step 1 X is good.

Proof. $\{U(x) \mid x \in X\}$ is an open cover of X and therefore has a finite subcover $\{U(x_1), U(x_2), \dots, U(x_s)\}$. By Step 2, each $U(x_i)$ is good, for $i = 1, \dots, s$, and so

$$X = \bigcup_{i=1}^{s} U(x_i)$$

is good by Step 1. ■

This finishes the proof. □

Theorem 3.3.3 (Heine–Borel).

A subspace $A \subseteq \mathbf{R}^n$ is compact if and only if it is closed and bounded.

We can immediately prove this in dimension 1 as if $A \subseteq \mathbf{R}$ is closed and bounded, then $A \subseteq [a, b]$ for some closed interval (since A is bounded), and since we know that $[a, b]$ is compact A is a closed subset of a compact set and therefore compact.

Proof. "$\Rightarrow$" This follows immediately from Propositions 3.1.8 and 3.1.9.
"$\Leftarrow$" Let $A \subseteq \mathbf{R}^n$ be bounded and closed. Since A is bounded, $A \subseteq [a, b]^n$ for some $a, b \in \mathbf{R}$ (Exercise). Since $[a, b]$ is compact (in $\mathbf{R}$) so is $[a, b]^n$ by induction on n and Theorem 3.3.1. Also, A is closed in $[a, b]^n$. Therefore, A is compact as a closed subspace of a compact space by Proposition 3.2.6. □

3.4 Equivalence of norms on $\mathbf{R}^n$

This section will be devoted to proving the following theorem:

Theorem 3.4.1.

Any two norms on $\mathbf{R}^n$ give Lipschitz equivalent metrics.

In order to prove this we will show that any norm $\|\cdot\|$ is equivalent to the taxicab norm $\|\cdot\|_1$ on $\mathbf{R}^n$, i.e. there exist $h, k \in (0, +\infty)$ such that

$$h\|x\|_1 \leq \|x\| \leq k\|x\|_1, \quad \forall x \in \mathbf{R}^n. \tag{3.1}$$

This will prove that any metric coming from a norm on $\mathbf{R}^n$ is Lipschitz equivalent to the taxicab metric on $\mathbf{R}^n$ and therefore, since Lipschitz equivalence is an equivalence relation, that all metrics on $\mathbf{R}^n$ coming from norms are Lipschitz equivalent.

Let $\boldsymbol{e}_i \in \mathbf{R}^n$ be the standard ith basis vector, i.e. its ith coordintate is 1 and the rest are 0.

Let $\|\cdot\|$ be a norm on $\mathbf{R}^n$ and let $k := \max\{\|\boldsymbol{e}_1\|, \dots, \|\boldsymbol{e}_n\|\}$. Then for any $x \in \mathbf{R}^n$,

$$\|x\| = \left\|\sum_{i=1}^{n} x_i \boldsymbol{e}_i\right\| \leq \sum_{i=1}^{n} |x_i| \cdot \|\boldsymbol{e}_i\| \leq k\|x\|_1, \tag{3.2}$$

which gives us the second inequality in Equation (3.1).

Lemma 3.4.2. *For any norm* $\|\cdot\|$ *we have* $\|x \pm y\| \geq |\|x\| - \|y\||$.

We will leave the proof of this as an exercise.

Lemma 3.4.3. *The function* $\|\cdot\| \colon \mathbf{R}^n \to \mathbf{R}$ *is continuous with respect to the taxicab norms.*

Proof. Let $x \in \mathbf{R}^n$ and $\varepsilon > 0$. Then take $\delta = {}^{\varepsilon}/_{k}$, where k is as in Equation (3.2). If $\|x - y\|_1 < \delta$, then

$$|\|x\| - \|y\|| \leq \|x - y\| \leq k\|x - y\|_1 < \varepsilon.$$

□

Now $S := \{x \in \mathbf{R}^n \mid \|x\|_1 = 1\} = \partial B_1(0; d_1)$ is a compact set by the Heine–Borel Theorem as it is closed (being the boundary of a subset) and bounded (being a subset of $\overline{B}_1(0; d_1)$). Therefore, the continuous function $\|\cdot\| \colon \mathbf{R}^n \to \mathbf{R}$, when restricted to S, attains its lower bound h on S and since $0 \notin S$, we have $h > 0$. Hence,

$$\|x\| \geq h > 0, \quad \text{for all } x \in S.$$

Lemma 3.4.4. *For every* $x \in \mathbf{R}^n$, $\|x\| \geq h\|x\|_1$.

Proof. If $x = 0$, then the statement holds. Suppose that $x \neq 0$ and set $\lambda := {}^{1}/_{\|x\|_1} > 0$. Then

$$\|\lambda x\|_1 = \lambda\|x\|_1 = 1 \implies \lambda x \in S.$$

So

$$h \leq \|\lambda x\| = \lambda\|x\| \implies \|x\| \geq h\|x\|_1.$$

□

We have therefore proven the first inequality in Equation (3.1) and therefore the theorem.

3.5 Uniform Continuity

Definition 3.5.1 (Uniform Continuity).

A function $f \colon (X, d_X) \to (Y, d_Y)$ is **uniformly continuous** if

$$\forall \varepsilon > 0, \exists \delta > 0 \quad \text{such that} \quad d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

Remark 3.5.2.

In the ordinary definition of continuity, at a point x in the domain, δ depends on x as well as on ε. In the definition of uniform continuity, δ only depends on

ε. Ordinary continuity is a local property whereas uniform continuity is a global property.

Example 3.5.3. The function $f\colon (0,1) \to \mathbf{R}$, given by $f(x) = {}^1\!/x$, is continuous but not uniformly continuous since, for $\varepsilon = 1$, and any $\delta > 0$, if we let $x = \min\{{}^1\!/2, \delta\}$ and $y = {}^x\!/2$, then we have $|x - y| = {}^x\!/2 < \delta$ but $|f(x) - f(y)| = |{}^1\!/x - {}^2\!/x| = {}^1\!/x \geq 1 = \varepsilon$.

Proposition 3.5.4. *Let $f\colon (X, d_X) \to (Y, d_Y)$ be continuous with X compact. Then f is uniformly continuous.*

Proof. Let $\varepsilon > 0$. Then for each $x \in X$, since f is continuous at x, there exists $\delta(x) > 0$ such that

$$d_X(x, y) < 2\delta(x) \implies d_Y(f(x), f(y)) < {}^\varepsilon\!/2.$$

The collection $\{B_{\delta(x)}(x) \mid x \in X\}$ is an open cover of X and so, since X is compact, has a finite subcover $\{B_{\delta(x_1)}(x_1), \dots, B_{\delta(x_n)}(x_n)\}$, for some points $x_1, \dots, x_n \in X$. Let $\delta = \min\{\delta(x_1), \dots, \delta(x_n)\}$. Then for $x, y \in X$ with $d_X(x, y) < \delta$ we have

1. there exists $1 \leq i \leq n$ such that $d_X(x, x_i) < \delta(x_i)$;
2. $d_X(y, x_i) \leq d_X(y, x) + d_X(x, x_i) < \delta + \delta(x_i) \leq 2\delta(x_i)$.

Now 1 implies that $d_Y(f(x), f(x_i)) < {}^\varepsilon\!/2$ and 2 implies that $d_Y(f(y), f(x_i)) < {}^\varepsilon\!/2$ so

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(x_i)) + d_Y(f(x_i), f(y)) < \varepsilon.$$

□

Of course a function does not need to have a compact domain to be uniformly continuous; any identity function $\mathrm{Id}_X\colon (X, d) \to (X, d)$ is uniformly continuous.

Definition 3.5.5 (Uniformly equivalent metrics).

Two metrics d and d' on a set X are **uniformly equivalent** if both $\mathrm{Id}\colon (X, d) \to (X, d')$ and $\mathrm{Id}\colon (X, d') \to (X, d)$ are uniformly continuous.

So if (X, d) is compact and is topologically equivalent to (X, d'), then d and d' are uniformly equivalent.

Definition 3.5.6 (Uniform Equivalence).

Given metric spaces (X, d_X) and (Y, d_Y), a function $f\colon X \to Y$ is a **uniform equivalence** if f is a bijection where both f and f^{-1} are uniformly continuous.

3.6 Sequential Compactness

In this section, we introduce a variation of the notion of compactness, which for metric spaces will turn to be equivalent to compactness.

Definition 3.6.1 (Sequentially Compact)**.**

A subset $W \subseteq (X, d)$ is **sequentially compact** if every sequence (w_n) in W has a subsequence that converges to a point in W.

Example 3.6.2. The real line $\mathbf{R}$ is not sequentially compact since the sequence $x_n = n$, for all $n \in \mathbf{N}$, does not have a convergent subsequence.

Lemma 3.6.3. *Let (x_n) be a sequence in X and let $x \in X$. If, for every $\varepsilon > 0$, $B_\varepsilon(x)$ contains x_n for infinitely many values of $n \in \mathbf{N}$, then (x_n) contains a subsequence that converges to $x \in X$.*

Proof. Let n_1 be such that $x_{n_1} \in B_1(x)$ and proceed inductively by choosing integers $n_1 < \cdots < n_r$ such that $x_{n_j} \in B_{1/j}(x)$, for each $1 \leq j \leq r$. Since $B_{1/r+1}(x)$ contains x_n for infinitely many n, there are infinitely many that satisfy $n_r < n$. Choose n_{r+1} to be one of these. Then $x_{n_r} \in B_{1/r}(x)$ for every $r \in \mathbf{N}$ and $x_{n_r} \to x$, as $r \to \infty$, in X. $\square$

Example 3.6.4. In $\mathbf{R}$, the sequence $x_n = \begin{cases} m, & \text{if } n = 2m-1; \\ 1/m, & \text{if } n = 2m, \end{cases}$ has a convergent subsequence $x_{2m} \to 0$, and $B_\varepsilon(0)$ contains every x_{2m} for $m > 1/\varepsilon$, of which there are infinitely many.

The contrapositive of Lemma 3.6.3 is often useful.

Lemma 3.6.5. *Suppose that (x_n) contains no convergent subsequence. Then, for every $x \in X$, there exists $\varepsilon(x) > 0$, such that $B_{\varepsilon(x)}(x)$ contains x_n for at most finitely many $n \in \mathbf{N}$.*

Example 3.6.6.

1. In $\mathbf{R}$, the sequence $(x_n = n)$ has no convergent subsequences. Given any $x \in \mathbf{R}$, the $\varepsilon(x)$ of Lemma 3.6.5 can be taken to be $1/2$, in which case $B_{1/2}(x)$ contains at most one element of x_n.

2. In the interval $(0,1] \subseteq \mathbf{R}$, the sequence $(1/n)_{n \geq 1}$ contains no convergent subsequences. Any $x \in (0,1]$ determines a unique integer $N(x)$ for which
$$1/(N(x)+1) \leq x \leq 1/N(x).$$
Choosing $\varepsilon(x) := 1/(N(x)+1) - 1/(N(x)+2)$ ensures that $B_{\varepsilon(x)}(x)$ contains at most one element of $(1/n)$.

Proposition 3.6.7. *If a subspace $W \subseteq X$ is compact, then it is sequentially compact.*

Proof. Let (w_n) be a sequence in W, and suppose that it has no convergent subsequence. So by Lemma 3.6.5, every $x \in W$ determines $\varepsilon(x) > 0$ such that $B_{\varepsilon(x)}(x)$ contains w_n for at most finitely many $n \in \mathbf{N}$. But the collection $\{B_{\varepsilon(x)}(x) \mid x \in W\}$ is an open cover for W and so therefore has a finite subcover $\{B_{\varepsilon(x_i)}(x_i) \mid 1 \leq i \leq n\}$ for some set of points $x_1, \ldots, x_n \in W$. But each $B_{\varepsilon(x_i)}(x_i)$ can only contain finitely many points of (w_n), which implies that W contains only finitely many terms of the sequence (w_n) which is a contradiction. $\square$

Definition 3.6.8 (Cauchy Sequence)**.**

A sequence (x_n) in (X, d) is **Cauchy** if

$$\forall \varepsilon > 0, \exists N \in \mathbf{N} \quad \text{such that} \quad m, n \geq N \implies d(x_m, x_n) < \varepsilon.$$

Proposition 3.6.9. *Every convergent sequence is Cauchy.*

Proof. Suppose (x_n) be a sequence in (X, d) such that $\lim_n x_n = x \in X$. Let $\varepsilon > 0$. Then there exists $n \in \mathbf{N}$ such that

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon, \quad \forall m, n \geq N.$$

□

Example 3.6.10. Not every Cauchy sequence is necessarily convergent. Consider the metric space $\mathbf{Q}$ with the subspace metric coming from the standard metric on $\mathbf{R}$. Then the $x_1 = 1, x_2 = 1.4, x_3 = 1.41, x_4 = 1.414, \dots$ (that is x_n is the first n decimal digit of $\sqrt{2}$) is Cauchy but does not converge in $\mathbf{Q}$.

Definition 3.6.11 (ε-net)**.**

Given $\varepsilon > 0$, an **ε-net** for a metric space X is a subset $S \subseteq X$ such that $X \subseteq \bigcup_{x \in S} B_\varepsilon(x)$.

Of course, for any $\varepsilon > 0$, we can always find an ε-net by taking $S = X$ but the interesting ones are the finite ones.

Proposition 3.6.12. *If (X, d) is sequentially compact, then given any $\varepsilon > 0$, there exists a finite ε-net for X.*

Proof. Suppose that for some $\varepsilon > 0$ there is no finite ε-net. We shall get a contradiction by constructing a sequence (x_n) in X with no convergent subsequence.

Let x_1 be a point in X, and suppose inductively that $x_1, x_2, \dots, x_r$ have been chosen in X such that $d(x_i, x_j) \geq \varepsilon$, for all $i, j \leq r$ with $i \neq j$. Then $\{x_1, \dots, x_r\}$ is not an ε-net and so there exists a point $x_{r+1} \in X$ such that $d(x_i, x_{r+1}) \geq \varepsilon$, for all $i \in \{1, \dots, r\}$. This completes the inductive step of constructing a sequence (x_n) for which $d(x_m, x_n) \geq \varepsilon$, for all $m \neq n$. This sequence clearly has no Cauchy subsequence and therefore no convergent subsequence. □

Definition 3.6.13 (Totally Bounded)**.**

A metric space is **totally bounded** if it has a finite ε-net, for every $\varepsilon > 0$.

Definition 3.6.14 (Lebesgue Number)**.**

Given an open cover $\mathscr{U}$ of a metric space X, a fixed real number $\varepsilon > 0$ is called a **Lebesgue number** for $\mathscr{U}$ if, for any $x \in X$, there exists $U(x) \in \mathscr{U}$ such that $B_\varepsilon(x) \subseteq U(x)$.

If ε is a Lebesgue number for $\mathscr{U}$, then so is any ε' such that $0 < \varepsilon' \leq \varepsilon$. There may not be a Lebesgue number for a given open cover $\mathscr{U}$. For example let $X = (0, 1)$, and $\mathscr{U} = \{(1/n, 1) \mid n \geq 2\}$. Then for any $\varepsilon > 0$, take $n > 1/\varepsilon$ and $x = 1/n$. There is no $(1/m, 1)$ such that $B_\varepsilon(x) \subseteq (1/m, 1)$, since $B_\varepsilon(x) = B_\varepsilon(1/n) = (0, 1/n + \varepsilon)$. So there does not exist a Lebesgue number for this open cover.

Lemma 3.6.15. *There exists a Lebesgue number for any open cover of a sequentially compact metric space X.*

Proof. Suppose that $\mathscr{U}$ is an open cover for which there does not exist a Lebesgue number. Then for any $n \in \mathbf{N}$, there exists some point $x_n \in X$ such that $B_{1/n}(x_n) \subseteq U$ is false for every $U \in \mathscr{U}$. By sequential compactness (x_n) has a subsequence (x_{n_r}) that converges to a point $x \in X$. Since $\mathscr{U}$ covers X, there exists $U_x \in \mathscr{U}$ such that $x \in U_x$. Since U_x is open, $B_{2/m}(x) \subseteq U_x$, for some $m \in \mathbf{N}$. Now, $B_{1/m}(x)$ contains x_{n_r} for all $r \geq R$, for some $R \in \mathbf{N}$. Fix $r \geq R$ so that $n_r \geq m$, and write $s = n_r$. Then $B_{1/s}(x_s) \subseteq B_{2/m}(x)$ since

$$d(x_s, y) < 1/s \implies d(x, y) \leq d(x, x_s) + d(x_s, y) < 1/m + 1/s \leq 2/m.$$

Therefore $B_{1/s}(x_s) \subseteq U_x$ contradicting the choice of x_s. So there exists a Lebesgue number for $\mathscr{U}$. ☐

Theorem 3.6.16.

Any sequentially compact metric space is compact.

Proof. Let $\mathscr{U}$ be an open cover of a sequentially compact metric space X. By Lemma 3.6.15 there exists a Lebesgue number ε for $\mathscr{U}$. By Proposition 3.6.12 there exists a finite ε-net $\{x_1, \dots, x_n\}$, where ε is the same Lebesgue number. Then $B_\varepsilon(x_i) \subseteq U_i$, for some $U_i \in \mathscr{U}$ by the definition of Lebesgue number. Since

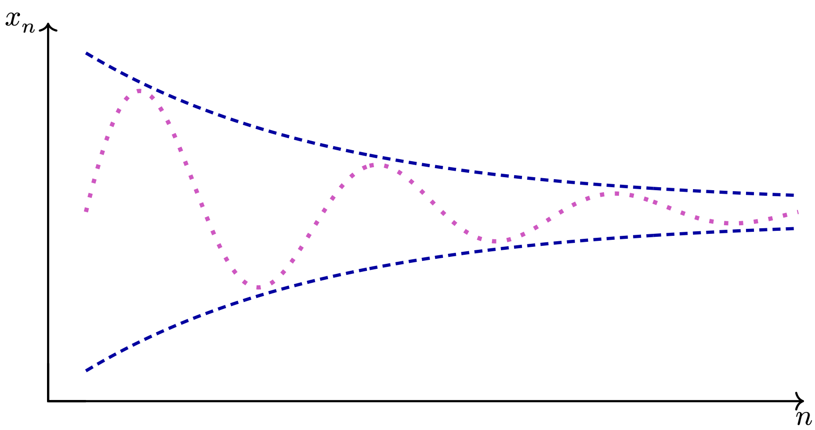
$$X \subseteq \bigcup_{i=1}^{n} B_\varepsilon(x_i) \subseteq \bigcup_{i=1}^{n} U_i,$$

we have a finite subcover $\{U_1, \dots, U_n\}$ of $\mathscr{U}$ for X. ☐

Corollary 3.6.17. *A metric space is compact if and only if it is sequentially compact.*

CHAPTER 4

COMPLETENESS



4.1 Definitions

Recall (Definition 3.6.8) that a sequence (x_n) in a metric space (X, d) is **Cauchy** if

$$\forall \varepsilon > 0, \exists N \in \mathbf{N} \quad \text{such that} \quad m, n \geq N \implies d(x_m, x_n) < \varepsilon.$$

Definition 4.1.1.

A metric space X is **complete** if every Cauchy sequence in X converges to a point in X.

Example 4.1.2.

1. $\mathbf{R}$ is complete by work done in Analysis 1.
2. $\mathbf{Q}$ is not complete: any sequence in $\mathbf{Q}$ which converges to $\sqrt{2} \in \mathbf{R}$ does not converge to any point in $\mathbf{Q}$.
3. $(0, 1)$ is not complete: the sequence $({}^1/n)_{n \geq 2}$ is Cauchy in $(0, 1)$ but does not converge in $(0, 1)$.

Remark 4.1.3.

From these examples we see that completeness is not a topological property. However, it is preserved under uniform equivalence (and therefore Lipschitz equivalence) [Exercise].

Proposition 4.1.4.

1. *A complete subspace W of a metric space X is closed in X.*
2. *A closed subspace W of a complete metric space X is complete.*

Proof. "1" Suppose that W is complete and that $x \in \overline{W}$. So there is a sequence (w_n) in W such that $\lim_n w_n = x$. Since (w_n) is convergent it is Cauchy, so by the completeness of W, (w_n) converges to a point in W. By the uniqueness of limits of sequences in metric spaces, this point must be x and so $x \in W$.

"2" Suppose that X is complete and that W is closed in X, i.e. $W = \overline{W}$. Let (w_n) be a Cauchy sequence in W. By the completeness of X, (w_n) converges to a point $x \in X$ and so $x \in \overline{W}$. So W is complete. □

Lemma 4.1.5. *If a Cauchy sequence (x_n) in a metric space (X, d) has a subsequence (x_{n_r}) that converges to x, then (x_n) converges to x.*

Proof. Let $\varepsilon > 0$. Since (x_n) is Cauchy,

$$\exists N \in \mathbf{N} \quad \text{such that} \quad m, n \geq N \implies d(x_m, x_n) < {}^{\varepsilon}/2.$$

Since (x_{n_r}) converges to x,

$$\exists R \in \mathbf{N}, \quad \text{such that} \quad r \geq R \implies d(x_{n_r}, x) < {}^{\varepsilon}/_2.$$

For any $n \geq N$, choose $r \geq R$ such that $n_r \geq N$, and then

$$d(x_n, x) \leq d(x_n, x_{n_r}) + d(x_{n_r}, x) < \varepsilon.$$

So (x_n) converges to x. □

Proposition 4.1.6. *Any compact metric space is complete.*

Proof. Suppose that X is compact and that (x_n) is a Cauchy sequence in X. By compactness, (x_n) has a convergent subsequence (x_{n_r}), converging to a point $x \in X$. Then (x_n) converges to x by Lemma 4.1.5. □

Example 4.1.7 ($\mathbf{R}^n$ is complete)**.** We will prove this with the Euclidean metric d_2 on $\mathbf{R}^n$. Suppose that $(x^r)_{r\geq 1}$ is a Cauchy sequence in $\mathbf{R}^n$ where

$$x^r = (x_1^r, x_2^r, \dots, x_n^r) \in \mathbf{R}^n.$$

For each $i \in \{1, \dots, n\}$, (x_i^r) is a Cauchy sequence in $\mathbf{R}$, since $|x_i^r - x_i^s| \leq d_2(x^r, x^s)$. So (x_i^r) converges to x_i say. We shall prove that (x^r) converges to $x = (x_1, \dots, x_n) \in \mathbf{R}^n$. Given $\varepsilon > 0$, for each i there exists $N_i \in \mathbf{N}$ such that $r \geq N_i$ implies that $|x_i^r - x_i| < {}^{\varepsilon}/_{\sqrt{n}}$. Then for all $r \geq \max\{N_1, \dots, N_n\}$,

$$d_2(x^r, x) \leq \sqrt{\sum_{i=1}^{n} (x_i^r - x_i)^2} < \varepsilon.$$

So (x^r) converges to $x \in \mathbf{R}^n$.

Remark 4.1.8.

The same proof shows that any finite product of complete metric spaces is complete.

4.2 Contractions

Definition 4.2.1 (Contraction)**.**

A function $f\colon X \to X$, where (X, d) is a metric space, is a **contraction** if there exists a constant $K < 1$ such that

$$d(f(x), f(y)) \leq K d(x, y), \quad \forall x, y \in X.$$

Remark 4.2.2.

Obviously, $K \geq 0$ and $K = 0$ if and only if f is a constant function.

Lemma 4.2.3. *Any contraction is uniformly continuous.*

Proof. Using the above notation, let $\varepsilon > 0$ and $\delta = \varepsilon/K$. For any $x, y \in X$ satisfying $d(x, y) < \delta$, we have

$$d(f(x), f(y)) \leq Kd(x, y) < \varepsilon,$$

as required. $\square$

Definition 4.2.4.

Given a set S, a **self-map** of S is a function $f\colon S \to S$. A **fixed point** of a self-map f of S is a point $p \in S$ such that $f(p) = p$.

Theorem 4.2.5 (Banach's Contraction Mapping Theorem).

If $f\colon X \to X$ is a contraction of a complete metric space X, then f has a unique fixed point $p \in X$.

Proof. We start by proving the existence of a fixed point. Choose $x_1 \in X$ arbitrarily and let $x_{n+1} = f(x_n)$, for $n \geq 1$. We shall prove that (x_n) is a Cauchy sequence. First note that $d(x_r, x_{r+1}) \leq K^{r-1} d(x_1, x_2)$, for all $r \geq 1$. This follows easily by induction from the contraction condition. Now for $m > n$, by repeated use of the triangle inequality,

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n).$$

So

$$\begin{aligned} d(x_m, x_n) &\leq (K^{m-2} + K^{m-3} + \cdots + K^{n-1}) d(x_1, x_2) \\ &= K^{n-1}(K^{m-n-1} + K^{m-n-2} + \cdots + 1) d(x_1, x_2) \\ &= \frac{K^{n-1}(1 - K^{m-n})}{1 - K} d(x_1, x_2) \\ &< \frac{K^{n-1}}{1 - K} d(x_1, x_2). \end{aligned}$$

Since $0 \leq K < 1$, $K^{n-1} \to 0$ as $n \to \infty$. So (x_n) is a Cauchy sequence and since X is complete it converges to some $p \in X$. By Lemma 4.2.3 f is continuous and so $f(x_n) \to f(p)$, as $n \to \infty$. But $f(x_n) = x_{n+1} \to p$ as $n \to \infty$. So $f(p) = p$.

We now proceed to show the uniqueness of the fixed point. If $f(p) = p$ and $f(q) = q$, then

$$d(p, q) = d(f(p), f(q)) \leq Kd(p, q).$$

Since $K < 1$, this is a contradiction unless $p = q$. So the fixed point is unique. $\square$

A remarkable aspect of the proof of the Contraction Mapping Theorem is that, not only does it prove the existence and uniqueness of the fixed point, it tells you how to find it: choose a random element $x_1 \in X$ and set $x_{n+1} = f(x_n)$. Then the unique fixed point p is the limit of the sequence (x_n), i.e. $p = \lim_n x_n$.

Example 4.2.6.

1. The completeness condition in the Contraction Mapping Theorem is necessary: the open interval $(0,1)$ is not a complete metric space. Consider $f\colon (0,1) \to (0,1)$; $f(x) = x/2$. This is a contraction with $K = 1/2$ but it has no fixed point since $0 \notin (0,1)$.
2. As a condition for a contraction it is not enough to say that $d(f(x), f(y)) < d(x,y)$, for all $x \neq y$ in X: the function $f\colon [1,\infty) \to [1,\infty)$, $f(x) = x + 1/x$ satisfies this condition, and $[1,\infty)$ is complete, but f has no fixed point since $f(x) = x \Rightarrow 1/x = 0$.
3. The statement that f is a contraction depends heavily on the choice of metric on X: it is possible that two metrics d and d' are Lipschitz equivalent on X, i.e. (X,d) is complete if and only if (X,d') is complete and that f is a d-contraction but not a d'-contraction.

Example 4.2.7 (Approximation Procedure)**.** Consider the function

$$g(x) = \frac{1}{2}\left(x + \frac{2}{x}\right).$$

This is a self-map of the interval $[1,2]$ because

$$1 \le x \le 2 \quad \Longrightarrow \quad 1/2 \le x/2 \le 1 \quad \text{and} \quad 1/2 \le 1/x \le 1,$$

so $1 \le g(x) \le 2$.

If $x, y \in [1,2]$ then

$$|g(x) - g(y)| = \left|\frac{x-y}{2} + \frac{y-x}{xy}\right| = |x-y| \cdot \left|\frac{1}{2} - \frac{1}{xy}\right| \le \frac{1}{2}|x-y|.$$

Therefore, $|g(x) - g(y)| \le |x-y|/2$, and g is a contraction with $K = 1/2$. Moreover, $[1,2]$ is complete as it is a closed subspace of the complete space $\mathbf{R}$. So by the Contraction Mapping Theorem g has a unique fixed point $p \in [1,2]$.

The fixed point must satisfy

$$p = g(p) = (p + 2/p)/2.$$

Therefore, $p^2 = 2$, and $p = +\sqrt{2}$.

Remember that we can pick a starting point $x_1 \in [1,2]$ at random and the sequence $(x_1, g(x_1), g^2(x_1), \dots)$ converges to the fixed point $\sqrt{2}$ for *every* $x_1 \in [1,2]$. If we choose $x_1 = 1$ we get

$$(1, 3/2, 17/12, 577/408, \dots) = (1, 1.5, 1.416666666, 1.414215686, \dots)$$

and if we choose $x_1 = 7/5$, then

$$(7/5, 99/70, 19601/13860, \dots) = (1.4, 1.414285714, 1.414213564, \dots)$$

to 9 decimal places. This is not bad as the correct answer is $1.414213562\dots$.

Remark 4.2.8.

This procedure can be generalised to compute the decimal expansion of $\sqrt{a}$ by considering

$$g(x) = \frac{1}{2}\left(x + \frac{a}{x}\right).$$

4.3 Function Spaces

Let $A \subseteq \mathbf{R}$.

Definition 4.3.1 (Pointwise Convergence)**.**

A sequence $(f_n : A \to \mathbf{R})$ **converges** to $f : A \to \mathbf{R}$ **pointwise** on A if $\lim_n f_n(x) = f(x)$, for all $x \in A$.

Add an entry for convergence in the index?

Example 4.3.2. Define $f_n(x) = x^n$ on $A = [0, 1]$. Then

$$\lim f_n(x) = f(x) := \begin{cases} 0, & x \in [0, 1); \\ 1, & x = 1. \end{cases}$$

Note that whilst each f_n is continuous, f is not.

Definition 4.3.3.

A sequence $(f_n : A \to \mathbf{R})$ **converges** to $f : A \to \mathbf{R}$ **uniformly** on A if

$$\forall \varepsilon > 0, \exists N \in \mathbf{N} \quad \text{such that} \quad n \geq N \implies |f_n(x) - f(x)| < \varepsilon, \quad \forall x \in A.$$

We write $f_n \to f$ uniformly on A and call f the **uniform limit**.

indexing?

Obviously, if $f_n \to f$ uniformly, then $f_n \to f$ pointwise on A. Uniform convergence is stronger in that N is independent of x, while for pointwise convergence we can use a different N for each x.

Remark 4.3.4.

1. Suppose that a sequence (a_n) in $\mathbf{R}$ converges to a. Then for any $A \subseteq \mathbf{R}$ let $f_n : A \to \mathbf{R}$ be the constant function $f_n(x) = a_n$ and $f : A \to \mathbf{R}$ be the constant function $f(x) = a$. Then $f_n \to f$ uniformly on A.
2. The sequence from Example 4.3.2 does not converge uniformly: let $\varepsilon = 1/2$ and for any $N \in \mathbf{N}$ choose $x \in (0, 1)$ such that $x^N = 1/2$ which exists by the

Intermediate Value Theorem. Then

$$|f_N(x) - f(x)| = |x^N - 0| = x^N = 1/2 = \varepsilon.$$

3. Warning: the convergence of $f_n(x) = x^n$ is not uniform on $[0,1)$ either – we can use the same proof. But if we take $A = [0,a]$, for $a \in (0,1)$, then the sequence does converge uniformly on $[0,a]$.

Proposition 4.3.5. *Let (f_n) be a sequence in $\mathscr{B}[a,b]$ and $f \in \mathscr{B}[a,b]$. Then*

$$\lim_n f_n = f \in \mathscr{B}[a,b] \iff f_n \longrightarrow f \text{ uniformly on } [a,b]$$

Proof. In $\mathscr{B}[a,b]$, $\lim f_n = f$ if and only if $d_{\text{sup}}(f_n, f) \to 0$, as $n \to \infty$.

"$\Rightarrow$" To prove that $f_n \to f$ uniformly on $[a,b]$ let $\varepsilon > 0$, and then there exists $N \in \mathbf{N}$ such that $0 \leq d_{\text{sup}}(f_n, f) < \varepsilon$. Since $|f_n(x) - f(x)| \leq d_{\text{sup}}(f_n, f)$, for all $x \in [a,b]$, by definition, we get that $|f_n(x) - f(x)| < \varepsilon$, for all $n \geq N$ and all $x \in [a,b]$

"$\Leftarrow$" Suppose that $f_n \to f$ uniformly on $[a,b]$ and that $\varepsilon > 0$. Then there exists $N \in \mathbf{N}$ such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon/2, \quad \text{for all } x \in [a,b].$$

Then

$$n \geq N \implies 0 \leq d_{\text{sup}}(f_n, f) = \sup_{x \in [a,b]} |f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon.$$

□

Definition 4.3.6 (Uniformly Cauchy).

A sequence (f_n) of functions $f_n \colon A \to \mathbf{R}$ is **uniformly Cauchy** on A if for all $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that

$$m, n \geq N \implies |f_m(x) - f_n(x)| < \varepsilon, \quad \forall x \in A.$$

Unif. Cauchy -> Cauchy in index

Theorem 4.3.7 (Cauchy's Criteria for Uniform Convergence).

Let (f_n) be a sequence of real-valued functions defined on $A \subseteq \mathbf{R}$. Then (f_n) converges uniformly on A if and only if (f_n) is uniformly Cauchy on A.

Proof. "$\Rightarrow$" Given $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that $n \geq N$ implies $|f_n(x) - f(x)| < \varepsilon/2$, for all $x \in A$. So for $m, n \geq N$,

$$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| < \varepsilon, \quad \forall x \in A.$$

"$\Leftarrow$" Suppose that (f_n) is uniformly Cauchy on A. Then for any $x \in A$, $(f_n(x))$ is a Cauchy sequence in $\mathbf{R}$ and therefore converges to a real number $f(x)$. Let $\varepsilon > 0$ and choose $N \in \mathbf{N}$ such that $m, n \geq N$ implies that $|f_m(x) - f_n(x)| < \varepsilon/2$, for all $x \in A$. Now keep n fixed and let $m \to \infty$ to get

$$n \geq N \implies |f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon, \quad \forall x \in A.$$

So $f_n \to f$ uniformly on A. $\square$

Corollary 4.3.8. *The metric space $\mathscr{B}[a, b]$ is complete.*

Proof. Let (f_n) be a Cauchy sequence in $\mathscr{B}[a, b]$. Then (f_n) is uniformly Cauchy on $[a, b]$ since for $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that

$$m, n \geq N \implies |f_m(x) - f_n(x)| \leq d_{\sup}(f_m, f_n) < \varepsilon, \quad \forall x \in [a, b].$$

So (f_n) converges uniformly on $[a, b]$, by Cauchy's Criteria for Uniform Convergence and hence is convergent in $\mathscr{B}[a, b]$ by Proposition 4.3.5. $\square$

Theorem 4.3.9.

If $f_n \colon [a, b] \to \mathbf{R}$ is continuous at $c \in [a, b]$, for each $n \in \mathbf{N}$, and $f_n \to f$ uniformly on $[a, b]$, then f is continuous at c.

Proof. Given $\varepsilon > 0$, uniform convergence implies that there exists $N \in \mathbf{N}$ such that

$$n \geq N \implies |f_n(x) - f(x)| < \varepsilon/3, \quad \forall x \in [a, b].$$

Then f_N is continuous at c so $\exists \delta > 0$ such that

$$x \in B_\delta(c) \cap [a, b] \implies |f_N(x) - f_N(c)| < \varepsilon/3.$$

So for any such x,

$$|f(x) - f(c)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| < \varepsilon.$$

$\square$

Corollary 4.3.10. *The metric space $\mathscr{C}[a, b]$ is complete.*

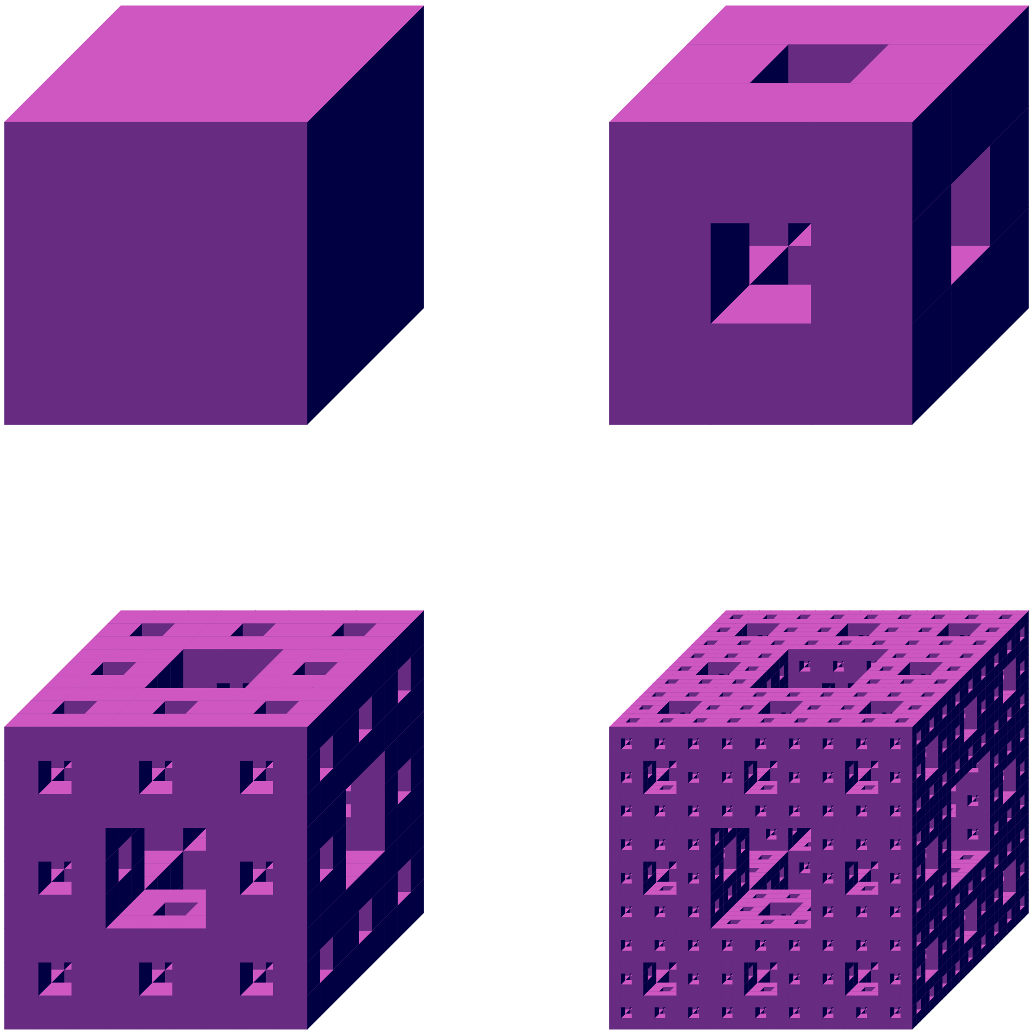
Proof. Let $\mathscr{C}_c$ be the subspace of functions in $\mathscr{B}[a, b]$ that are continuous at c. The previous theorem tells us that $\mathscr{C}_c$ is closed in the complete space $\mathscr{B}[a, b]$. Hence, $\mathscr{C}_c$ is complete. Observe that

$$\mathscr{C}[a, b] = \bigcap_{c \in [a,b]} \mathscr{C}_c,$$

which implies that $\mathscr{C}[a, b]$ is also closed in $\mathscr{B}[a, b]$ and hence complete. $\square$

CHAPTER 5

APPLICATIONS



5.1 Initial Value Problem

An **initial value problem** (**IVP**) is an ordinary differential equation of the form

$$\frac{dy}{dt} = f(t, y(t)),$$

with $f\colon U \subseteq \mathbf{R}^2 \to \mathbf{R}$, where U is an open set of $\mathbf{R}^2$, together with a point $(t_0, y_0) \in U$ which is called the **initial condition**.

A **solution** to an IVP is a function y that is a solution to the differential equation and satisfies $y(t_0) = y_0$.

We know how to find solutions to simple ODEs: for example, suppose

$$\frac{dy}{dt} = 2y \implies \int \frac{dy}{y} = \int 2\,\mathrm{d}t \implies |y| = \mathrm{e}^{C_1}\mathrm{e}^{2t}.$$

Let $C = \pm\mathrm{e}^{C_1}$ so $y = C\mathrm{e}^{2t}$. When we have an initial condition we can find C, for example, if $y(0) = 19$ we get $C = 19$ and the solution is $y(t) = 19\mathrm{e}^{2t}$.

Theorem 5.1.1 (Picard, Lindelof, Lipschitz and Cauchy[1])**.**

Given an IVT, suppose that

$$f\colon U \to \mathbf{R}$$

is continuous and Lipschitz continuous in y, that is, there exists $L > 0$ such that

$$|f(t, y_1(t)) - f(t, y_2(t))| \leq L|y_1(t) - y_2(t)|, \quad \forall t.$$

Then there is an $\varepsilon > 0$, such that, there exists a unique solution $y(t)$ to the IVP on the interval $[t_0 - \varepsilon, t_0 + \varepsilon]$.

The idea of the proof is to integrate both sides of the differential equation:

$$y(t) - y(t_0) = \int_{t_0}^{t} f(s, y(s))\,\mathrm{d}s,$$

and set $\varphi_0(t) = y_0$ to be the constant function. Then we perform what is called **Picard iteration** by setting:

$$\varphi_{k+1}(t) := y_0 + \int_{t_0}^{t} f(s, \varphi_k(s))\,\mathrm{d}s.$$

We see that φ_k satisfies the initial condition, for all $k \in \mathbf{N}$. The sequence (φ_n) will be Cauchy in $\mathcal{C}[t_0 - \varepsilon, t_0 + \varepsilon]$, which is complete and therefore (φ_n) converges to φ which satisfies

$$varphi(t) = y_0 + \int_{t_0}^{t} f(s, \varphi(s))\,\mathrm{d}s \implies \frac{d\varphi}{dt} = f(t, \varphi(t)) \quad \text{and} \quad \varphi(t_0) = y_0,$$

[1] Charles Émile Picard (1856–1941), Ernst Leonard Lindelöf (1870–1946), Rudolf Otto Sigismund Lipschitz (1832–1903) and Augustin Louis Cauchy (1789–1857).

as the fixed point of a contraction. So setting $y = \varphi(t)$ we have a solution to the IVP.

Add link to the proof

Example 5.1.2. Let $y(t) = \tan(t)$ and $t_0 = 0$. Then

$$\frac{dy}{dt} = 1 + y^2, \quad y_0 = 0 \quad \text{and} \quad \varphi_0(t) = 0.$$

Then

$$\varphi_{k+1}(t) = \int_0^t (1 + (\varphi_k(s))^2)\,\mathrm{d}s$$

and $\varphi_n(t) \to y(t)$, as $n \to \infty$. Calculating the first few Picard iterates:

$$\varphi_1(t) = \int_0^t (1 + 0^2)\,\mathrm{d}s = t; \quad \varphi_2(t) = \int_0^t (1 + s^2)\,\mathrm{d}s = t + \frac{t^3}{3};$$

$$\varphi_3(t) = \int_0^t \left(1 + \left(s + \frac{s^3}{3}\right)^2\right)\mathrm{d}s = t + \frac{t^3}{3} + \frac{2t^5}{15} + \frac{t^7}{63}.$$

This is calculating the series expansion of tan.

Proof of Theorem 5.1.1. Let $a, b > 0$ be such that

$$C_{a,b} = I \times B = [t_0 - a, t_0 + a] \times [y_0 - b, y_0 + b] \subseteq U.$$

So $C_{a,b}$ is a compact subspace on which f is defined. Let

$$M := \sup_{(t,y)\in C_{a,b}} |f(t,y)|.$$

Consider the metric space $\mathcal{C} := \mathcal{C}(I, B)$ of continuous functions from I to B with the $d_{\sup}$ metric. This is a subspace of the complete metric space $\mathcal{C}[t_0 - a, t_0 + a]$ and it is closed. Indeed, for any sequence (f_n) in $\mathcal{C}$ that converges to a point in $f \in \mathcal{C}[t_0 - a, t_0 + a]$ and for any $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that

$$n \geq N \implies |f(t) - f_n(t)| \leq d_{sup}(f, f_n) < \varepsilon, \quad \forall t \in I.$$

So for any $\varepsilon > 0$, there is an $N \in \mathbf{N}$ such that

$$n \geq N \implies y_0 - b - \varepsilon \leq f_n(t) - \varepsilon \leq f(t) \leq f_n(t) + \varepsilon \leq y_o + b + \varepsilon, \quad \forall t \in I$$

and so $f(t) \in B$, for all $t \in I$. Therefore, $\mathcal{C}$ is closed in $\mathcal{C}[t_0 - a, t_0 + a]$ and therefore is complete.

Now we define the self-map $\Gamma\colon \mathcal{C} \to \mathcal{C}$ given by

$$\Gamma\varphi(t) := y_0 + \int_{t_0}^t f(s, \varphi(s))\,\mathrm{d}s.$$

First, we need to show that Γ is actually a self-map, that is, that $\Gamma\varphi \in \mathcal{C}$, for all $\varphi \in \mathcal{C}$. If $\varphi \in \mathcal{C}$, then $|\varphi(t) - y_0| \leq b$, for all $t \in I$. And so,

$$|\Gamma\varphi(t) - y_0| = \left|\int_{t_0}^{t} f(s,\varphi(s))\,\mathrm{d}s\right| \leq \left|\int_{t_0}^{t'} f(s,\varphi(s))\,\mathrm{d}s\right| \leq M|t' - t_0| \leq Ma,$$

where t' is some number in I where the maximum is achieved. So we need $a \leq b/M$.

Now we will prove that Γ is a contraction. Let $t \in I$ be such that

$$d_{\sup}(\Gamma\varphi_1, \Gamma\varphi_2) = |\Gamma\varphi_1(t) - \Gamma\varphi_2(t)|.$$

Then by the definition of Γ:

$$\begin{aligned}
|(\Gamma\varphi_1 - \Gamma\varphi_2)(t)| &= \left|\int_{t_0}^{t} f(s,\varphi_1(s)) - f(s,\varphi_2(s))\,\mathrm{d}s\right| \\
&\leq \int_{t_0}^{t} |f(s,\varphi_1(s)) - f(s,\varphi_2(s))|\,\mathrm{d}s \\
&\leq L\int_{t_0}^{t} |\varphi_1(s) - \varphi_2(s)|\,\mathrm{d}s \leq La \cdot d_{\sup}(\varphi_1,\varphi_2),
\end{aligned}$$

since f is Lipschitz continuous in y. So Γ is a contraction if a is chosen so that $a < 1/L$.

So Picard's operator Γ is a contraction on the complete metric space $\mathcal{C}$ so there exists a unique function $\varphi \in \mathcal{C}$ such that $\Gamma\varphi = \varphi$. This function is the unique solution of the IVP valid on $[t_0 - a, t_0 + a]$ where a satisfies the condition that $a < \min\{b/M, 1/L\}$. □

5.2 Cantor's Intersection Theorem

Theorem 5.2.1 (Cantor[2]).

A metric space X is complete if and only if for any sequence (V_n) of non-empty closed subsets of X, with $V_{n+1} \subseteq V_n$, for all $n \in \mathbf{N}$, such that $\operatorname{diam} V_n \to 0$, as $n \to \infty$, we have

$$\bigcap_{n\in\mathbf{N}} V_n \neq \varnothing.$$

Remark 5.2.2.

If $\operatorname{diam} V_n \to 0$, as $n \to \infty$, then $|\bigcap_{n\in\mathbf{N}} V_n| \in \{0,1\}$.

Proof. "⇒" Let $x_n \in V_n$. Then (x_n) is a Cauchy sequence since given $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that $\operatorname{diam} V_N < \varepsilon$, then for any $m,n \geq N$, $x_m, x_n \in V_N$, so $d(x_m,x_n) < \varepsilon$.

[2] Georg Ferdinand Ludwig Philipp Cantor (1845–1918).

By the completeness of X, (x_n) converges to some $x \in X$. Since $x_n \in V_m$, for all $n \geq m$, $x \in \overline{V_m} = V_m$. This is true for any m, so

$$\bigcap_{n \in \mathbf{N}} V_n = \{x\}.$$

"$\Leftarrow$" Suppose X is not complete and let (x_n) be a non-convergent Cauchy sequence in X. Let $V_n = \{x_r \mid r \geq n\}$. Then V_n is closed since, if it were not, there would be a sequence (v_j) in $V := V_n$ that converges to a point $x \in \overline{V} \setminus V$. Since $v_j = x_r$, for some $r \geq n$, (v_j) is a subsequence (x_{n_k}) of (x_n), which would imply that (x_n) converges to x as (x_n) is Cauchy (see Lemma 1.5 from the notes from Week 10). Clearly, $V_{n+1} \subseteq V_n$, for all $n \in \mathbf{N}$. Also, given $\varepsilon > 0$, there exists $N \in \mathbf{N}$ such that $m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon$, so $\operatorname{diam} V_N \leq \varepsilon$ and $\operatorname{diam} V_n \to 0$, as $n \to \infty$. Suppose that $x \in V_n$, for all $n \in \mathbf{N}$. Then $\lim x_n = x$ since, for any $n \in \mathbf{N}$, $d(x, x_n) \leq \operatorname{diam} V_n$, so $d(x, x_n) \to 0$, as $n \to \infty$. This contradicts our assumption and so, $\bigcap_{n \in \mathbf{N}} V_n = \varnothing$. □ ref.?

5.3 The Cantor Set

Definition 5.3.1.

The **Cantor Set** $K \subseteq \mathbf{R}$ is defined by the following inductive procedure. Let $K_0 = [0, 1]$ and define $K_1 \subseteq K_0$ by deleting its open middle third; so $K_1 = [0, 1/3] \cup [2/3, 1]$ is the union of two closed intervals, each of length $1/3$. Let K_n be the union of 2^n closed intervals, each of length $1/3^n$, and define K_{n+1} by deleting the open middle third of each; so K_{n+1} is the union of 2^{n+1} closed intervals, each of length $1/3^{n+1}$. Then

$$K := K_0 \cap K_1 \cap \cdots \cap K_n \cap \cdots = \bigcap_{n \in \mathbf{N}} K_n.$$

Observe, K is non-empty, because the endpoints 0 and 1 are never removed from $[0, 1]$. We want a way to describe all points in K.

There are different ways to describe expansions of real numbers. Let $a \in [0, 1]$.

Decimal expansion

$$a = 0.a_1 a_2 \ldots a_n \ldots := \sum_{n=1}^{\infty} \frac{a_n}{10^n}, \quad a_n \in \{0, \ldots, 9\}.$$

e.g. $1/2 = 0.5 = 0.4999\ldots$ and $1 = 0.999\ldots$.

Binary expansion

$$a = 0.a_1 a_2 \ldots a_n \ldots := \sum_{n=1}^{\infty} \frac{a_n}{2^n}, \quad a_n \in \{0, 1\}.$$

e.g. $1/2 = 0.1 = 0.0111\ldots$ and $1 = 0.111\ldots$.

Ternary expansion

$$a = 0.a_1 a_2 \dots a_n \dots := \sum_{n=1}^{\infty} \frac{a_n}{3^n}, \quad a_n \in \{0, 1, 2\}.$$

Ternary expansions are unique except that $0.a_1 \dots a_{n-1} a_n 222 \dots = 0.a_1 \dots a_{n-1} a'_n 000 \dots$ whenever $a_n \neq 2$ and $a'_n = a_n + 1$. So $0 = 0.000\dots$ and $1 = 0.222\dots$. We also know that $1/3, 2/3 \in K$ and

$$1/3 = 0.0222\dots = 0.1000\dots \quad \text{and} \quad 2/3 = 0.2000\dots = 0.1222\dots .$$

We know that $1/2 \notin K$ and $1/2 = 0.111\dots$.

Theorem 5.3.2.

The Cantor Set K consists of all numbers $a \in [0, 1]$ which have a ternary expansion containing only 0s and 2s.

Proof. The middle third of K_0 consists of exactly a whose ternary expansions have $a_1 = 1$. So $a \in K_1$ if and only if it has an expansion $a_1 \neq 1$. This includes $1/3$ and $2/3$.

The middle third of $[0, 1/3] \subseteq K_1$ consists exactly of those a which have a ternary expansion with $a_1 = 0$ and $a_2 = 1$. Similarly, the middle third of $[2/3, 1]$ consists of those a which have a ternary expansion with $a_1 = 2$ and $a_2 = 1$. So $a \in K_2$ if and only if it has an expansion satisfying $a_1, a_2 \neq 1$.

Proceeding inductively, the middle third of each of the 2^n closed intervals that make up K_n consists of those a whose ternary expansions have $a_1 = 0$ or $2, \dots, a_n = 0$ or 2 and $a_{n+1} = 1$. So $a \in K_{n+1}$ if and only if it has an expansion satisfying $a_{n+1} \neq 1$, and so creating 2^{n+1} closed intervals. This holds for all $n \geq 1$ and the result follows. $\square$

It might seem that the only numbers left in K are the endpoints of the closed intervals, i.e. numbers of the form $k/3^n$, but actually $1/4 \in K$ since its ternary expansion is $1/4 = 0.020202\dots$. However, points of the form $k/3^n$ are dense in K.

Corollary 5.3.3. *The Cantor Set is uncountable.*

Proof. Suppose elements of K can be listed $a_m = 0.a_{m_1} a_{m_2} \dots$, where $m \in \mathbf{N}$ and $a_{m_j} \in \{0, 2\}$, for all $j \in \mathbf{N}$. Then define

$$b := 0.b_1 b_2 \dots, \quad \text{where} \quad b_j = \begin{cases} 2, & \text{if } a_{j_j} = 0; \\ 0, & \text{if } a_{j_j} = 2, \end{cases}$$

for every $j \in \mathbf{N}$. Then $b \in K$ and $b_j \neq a_{j_j}$ which implies that $b \neq a_j$, for any $j \in \mathbf{N}$. So b is not on the list and K is uncountable. $\square$

The Cantor Set is, in fact, in bijection with $[0, 1]$. Let's compute the sums of the lengths of the intervals that are removed from $[0, 1]$ to create K:

$$\frac{1}{3} + \frac{2}{3^2} + \frac{2^2}{3^3} + \dots + \frac{2^{n-1}}{3^n} + \dots = \frac{1}{3} \cdot \frac{1}{1 - 2/3} = 1.$$

So the Cantor Set has length 0 but still the same number of elements as $[0,1]$!

Proposition 5.3.4. *The Cantor Set is closed.*

Proof. The closed interval $K_0 = [0,1]$ is closed and K_1 is obtained by removing an open interval, this implies that K_1 is closed. Assume K_n is closed. Then K_{n+1} is obtained by removing a finite number of open intervals and so K_{n+1} is also closed. So $K = \bigcap_{n\in\mathbf{N}} K_n$ is an infinite intersection of closed sets and is therefore closed. $\square$

Corollary 5.3.5. *The Cantor Set is compact and complete.*

Proof. Clearly K is bounded, and since it is also closed, by the Heine–Borel Theorem, it is compact. Also, every compact metric space is complete. $\square$

It is a self-similar fractal in that $K \cap [0, 1/3^n]$ (and every other similar self-interval) is in bijection with K; as can be seen:

$$\begin{aligned} K \cap [0, 1/3^n] &\longrightarrow K \\ 0.0\ldots 0a_{n+1}a_{n+2}\ldots &\longmapsto 0.a_{n+1}a_{n+2}\ldots \end{aligned}$$

Proposition 5.3.6. *The Cantor Set in nowhere dense and $\partial K = K$.*

Proof. Since $\partial K = \overline{K}\backslash K^\circ$ and $\overline{K} = K$ we just need to show that $K^\circ = \varnothing$. Let $a \in K$ have ternary expansion $0.a_1a_2\ldots$ and let $\varepsilon > 0$. Choose $N \in \mathbf{N}$ such that $b = 0.0\ldots 0b_Nb_{N+1}\ldots$ must satisfy $0 < b < \varepsilon$. Then $0.a_1a_2\ldots a_{N-1}111\ldots$ lies in $B_\varepsilon(a) = (a-\varepsilon, a+\varepsilon)$ but not in K. Therefore, $K^\circ = \varnothing$.

Now $(\overline{K})^\circ = \varnothing$, so K is nowhere dense. $\square$

Example 5.3.7. It is interesting to compare the Cantor Set K with $\mathbf{Q}\cap[0,1] \subseteq \mathbf{R}$ which is countable, not closed and therefore not compact and $\partial(\mathbf{Q}\cap[0,1]) = [0,1] \neq \mathbf{Q}\cap[0,1]$.

5.4 Iterated Function Systems

5.4.1 The Hausdorff Metric

Let (X,d) be a metric space with a subset $B \subseteq X$ and a point $x \in X$. The **distance between x and B** is defined to be

$$d(x,B) := \inf\{d(x,b) \mid b \in B\}.$$

It is easy to see that if $x \in B$, then $d(x,B) = 0$, but the converse is not true in general (can you think of an example?). We can also show (Exercise) that if B is compact and non-empty, then there exists $b^* \in B$ such that $d(x,B) = d(x,b^*)$.

If $A, B \subseteq X$ are non-empty compact subspaces, then we define the **distance from A to B** by

$$d(A,B) := \sup\{d(a,B) \mid a \in A\}.$$

Example 5.4.1. It is not true that $d(A,B)$ and $d(B,A)$ are equal in general. As subspaces of $\mathbf{R}$ we see that for $A = [0,3]$ and $B = [0,1] \cup [2,3]$ that

$$d(A,B) = \sup_{a \in A} d(a,B) = \sup_{a \in A} \begin{cases} 0, & a \in B; \\ \min\{a-1, 2-a\}, & a \notin B. \end{cases} = \frac{1}{2}.$$

But $d(B,A) = 0$ as $B \subseteq A$.

Let $\mathscr{H}$ be the set of all non-empty compact subsets of X and define

$$d_H(A,B) := \max\{d(A,B), d(B,A)\}.$$

Exercise 5.4.2. If $A \subseteq X$ and $\varepsilon > 0$, the **ε-neighbourhood**[3] of A is given by

$$A_\varepsilon := \{x \in X \mid d(x,A) < \varepsilon\}.$$

Show that $d_H(A,B) < \varepsilon$ if and only if $A \subseteq B_\varepsilon$ and $B \subseteq A_\varepsilon$.

Proposition 5.4.3. *The pair $(\mathscr{H}, d_H)$ forms a metric space.*

This is known as the **Hausdorff space**[4] associated to (X,d), which measures how far two compact subspaces of (X,d) are from each other.

Proof. For (M1) and (M2), we see that $d_H(A,B) = 0$ if and only if $A \subseteq B$ and $B \subseteq A$. So $d_H(A,B) = 0$ if and only if $A = B$. The axiom (M3) is automatically satisfied as the definition is symmetric. For the triangle inequality (M4) it is enough to show that $d(A,B) \leq d(A,C) + d(C,B)$ because interchanging the role A and B gives the desired result. For any $a \in A$ we have

$$\begin{aligned} d(a,B) &= \inf_{b \in B} d(a,b) \\ &\leq \inf_{b \in B} d(a,c) + d(c,b), && \forall c \in C \\ &= d(a,c) + d(c,B), && \forall c \in C \\ &\leq d(a,c) + d(C,B), && \forall c \in C. \end{aligned}$$

Minimising over C gives

$$d(a,B) \leq d(a,C) + d(C,B), \quad \forall a \in A.$$

Maximising over A on the right side gives

$$d(a,B) \leq d(A,C) + d(C,B), \quad \forall a \in A.$$

Then maximising over A on the left side gives what we are looking for. □

[3]index?

[4]Felix Hausdorff (1868–1942).

Theorem 5.4.4 (Federer,[5] 1969).

The associated Hausdorff metric space $(\mathscr{H}, d_H)$ of a complete metric space (X, d) is complete.

Sketch Proof. Suppose (A_n) is a Cauchy sequence in $(\mathscr{H}, d_H)$. Define A to be the set of all $x \in X$ with the property that there exists a sequence of points $x_n \in A_n$ with $x_n \to x$. It is fairly easy to prove that A is compact and non-empty and is the limit of (A_n) in the Hausdorff metric. □

5.4.2 Lipschitz Constant

Let $f\colon X \to X$ be a self-map of X. Then we define the **Lipschitz constant**

$$\operatorname{Lip} f := \sup_{x \neq y} \frac{d(f(x), f(y))}{d(x, y)} \tag{5.1}$$

if the supremum exists, otherwise, we set $\operatorname{Lip} f = +\infty$.

Lemma 5.4.5. *If* $\operatorname{Lip} f = K \in \mathbf{R}$, *then* $K \geq 0$, *and*

$$d(f(x), f(y)) \leq K d(x, y), \quad \text{for all } x, y \in X,$$

and, moreover, $\operatorname{Lip} f$ *is the least such* K.

Proof. If $\operatorname{Lip} f \in \mathbf{R}$, then, since it is the supremum of a set of non-negative numbers, $K \geq 0$. Also, for $x \neq y$,

$$\frac{d(f(x), f(y))}{d(x, y)} \leq \operatorname{Lip} f = K,$$

which gives the inequality as the case $x = y$ is trivial.

Moreover, $\operatorname{Lip} f$ is the least such K otherwise it would contradict it being the supremum as defined in Equation (5.1). □

Therefore, if $\operatorname{Lip} f < 1$, we know that f is a contraction.

Lemma 5.4.6. *Let (X, d) be a metric space with associated Hausdorff space $(\mathscr{H}, d_H)$. Suppose that $f\colon X \to X$ is a self-map. Then:*

1. $d_H(f(A_1), f(A_2)) \leq \operatorname{Lip} f \cdot d_H(A_1, A_2)$;
2. $d_H(A_1 \cup A_2, B_1 \cup B_2) \leq \max\{d_H(A_1, B_1), d_H(A_2, B_2)\}$,

where $A_1, A_2, B_1, B_2 \in \mathscr{H}$.

[5] Herbert Federer (1920–2010).

Proof. "1" We have

$$\begin{aligned} d(f(A_1), f(A_2)) &= \sup_{a\in A_1} d(f(a), f(A_2)) = \sup_{a\in A_1} \inf_{a'\in A_2} d(f(a), f(a')) \\ &\le \sup_{a\in A_1} \inf_{a'\in A_2} Kd(a,a') = Kd(A_1, A_2), \end{aligned}$$

where $K = \operatorname{Lip} f$. Similarly, $d(f(A_2), f(A_1)) \le Kd(A_2, A_1)$. This implies the result.

"2" It is easy to see that

$$\begin{aligned} d(A_1\cup A_2, B) = \sup_{a\in A_1\cup A_2} d(a,B) &= \max\{\sup_{a\in A_1} d(a,B), \sup_{a\in A_2} d(a,B)\} \\ &= \max\{d(A_1,B), d(A_2,B)\}. \end{aligned}$$

for any $B \in \mathscr{H}$, which implies

$$d(A_1\cup A_2, B_1\cup B_2) = \max\{d(A_1, B_1\cup B_2), d(A_2, B_1\cup B_2)\}. \tag{5.2}$$

We also have

$$\begin{aligned} d(A, B_1\cup B_2) &= \sup_{a\in A} d(a, B_1\cup B_2) = \sup_{a\in A} \inf_{b\in B_1\cup B_2} d(a,b) \\ &= \sup_{a\in A} \min\{d(a,B_1), d(a,B_2)\} \le \min\{d(A,B_1), d(A,B_2)\}, \end{aligned}$$

which implies

$$d(A, B_1\cup B_2) \le d(A, B_i), \quad \text{for } i = 1, 2. \tag{5.3}$$

Substituting Equation (5.3) into Equation (5.2) gives

$$d(A_1\cup A_2, B_1\cup B_2) \le \max\{d(A_1,B_1), d(A_2,B_2)\}.$$

So

$$\begin{aligned} d_H(A_1\cup A_2, B_1\cup B_2) &= \max\{d(A_1\cup A_2, B_1\cup B_2), d(B_1\cup B_2, A_1\cup A_2)\} \\ &\le \max\{d(A_1,B_1), d(A_2,B_2), d(B_1,A_1), d(B_2,A_2)\} \\ &= \max\{d_H(A_1,B_1), d_H(A_2,B_2)\}. \end{aligned}$$

□

5.4.3 Iterated Function Systems

An **iterated function system** (**IFS**) is a finite set of contractions on a complete metric space (X, d), that is,

$$\{f_i \colon X \implies X \mid i = 1, 2, \dots, N\},$$

where $N \in \mathbf{N}$ and each f_i is a contraction on a complete metric space X.

We are interested in subsets $K \subseteq X$ that are **invariant** under an IFS, that is,

$$K = \bigcup_{i=1}^{N} f_i(K) = f_1(K) \cup f_2(K) \cup \cdots \cup f_N(K).$$

Given a compact subset $A \subseteq X$ we define

$$f(A) := \bigcup_{i=1}^{N} f_i(A) = f_1(A) \cup f_2(A) \cup \cdots \cup f_N(A). \tag{5.4}$$

This induces a self-map $f\colon \mathscr{H} \to \mathscr{H}$ as all contractions are continuous, the continuous image of compact spaces are compact, and finite unions of compact spaces are compact.

Theorem 5.4.7 (Hutchinson,[6] 1981)**.**

Given an iterated function system on $\mathbf{R}^n$, there exists a unique non-empty compact invariant subset.

Proof. Let $\{f_i\colon \mathbf{R}^n \to \mathbf{R}^n \,|\, i = 1, 2, \dots, N\}$ be an IFS on $\mathbf{R}^n$ and let K_i be the contraction constant for f_i. Since $\mathbf{R}^n$ is complete we know by Theorem 5.4.4 that its associated Hausdorff space $(\mathscr{H}, d_H)$ is also complete. We will show that the map $f\colon \mathscr{H} \to \mathscr{H}$ given in Equation (5.4) is a contraction.

Let $K = \max\{K_i \mid i = 1, 2, \dots, N\}$. Then, for $A, B \in \mathscr{H}$,

$$\begin{aligned}
d_H(f(A), f(B)) &= d_H\left(\bigcup_{i=1}^{N} f_i(A), \bigcup_{i=1}^{N} f_i(B)\right) \\
&\leq \max\{d_H(f_i(A), f_i(B)) \mid i = 1, 2, \dots, N\} && \text{by Lemma 5.4.6(2)} \\
&\leq \max\{K_i d_H(A, B) \mid i = 1, 2, \dots, N\} && \text{by Lemma 5.4.6(1)} \\
&\leq K d_H(A, B).
\end{aligned}$$

So f is a contraction on the complete metric space $(\mathscr{H}, d_H)$ and so there is a unique point $C \in \mathscr{H}$ such that $f(C) = C$, that is, there is a unique non-empty compact invariant subspace C of $\mathbf{R}^n$. $\square$

Obviously, if there is only one contraction in our IFS we have just one fixed point $S = \{p\}$. As an example, let $f\colon \mathbf{R} \to \mathbf{R}$ be the contraction given by $f(x) = x/2$, which has 0 as its unique fixed point. Hutchinson's Theorem tells us that for *any* compact subset $S_0 \subseteq \mathbf{R}$, if we define $S_{k+1} := f(S_k)$, that the sequence (S_n) converges to $\{0\}$, as $n \to \infty$. Taking $S_0 = \{1\}$ gives us $S_n = \{1/2^n\}$. Whilst taking $S_0 = [-1, 1]$ gives us $S_n = [-1/2^n, 1/2^n]$.

5.4.4 The Chaos Game

It becomes much more interesting when an IFS consists of several contractions. Our unique invariant subsets often take the form of interesting *fractals*!

Our first such example will consist of two contractions on $\mathbf{R}$:

$$\{f_1(x) = \frac{x}{3}, f_2(x) = 1 + \frac{(x-1)}{3}\}.$$

[6] John Edward Hutchinson (1946).

Then our unique compact invariant set, which we will denote by C, is the famous Cantor Set! We can generate this set by starting with a compact subset $S_0 \subseteq \mathbf{R}$ and then define $S_{n+1} = f(S_n)$ as before.

Exercise 5.4.8. Calculate S_i, for $i = 0, 1, \dots, 3$, when $S_0 = [0, 1]$ and when $S_0 = \{0, 1\}$.

The *Chaos Game* comes about when we want to plot an IFS using a computer. This is usually done in $\mathbf{R}^2$. We begin by choosing a point $x_0 \in S \subseteq \mathbf{R}^2$ (we can take it as one of the fixed points of the contractions in our IFS). Then we *randomly* choose one of our contractions f_i from our IFS and apply this to x_0 to obtain $x_1 = f_i(x_0)$. We program the computer to do this for a large number of iterations (e.g. 1,000,000) randomly choosing a contraction from our IFS every time. Plotting all of these points in $\mathbf{R}^2$ will converge to the invariant set S of our IFS *every* time, even though the choices of which contraction to use are random!

Another famous IFS is known as the **Sierpiński Triangle**[7] which is a fractal with the overall shape of an equilateral triangle; see the picture on the title page for an illustration.

Exercise 5.4.9. Do some research and write a brief paragraph about the Sierpiński Triangle. What three contractions are used to generate it? Go to the website `https://scratch.mit.edu/projects/170702288` and create some different Sierpiński triangles by moving the vertices of the triangle around (it takes a bit of time to generate a decent picture).

One of the most fascinating things about the fractals coming from iterated function systems is that they can often be used to model objects in nature. A famous example of this is **Barnsley's Fern** which is generated by the following four contractions of $\mathbf{R}^2$:

$$f_1(x) = \begin{pmatrix} 0 & 0 \\ 0 & 0.16 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad f_2(x) = \begin{pmatrix} 0.85 & 0.04 \\ -0.04 & 0.85 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1.6 \end{pmatrix},$$

$$f_3(x) = \begin{pmatrix} 0.2 & -0.26 \\ 0.23 & 0.22 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1.6 \end{pmatrix}, \quad f_4(x) = \begin{pmatrix} -0.15 & 0.28 \\ 0.26 & 0.24 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0.44 \end{pmatrix}.$$

Add picture

When we play the Chaos Game with the Barnsley Fern we randomly choose one of the four contractions of $\mathbf{R}^2$ with a certain probability p. For the standard Barnsley Fern

$$p(f_1) = 0.01, \quad p(f_2) = 0.85, \quad p(f_3) = 0.07, \quad p(f_4) = 0.07.$$

Exercise 5.4.10. Go to the website `https://www.chradams.co.uk/fern/maker.html` to see the standard Barnsley Fern and play around with this "fern generator". Ask it to plot a small number of points (e.g. 100) and then increase this by a factor of 10 each time to see the fern appear before your eyes. Play around with some of the values defining the contractions and their associated probabilities to generate your own unique fern!

[7] Wacław Franciszek Sierpiński (1882–1969).

5.5 Data Science and Clustering Algorithms

Data science, also sometimes called *big data*, is a field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from extremely large amounts of data. It is a form of statistics using data analysis, machine learning etc. as well as metric spaces!

A common technique for data analysis is *cluster analysis* which groups a set of objects together in such a way that objects in the same group (called a *cluster*) are more similar (in some sense) to each other than to those in other clusters. To do this we can use the notion of distance/metric: so that nearby objects are more similar than objects that are far away.

Applications

k-means clustering

We partition n observations into k clusters in which each observation belongs to a cluster with the nearest *mean*. We are assuming that n is very very large and that k is much more manageable. The idea is to minimise the distance between points in one cluster and a point which is the centre (mean) of another cluster. An algorithm to do this is:

1. Randomly select k points as the means;
2. Associate each point to its closest mean to create k clusters;
3. The *total cost* is the sum of the distances of points in a cluster to their means;
4. For each mean m and for each non-mean p:
   a) swap m and p, associate each point to its closest mean and recompute the total cost;
   b) if the total cost increases, then undo the swap.

There has been recent work (2016) exploring the mathematical theory of clustering in metric spaces. You will be able to find the pdf file for the relevant readable paper on the Learning Mall. We will now try to explain the main results and you will be able to find the relevant proofs in the paper.

Put it on LMO

Let (X, d) be a metric space where X is finite, i.e. $|X| = n$ for some $n \in \mathbf{N}$.

Definition 5.5.1 (Relative Distance).

For points $x, y \in X$, the **relative distance** from x to y is given by

$$\mathrm{RD}(x \mid y) := \frac{1}{n} \sum_{z \in X} (d(x, y) - d(x, z)).$$

The **relative distance** from a random point to y is

$$\mathrm{RD}(y) := \frac{1}{n} \sum_{z \in X} \mathrm{RD}(z \mid y).$$

Remark 5.5.2.

Note that $\mathrm{RD}(x \mid y)$ is not symmetric and may be negative.

Let $S_1, S_2 \subseteq X$ be two subsets. Then the **relative distance** from S_1 to S_2 is

$$\mathrm{RD}(S_1 \mid S_2) := \frac{1}{|S_1| \cdot |S_2|} \sum_{x \in S_1, y \in S_2} \mathrm{RD}(x \mid y).$$

Definition 5.5.3 (Cohesion).

The **cohesion** between two points $x, y \in X$ is given by

$$\gamma(x, y) := \mathrm{RD}(y) - \mathrm{RD}(x \mid y).$$

The points x and y are **cohesive** if $\gamma(x, y) \geq 0$ and are **incohesive** if $\gamma(x, y) \leq 0$.

Proposition 5.5.4. *For all $x, y \in X$:*

1. $\gamma(x, y) = \gamma(y, x)$;
2. $\gamma(x, x) \geq 0$;
3. $\gamma(x, x) \geq \gamma(x, y)$;
4. $\sum_{y \in X} \gamma(x, y) = 0$.

Again, for two subsets $S_1, S_2 \subseteq X$ we define

$$\gamma(S_1, S_2) := \sum_{x \in S_1, y \in S_2} \gamma(x, y),$$

with S_1 and S_2 **cohesive** if $\gamma(S_1, S_2) \geq 0$ and **incohesive** if $\gamma(S_1, S_1) \leq 0$.

Definition 5.5.5 (Cluster).

Suppose $S \neq \varnothing$, $S \subseteq X$, is a **cluster** if $\gamma(S, S) \geq 0$, that is, S is cohesive to itself.

Theorem 5.5.6.

Let S be a non-empty subset of X with $S \neq X$. Then the following are equivalent:

I. S is a cluster;

II. $X \setminus S$ *is a cluster;*

III. $\gamma(S, X \setminus S) \leq 0$*;*

IV. $\gamma(S, S) \geq \gamma(S, X \setminus S)$*.*

An algorithm for creating clusters is given by first putting each element into its own cluster, i.e. $S_i = \{x_i\}$ for $i = 1, \ldots, n$. Now compute $\gamma(S_i, S_j) = \gamma(x_i, x_j)$, for all i, j. If there exists i, j such that $\gamma(S_i, S_j) > 0$ then merge S_i and S_j into a new set $S_k = S_i \cup S_j$. Keep doing this process until it finishes.

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